



FITTER-EU

DELIVERABLE 4.2

Methodology of the co-creation process and report on mitigation measures



**Funded by
the European Union**

Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union. Neither the European Union nor the granting authority can be held responsible for them.

PROJECT ACRONYM	FITTER-EU
Project ref. number	101132546
Document title	Methodology of the co-creation process and report on mitigation measures
Document type	R — Document, report
Due date of deliverable	28/02/2026
Submission date	20/02/2026
Status	Draft/Final
Dissemination level	SEN - Sensitive
Language	English
Organisation responsible of deliverable	SV
Author(s)	Maria Sangiuliano, Oriana Salomón Balsamo
With contributions by	FGB – Silvia Sansonetti, Roberta Paoletti SFC – Bence Juhász TU Dublin – Sara Clavero, Maria Chiara Leva TFC – Mary Corbett, Sushmita Puhan UCM – Paloma Piqueiras, Olga Kolotouchkina UNINOVA – Joao Sarraipa, André Rocha WECF – Pia Wieser, Kiara Groneweg, Franzi Wiedmann

REVISION HISTORY			
Version	Date	Modified by	Comments
First draft version delivered for revision	16/02/2026	SV (Oriana Salomón Balsamo, Maria Sangiuliano)	
Second draft version – review-	18/02/2026	EUT (Laura Armayones), TU Dublin (Sara Clavero) TFC (Mary Corbett)	
Final version	20/02/2026	SV (Oriana Salomon)	

Summary

This deliverable presents the methodology and results of Task 4.2 “Co-creating intersectional mitigation plans” implemented within FITTER-EU. The objective of the task was to develop a structured co-creation process to identify and propose mitigation measures addressing the negative effects of twin transition policies on disadvantaged groups (DGs) in the energy, housing and transport sectors.

The co-creation process was implemented through national laboratories in six country case studies: Germany, Hungary, Ireland, Italy, Portugal and Spain. In each country, a series of in-presence and online sessions were organised between September and December 2025, involving intermediary organisations representing DGs, policymakers, policy experts and civil society actors. The methodology was grounded in the just transition framework developed in WP2 and the risk and scenario analysis carried out in WP3. It combined a systemic approach with structured facilitation tools, including system mapping exercises to analyse the twin transition negative effects in three sectors —housing, energy and transport— to move towards the co-creation of policy proposals —‘mitigation measures’— aimed at promoting a just transition.

As a result of the process, a portfolio of mitigation measures was developed in each country, including both adaptations of existing policies and new policy proposals. These measures were analysed in terms of their transformative potential and feasibility and classified accordingly. The outputs of the co-creation labs are structured to feed into the FITTER Digital Platform developed under WP5, contributing to the project’s objective of supporting anticipatory governance for a fair and inclusive twin transition.

Table of contents

Acronyms.....	7
1. Introduction.....	9
1.1. Overview of FITTER-EU	9
1.2. Purpose and scope of the Deliverable	9
1.3. Structure of the Deliverable.....	10
1.4. Link with other work packages (WP)s and Deliverables	10
2. The co-creation methodology.....	11
2.1. Conceptual framework and alignment with other work packages	11
2.2. A systemic approach to address twin transition problems	11
2.3. Design and structure of the co-creation process	13
2.3.1. First in-presence lab.....	16
2.3.2. Second in-presence lab	20
2.3.3. Third in-presence lab	23
2.4. Operational process to develop the methodological guidelines	25
2.5. Limitations in the methodology	26
3. The implemented co-creation labs.....	29
3.1. Overview of the FITTER-EU lab sessions and participants.....	29
3.2. Deviations from the methodology	35
3.2.1. Germany.....	36
3.2.1. Hungary.....	37
3.2.2. Ireland.....	39
3.2.3. Italy	40
3.2.4. Spain.....	40
3.2.5. Portugal.....	41
4. Twin transition challenges identified in the labs	43
4.1. Energy sector	43
4.1.1. Germany.....	44
4.1.2. Hungary.....	45
4.1.3. Ireland.....	46
4.1.4. Italy	46
4.1.5. Spain.....	47
4.1.6. Portugal.....	48
4.2. Housing sector	49
4.2.1. Germany.....	50
4.2.2. Ireland.....	51
4.2.3. Portugal.....	52
4.3. Transport sector.....	52

4.3.1.	Hungary.....	54
4.3.2.	Italy	55
4.3.3.	Spain.....	55
5.	<i>Results of the co-creation process.....</i>	57
5.1.	Overall analysis of the co-created policies.....	58
5.2.	Crosscutting and Lab's specific emerging approaches to just twin transitions.....	62
5.2.1.	Sectoral overview: energy.....	62
5.2.2.	Sectoral overview: housing	64
5.2.3.	Sectoral overview: transport	65
	<i>List of references</i>	68
	<i>Annex I: List of organisations participating in the national labs.....</i>	69
	<i>Annex II: List of organisations participating in the national labs.....</i>	70
	<i>Annex III: The co-created policy portfolios</i>	72
6.	<i>German policy portfolio</i>	72
6.1.	Energy	72
6.1.1.	Adjustments in existing policies.....	72
6.1.2.	New policies	74
6.2.	Housing.....	75
6.2.1.	Adjustments in existing policies.....	75
6.2.2.	New policies	80
7.	<i>Hungarian policy portfolio.....</i>	82
7.1.	Adjustments in existing policies	82
7.2.	New policies.....	89
8.	<i>Irish policy portfolio</i>	95
8.1.	Energy	95
8.1.1.	Adjustments in existing policies.....	95
8.2.	Housing.....	97
8.2.1.	Adjustments in existing policies	97
8.3.	Energy and housing	99
8.3.1.	New policies	99
9.	<i>Italian policy portfolio</i>	103
9.1.	Energy	103
9.1.1.	Adjustments in existing policies.....	103
9.1.2.	New policies	110
9.2.	Transport.....	114
9.2.1.	Adjustments in existing policies.....	114
9.2.2.	New policies	122
10.	<i>Spanish policy portfolio</i>	125
10.1.	Energy.....	125
10.1.1.	Adjustments in existing policies	125

10.1.2.	New policies	126
10.2.	Transport	130
10.2.1.	Adjustments in existing policies	130
10.2.2.	New policies	131
11.	<i>Portuguese policy portfolio</i>	137
11.1.	Energy	137
11.1.1.	Adjustments in existing policies	137
11.1.2.	New policies	137
11.2.	Housing	138
11.2.1.	New policies	138
11.3.	Cross-cutting: Energy and Housing	139
11.3.1.	New policies	139

Acronyms

AI	Artificial Intelligence
ALU	Urban Logistics Areas (Italy)
API	Application Programming Interface
ART	Italian Transport Regulatory Authority (Italy)
ATAC	Azienda per la Mobilità (Rome Public Transport Operator)
BEG	Federal Funding for Efficient Buildings (Germany)
BER	Building Energy Ratio
CapEx	Capital Expenditure
CER	Renewable Energy Community (Italy)
CfD	Contract for Difference
CfD/FiT	Contract for Difference / Feed-in Tariff
CG	Cluster Group
CSO	Civil Society Organisation
D	Deliverable
DE	Germany
DG	Disadvantaged Groups
DSO	Distribution System Operator
DSP	Decision Support System
ECON	Economic incentives
ENPE	National Strategy Against Energy Poverty (Spain)
EPC	Energy Performance Contract
ES	Spain
ESCO	Energy Service Company
ESG	Environmental, Social and Governance
EU	European Union
EV	Electro vehicle
GA	Grant Agreement
GEG	Building Energy Act (Germany)
GW	GigaWatt
HU	Hungary
HUC	Urban Consolidation Hubs
IE	Ireland
INFO	Information/education
ISEE	Equivalent Economic Situation Indicator (Italy)
IT	Italy
ITS	Istituti tecnici e superiori (
LV	Low voltage
MaaS	Mobility as a Service
NEET	Not in Education, Employment or Training
NGO	Non-Governmental Organisation
PNIRE	National Infrastructure Plan for Electric Vehicle Charging (Italy)

PNRR	National Recovery and Resilience Plan
PPA	Power Purchase Agreement
PT	Portugal
PV	Photo voltaic
REG	Regulatory
RES	Renewable Energy Sources
SERV	Public services
SLA	Service Level Agreement
SME	Small & Medium Enterprise
SMS	Short Message Service
SPID	Public Digital Identity System (Italy)
T	Task
VAT	Value Added Tax
VET	Vocational Education and Training
WP	Work Package
ZTL	Limited Traffic Zones

1. Introduction

1.1. Overview of FITTER-EU

The FITTER-EU project aims to contribute to existing research on the origins, dynamics and determinants of inequalities and enable anticipatory governance to support a fair and inclusive twin transition across Europe. The project innovates through the formulation and development of an ecosystem (i.e., the FITTER ecosystem) that pivotally includes a highly interactive and gamified Digital Platform powered by an Intelligent Predictive Decision Support System. Through the appliance of a co-creation methodological approach, the ecosystem aims to enable policymakers to predict which social groups may be at risk of being adversely affected by twin transition policies under different scenarios, with the option to simulate the implementation of these policies for the assessment of inequalities and social exclusion risks. The Digital Platform within the ecosystem aims to provide proposed mitigation measures and better practice guides that can be used to curb potential negative effects of policies on identified at-risk groups. In addition, the FITTER ecosystem incorporates wide platform connectivity capability, which helps to support the needs of the engaging stakeholders within the community of the FITTER Cluster Group (CG). The group includes key national and pan-European members such as policymakers and civil society organisations (CSO).

1.2. Purpose and scope of the Deliverable

This deliverable presents the methodology, implementation and results of Task (T) 4.2 “Co-creating intersectional mitigation plans” within the FITTER-EU project. The purpose of this document is to provide a comprehensive account of the co-creation process carried out in six national case studies and to report on the mitigation measures developed to address the negative effects of twin transition policies on disadvantaged groups (DGs).

T4.2 focuses on analysing and responding to the negative impacts related to the implementation of twin transition policies in three key sectors — energy, housing and transport. Building on the results of Work packages (WP) 2 and 3, and in complementarity with T4.1, the task translates the identified sectoral challenges and risks into concrete policy proposals through a structured participatory process. This process was implemented in Germany (DE), Spain (ES), Hungary (HU), Ireland (IE), Italy (IT) and Portugal (PT) through a series of co-creation labs involving civil society organisations, policymakers and other relevant stakeholders.

The deliverable details the design of the co-creation methodology, including the objectives and content of each lab session, the facilitation tools used to guide collective analysis, and the analytical framework applied to assess the proposed mitigation measures. Particular attention is given to how intersectional challenges were identified and prioritised, and how these informed the development of policy proposals aimed at promoting a fair and inclusive twin transition.

As a result of the co-creation process, six national policy portfolios were developed, comprising 82 mitigation measures, including both adjustments to existing policies and new policy proposals. These measures are presented and analysed in terms of their transformative potential and feasibility, and structured to enable their integration into the FITTER ecosystem and Digital Platform developed under WP5.

In addition to reporting on the outputs generated, this deliverable reflects on methodological limitations, implementation challenges and contextual differences across countries.

1.3. Structure of the Deliverable

This deliverable is structured into five chapters. Following the Introduction, Chapter 2 presents the conceptual framework underpinning the methodology, the structure and content of the lab sessions, and the tools used to facilitate collective analysis and policy design. It also includes a section outlining the limitations of the methodology.

Chapter 3 provides an overview of the co-creation labs as implemented in Germany, Hungary, Ireland, Italy, Portugal and Spain. It reports on participation trends, organisational profiles, sectoral focus and deviations from the common methodology. Country-level specificities and adaptations are described in order to contextualise the outputs generated in each case study.

Chapter 4 presents the intersectional challenges identified and prioritised during the first lab in the sectors studied in each national case study. These challenges were central to the analysis of the overall process and to the development of mitigation measures.

Finally, Chapter 5 presents the results of the co-creation process by analysing the 82 mitigation measures elaborated in the labs. The first annex provides links to the national reports, which were the main inputs for the elaboration of this deliverable. The second annex includes the complete list of organisations participating in the labs. The third annex presents the policy portfolios and the detailed proposals co-created in each lab.

1.4. Link with other work packages (WP)s and Deliverables

Deliverable 4.2 is closely interconnected with several components of the FITTER-EU project architecture:

- **WP2** provides the justice-based and intersectional foundation of the co-creation methodology, including the operational definition of fairness and inclusiveness and the normative framing of a just twin transition. WP2 also developed the stakeholder engagement strategy of the project in Deliverable D2.5 and the engagement strategy for DGs in D2.6, which were key for structuring the engagement strategy of the co-creation labs.
- **WP3** supplies the empirical basis for the co-creation process by identifying sectoral challenges — twin transition “hazards” and “negative impacts” — modelling transition pathways and determining the DGs affected by twin transition policies.
- **WP4**, and in particular Task 4.1, complements Task 4.2 by gathering survey-based evidence on perceived risks and negative impacts derived from the implementation of twin transition interventions, informing the prioritisation of challenges addressed in the labs.
- **WP5** integrates the mitigation measures generated through the co-creation process into the FITTER platform, aimed at supporting informed decision-making processes of policymakers to promote a just transition.
- **WP6** will validate the mitigation measures elaborated in the labs.
- **WP7** communication efforts have supported stakeholder engagement strategies for the co-creation labs.

2. The co-creation methodology

2.1. Conceptual framework and alignment with other work packages

The co-creation process established in FITTER-EU T4.2 “Co-creating intersectional mitigation plans” aims to propose a set of mitigation measures to address the negative impacts and risks resulting from the implementation of twin transition policies, as analysed in WP2 and WP3.

WP3 constitutes the starting point of this task, as it analyses the negative effects of twin transition policies, develops future scenarios, and identifies the affected population groups — the DGs. WP4 then shifts the focus towards a solution-oriented approach, examining potential mitigation measures to address the negative effects of transition policies.

Task 4.1 conducts a risk assessment of DGs’ perceptions of the negative effects of the transition, while Task 4.2 implements a co-creation process to propose policy solutions to mitigate these effects. Within this framework, co-creation is understood as a structured process for the collective analysis of problems and the participatory development of solutions. The umbrella for the conceptualization of WP4 methodology is the theoretical framework developed in WP2, determining the two pillars of the FITTER co-creation methodology:

- i. The FITTER approach for a ‘just’ transition (D2.1) starts from a “*Process that addresses intersectional inequalities, with fair burden-sharing, while controlling climate change and taking the dimensions of environmental, climate, and energy justice into account (adapted from Galgóczi 2023)*”. Within this framework, the subject of justice is the historically constituted and corporeally vulnerable rights-bearing individual, while the object of justice comprises the economic, environmental and social relations of production and distribution within each sector. The domain of justice is defined by the FITTER sectors.
- ii. A systems thinking approach in policy analysis to address twin transition negative effects by addressing the root causes of inequality with transformative policies instead of implementing traditional reactive policy processes.

Policy solutions emerging as a result of the co-creation process will be integrated in the FITTER platform developed in WP5 as a decision-support system to address the negative impacts of the twin transition on inequalities.

Within this framework, the methodology developed for the co-creation path pursues two main objectives:

- To elicit DGs/stakeholders input and feedback on the negative effects of the twin transition in terms of equality within the energy, transport, and housing sectors, providing specific policy responses –i.e. mitigation measures– to each country addressed by the project.
- To generate relevant inputs for the FITTER-EU platform and collect effective mitigation measures –in terms of policy responses– to counter the identified hazards and negative impacts co-designed with stakeholders and inspired by a system thinking/complexity-oriented approach.

2.2. A systemic approach to address twin transition problems

The negative effects on equality arising from the implementation of the twin transition are understood as *wicked problems*—multidimensional challenges characterised by “incomplete or contradictory information, differing views on the nature of the problem, or complex interactions with other issues.” (European Union Institute for Security Studies, 2015). Tackling such complex problems requires

acknowledging the underlying systemic dynamics and designing innovative solutions. Traditional policy approaches often fail to address interconnections across dimensions and the systemic nature of these challenges, which leads to ineffective outcomes (Joint Research Centre, 2025). For T4.2, implementing a systemic approach in the co-creation of policy solutions is essential to enable the delivery of a just twin transition.

The following table presents a set of useful definitions that have informed the methodological guidelines, (Matti et al., 2020).

Term	Definition
System	“In broad terms, anything that is not chaos. A system is formed by interlinked individual elements and their relationships. Changes in its elements produce changes in systems; thus, to understand system outcomes, it is necessary to propose an integrated analytical framework, including the analysis of the system’s elements and processes” (Matti et al., 2020).
Socio-technical systems	In socio-technical systems “Interactions within social structures need technical infrastructure, while the creation of new technologies serves to mobilise social systems” (Matti et al., 2020). Sectors analysed in FITTER-EU are socio-technical systems. It is not possible to control social systems because of the complexity and the multiple dynamics coexisting in them, while technical systems are planned in advance to produce reliable, predictable and specific outcomes. As a result, changes in socio-technical “are a matter of contingency and can only be understood retrospectively and not in advance” (Matti et al., 2020).
System innovation	Transition from one socio-technical system to another
Transformative changes	“Complex, long-term and messy processes in which dominant practices become replaced. Complex socio-technical systems such as cities normally do not change by themselves. Actors play a key role in shaping desirable transitions through transformative activities” (Matti et al., 2020).

Table 1: Operational definitions related to systems innovation

Within this framework, negative effects produced by twin transition policies are framed as intersectional challenges, which refer to the situations where multiple dimensions of vulnerability — such as income, age, gender, ethnicity, disability, among others — overlap and interact with each other. These intersecting factors do not simply add up but combine in ways that can compound disadvantages, making some groups disproportionately exposed to the negative effects of green and digital transition policies. Addressing intersectional challenges therefore requires recognising the complexity of how inequalities are produced and reinforced across different dimensions and ensuring that policy responses are sensitive to the diverse experiences and needs of affected groups.

On the other hand, mitigation measures to counter intersectional challenges are intended as transformative activities in terms of policy measures, which are able to trigger transformative changes. Therefore, the final outcome of the co-creation process is the proposal of a portfolio of policy interventions in each country case study —including the adaptation of existing policies and the proposal of new transformative policies— to address the equality-led challenges of twin transitions in the energy, housing and transport sectors.

The methodology elaborated by SV for the co-creation path of the FITTER-EU labs is inspired and adapted from two documents by EIT Climate KIC:

- Matti, C., Martín Corvillo, JM, Vivas Lalinde, I., Juan Agulló, B., Stamate, E., Avella, G., and Bauer A. (2020). *Challenge-led system mapping. A knowledge management approach*. Transitions Hub series. EIT Climate-KIC, Brussels.
- De Vicente Lopez, Javier and Matti, Cristian (2016). *Visual toolbox for system innovation. A resource book for practitioners to map, analyse and facilitate sustainability transitions*. Transitions Hub Series. Climate-KIC, Brussels.

2.3. Design and structure of the co-creation process

The FITTER-EU co-creation process was implemented in the six country case studies —Germany, Hungary, Ireland, Italy, Portugal and Spain— studying the policies of three sectors: energy, housing & built environment and transport and mobility. Each country focused on two specific sectors: energy was cross-cutting throughout the countries, while transport was analysed in Italy, Hungary and Spain; and housing in Germany, Ireland and Portugal.

In each country, the co-creation process was implemented through a laboratory format ('lab') comprising five meetings — three in person and two online — held between September and December 2025. Each lab involved 15–20 representatives of DGs and policymakers.

Regarding the former, in line with the stakeholder engagement strategy with DGs (D2.6), the co-creation process was conducted with intermediary organisations/policy makers representing or close to DGs rather than with individual persons belonging to disadvantaged populations. This decision was taken because the content analysis and development carried out within the laboratories and the time constraints required participants to have specific expertise and technical knowledge concerning the policy and political agenda of the sectors analysed in each country. It was also intended to mitigate potential power imbalances within the lab and to ensure broader representativeness of positions, avoiding reliance solely on the perspectives of particular individuals.

Intermediaries/policy makers involved in the lab were understood in a broad sense, involving experts and institutional representatives from the public administration, think tanks, research bodies and other organisations focusing on public policy research. No elected political representatives from national or local government were included among the targeted participants invited to the Labs.

Even if the co-creation process did not foresee the payment of expert fees, the reimbursement of travel expenses was comprehended. In addition, the following non-monetary incentives were considered to promote participation of stakeholders:

- Sharing of the comparative results of the co-creation processes carried out in the six project countries.
- Preparation of national and or transnational white papers, with opportunities for co-authorship and participation in international conferences.
- A transnational meeting (pending to be organised).
- Access to the Beta version of the FITTER platform.
- Participation in the final conference in Seville (November 2026), with reimbursement of travel expenses.

The following figure presents the overall flow of the co-creation process.

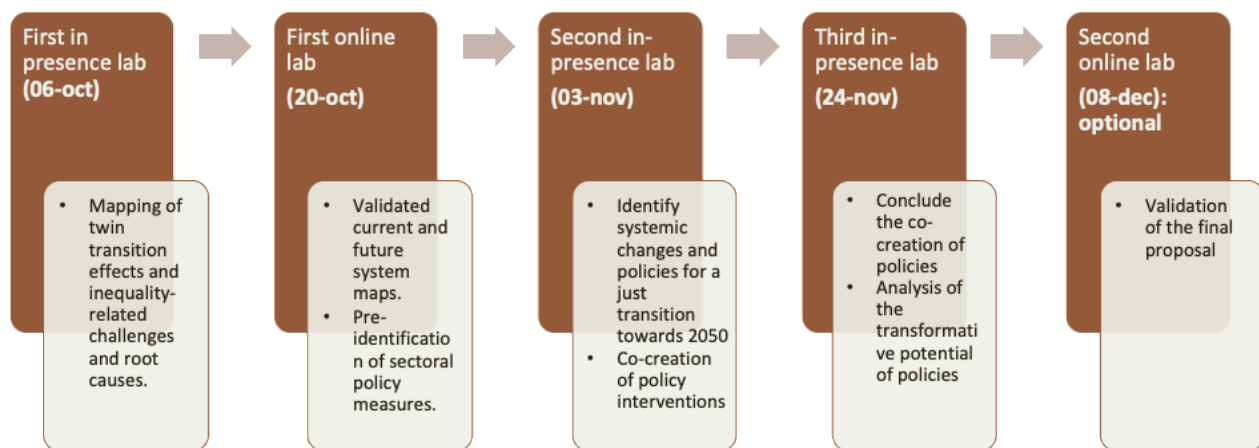


Figure 1: Overview of the co-creation process

All in-person labs were designed to last half a day, with the recommended format consisting of a morning session either followed or preceded by a networking lunch, aperitif, or dinner.

In-person meetings were central to the co-creation process, as they were the sessions in which content was developed and analysed, whereas the online sessions were dedicated to validating the outcomes. Below, the co-creation process is outlined in brief, focusing on the objectives, inputs, and outputs of each session. Further details on the specific methods used are provided in Section 2.4.

First in-presence lab

Objectives:

- Presenting the justice framework developed within the FITTER-EU project, along with the objectives and policy mechanisms selected for each sector.
- Guiding participants through a series of activities to reflect on and discuss the possible negative impacts and the challenges of the twin transitions.
- Discussing the systemic dynamics behind the identified problems, in line with the systems thinking framework.

Inputs

- The first lab is conceived as an introductory/ “diagnostic” session that builds on the preliminary FITTER-EU research results, and it aims at collaboratively identifying the negative impacts of the twin transition

Outputs:

- Identification of most relevant sectoral challenges —twin transition problems in each sector—
- First draft of two system maps reflecting the current situation in each sector, identifying the causes of behind the sectoral challenges identified, most relevant stakeholders and barriers and drivers for change.

Second in-presence lab

Objectives

- Identify and prioritise the systemic changes and policy measures required to move from the current system (validated during the first in-presence lab) towards a just twin transition by 2050.

- Start with the co-creation of policy measures.

Inputs

- Validated sectoral system maps.
- List of pre-identified twin transition and sectoral policy measures determining system dynamics.

Outputs

- First version of co-created policy measures, including adjustments of existing policy measures and new proposals of policies.

Third in-presence lab

Objectives

- Finalise the co-creation of policy measures proposed in the second in-presence lab.
- Analyse the transformative potential of policy proposals and classification of policy types according to their impact and feasibility.

Inputs

- List of policy proposals (adjustments in existing policies and new proposals) in each sector in relation to the sectoral challenges identified and first version of the detailed proposal of each policy.

Outputs

- Final proposal of mitigation measures (policy proposals) to address the sectoral challenges identified.
- Typology of policies determined according to their transformative potential.

Validation sessions

Two validation sessions were proposed as part of the methodology. The first took place after the initial in-person lab and aimed to validate the system maps co-created during that meeting. The second was held at the end of the entire co-creation process and focused on validating the final proposals developed in each lab, as well as the overall final proposal.

These sessions were primarily designed to involve policymakers and policy experts who had not participated in the co-creation process, although those involved in the labs could also be invited. Each session lasted one and a half hours and combined presentations of the labs' outcomes, group discussions, and plenary sessions structured around guiding questions to collect participants' feedback.

Transnational co-creation lab

As established in the Grant Agreement (GA), the original conceptualisation of the FITTER-EU co-creation process envisaged the organisation of an EU-level transnational FITTER Lab to scale up the results of the national co-creation labs: "to address the common challenges identified in the national cases under a European Union (EU) perspective, proposing transnational sectoral mitigation measures". As WP2/3 didn't foresee the development of an EU level case study, it was agreed that there wasn't enough background work ready to make the organisation of an EU level Lab meaningful and solid, so the session was not carried out.

At present, in coordination with WP5 and WP6, explorations are undergoing to position a transnational co-creation session with a dual objective: (i) to present further developments of the FITTER-EU platform to stakeholders as part of a feedback/accountability and restitution process; and (ii) to support the validation of the mitigation measures foreseen under WP6.

Preliminarily, it would be held online and expected to involve approximately 30 participants from the national labs, including members of intermediary organisations representing DGs and policymakers.

2.3.1. First in-presence lab

Proposed agenda

Time	Title of the activity
9:00-9:15	Welcome and signature of informed consent forms
9:15- 9:30	Introductions-getting to know each other
9:35: 9:45	Setting the scenes on the green and digital transitions
9:45 10:30	Disadvantaged and vulnerable groups: challenges/problems
10:30-11:15	The systemic dimensions of the challenges
11:15-11:30	Coffee Break
11:30-12:00	Plenary
12:00-13:15	The root causes of inequalities
13:15-13:30	Wrap up, feedback collection, next steps

Table 2: First in-presence lab agenda.

The agenda of each lab session was organised to have different sessions, combining sessions organised in working sector-oriented subgroups and plenary discussions.

The first meeting was conceived as a diagnostic session aimed at analysing the most relevant negative effects —framed as ‘sectoral challenges’— derived from twin transition policies in the two analysed sectors in each country case study, and to translate these insights into a first systemic representation of the sectoral dynamics underpinning inequalities.

Following an initial framing of the project objectives and the rationale of the co-creation process, the agenda started with an exercise aimed at encouraging participants to reflect on how twin transition policies may negatively affect different vulnerable groups. In the session between **9:45 and 10:30 “Disadvantaged and vulnerable groups: challenges/problems”**, the analytical work began with a guided brainstorming exercise based on the use of personas. The used template is presented in the following figure.

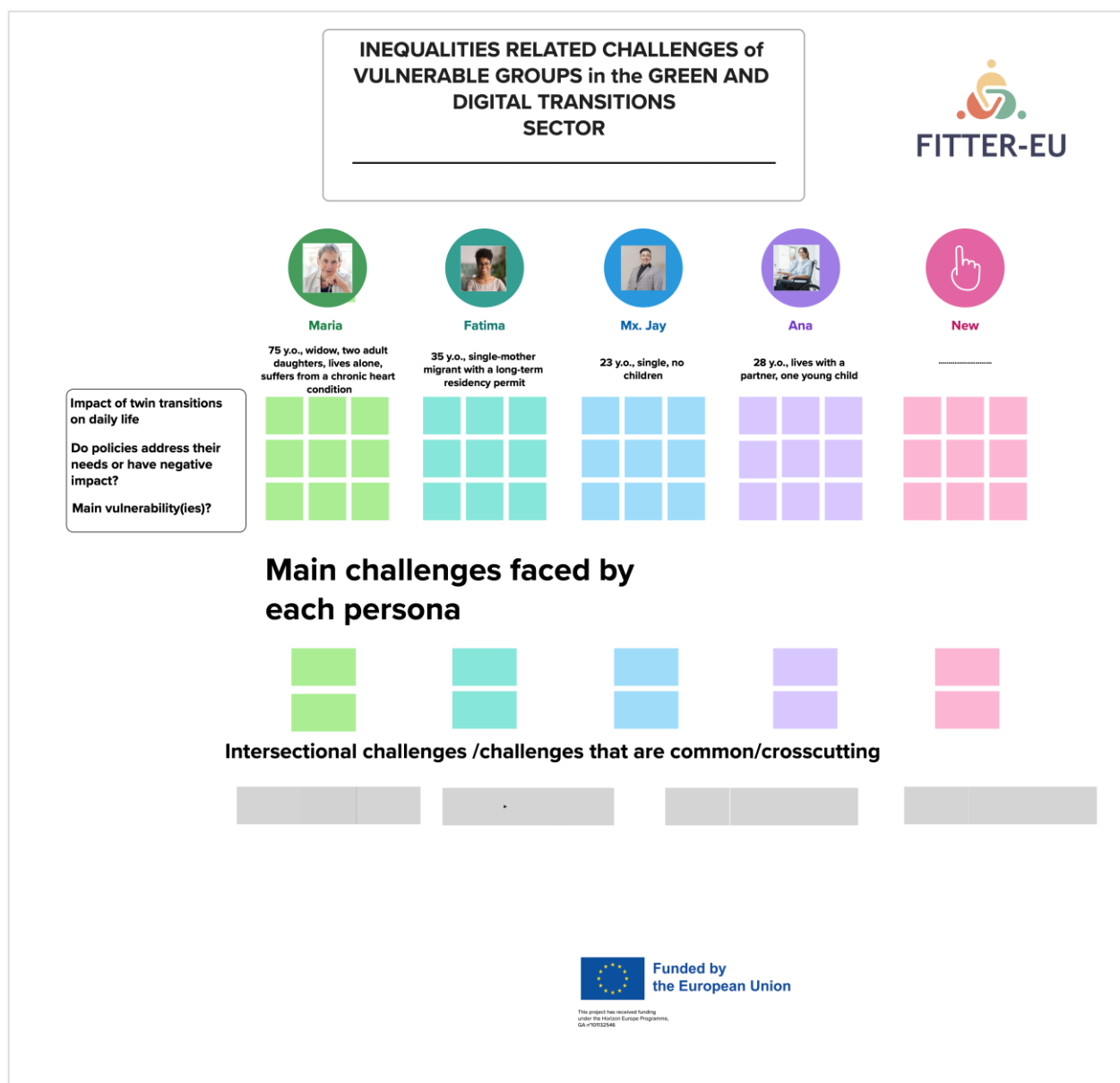


Figure 2: Facilitation template – First lab agenda session 9:45 and 10:30 “Disadvantaged and vulnerable groups: challenges/problems”

Fictional but realistic character profiles were provided to represent individuals potentially affected by green and digital transition policies. This technique enabled participants to adopt a situated perspective, reflecting on how specific socio-demographic characteristics (e.g. income level, age, employment status, housing conditions) interact with sectoral policies and technological change. The objective was not to generalise from anecdotal cases, but to stimulate structured reflection on differentiated impacts and intersectional vulnerabilities. Participants identified and clustered the main challenges faced by the personas, and discussed whether existing policies mitigated or exacerbated these challenges, including unintended effects. Particular attention was paid to avoiding stereotyping and to recognising the complexity of intersecting inequalities. After analysing individual challenges faced by the personas, cross-cutting elements were analysed to identify the most relevant sectoral challenges.

During the following session between **10:30-11:15 “The systemic dimensions of the challenges”** and building on the challenges identified through the personas, the session then shifted towards a systemic perspective using the “pentagonal problem” framework (De Vicente Lopez & Matti, 2016) (TIP Resource

Lab, 2022), see the template in Figure 3. This technique was used to start the analysis of systemic dimensions of sectoral challenges, unpacking them in five interrelated dimensions: (i) environment; (ii) society and culture; (iii) economy, markets and finance; (iv) policy and governance; and (v) science, technology and infrastructure. In each dimension, participants identified systemic factors underlying these challenges, mapping relevant dynamics within each dimension and exploring interconnections across dimensions. The exercise emphasised that observed inequalities are rarely attributable to single factors, but rather emerge from interacting institutional, economic, technological and socio-cultural configurations.

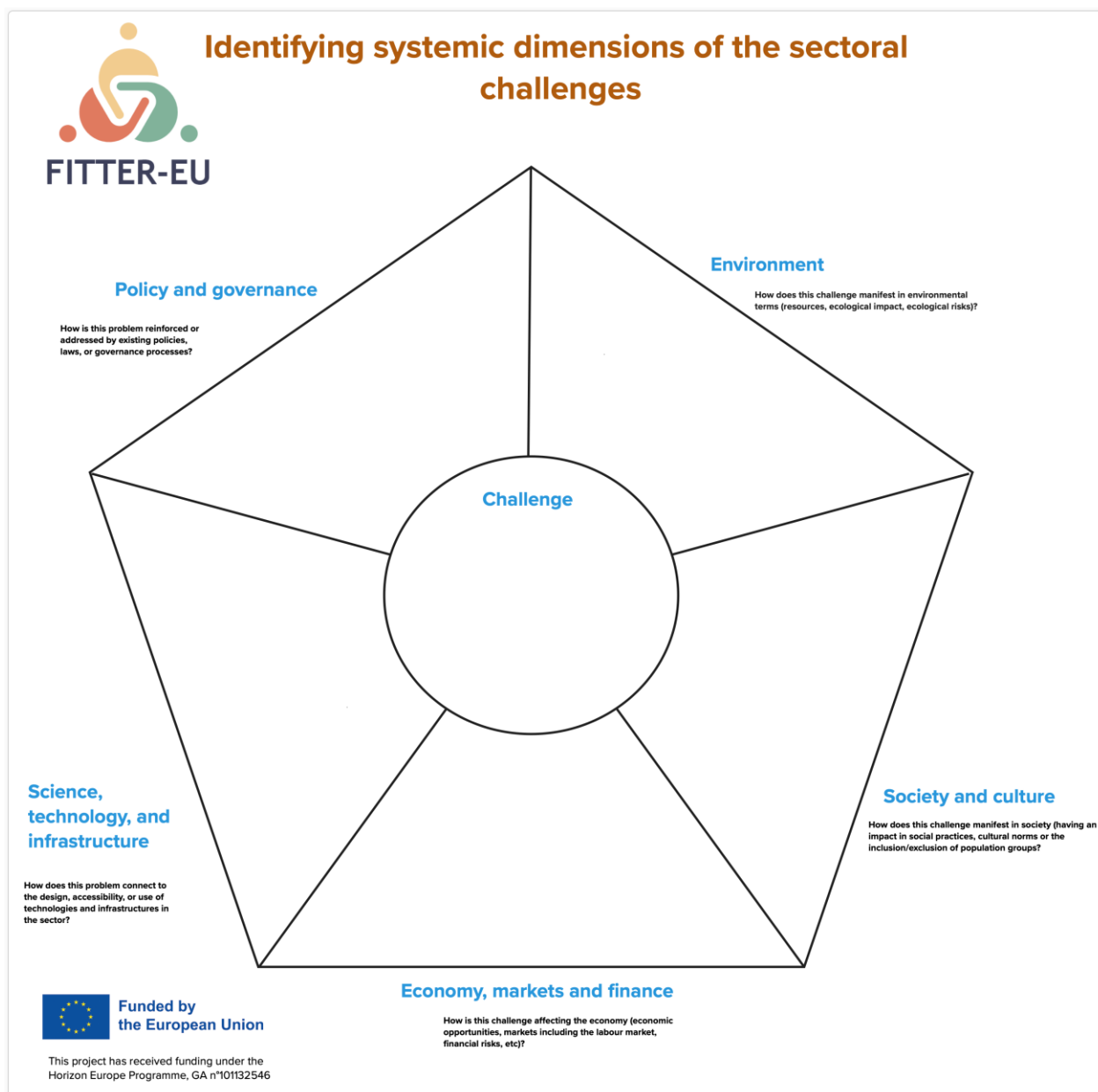


Figure 3: Facilitation template – First lab agenda session 10:30-11:15 “The systemic dimensions of the challenges”

A plenary discussion followed, during which each sectoral group presented its main challenges and systemic dimensions identified. The objective was to identify converging patterns and shared dynamics across sectors, thereby distinguishing sector-specific features from broader structural drivers of unjust transitions.

Finally, the last agenda session **12:00-13:15 “The root causes of inequalities”** (template in Figure 4) was intended to build the systemic maps in each sector identifying the most relevant actors, barriers and drivers behind the sectoral challenges identified and the related inequality dynamics. The exercise combined individual reflection and collective deliberation. Participants first identified key actors and structural elements within each systemic dimension and classified them as drivers, barriers or neutral elements with respect to a just transition. Subsequently, they collectively clustered similar elements and established causal links, feedback loops and reinforcing dynamics among them. The outcome of this process was a first draft of two sector-specific systems maps representing the current configuration of actors, structures and mechanisms contributing to inequality-related challenges. These maps served as an analytical baseline for subsequent phases of the co-creation process.

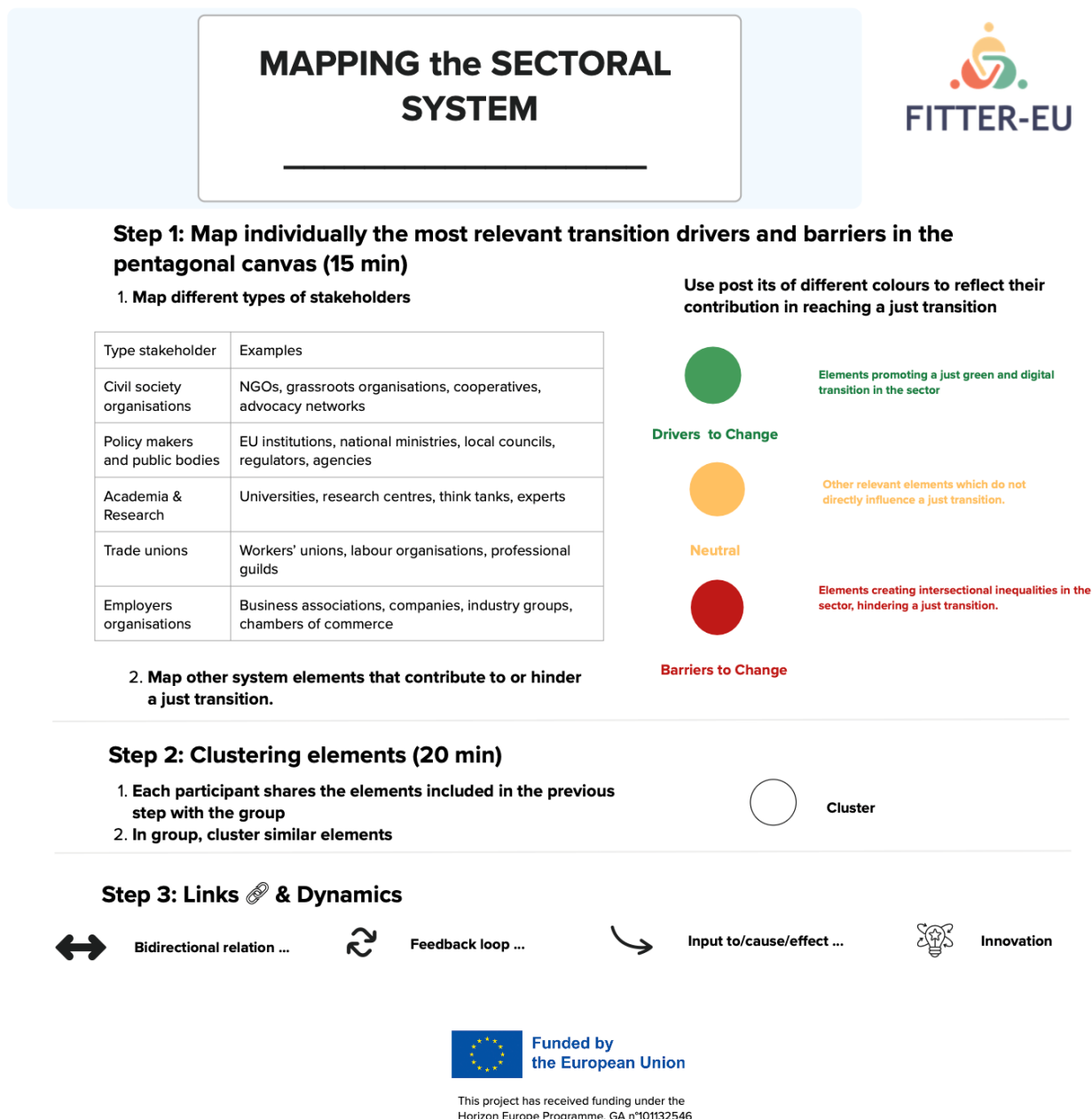


Figure 4: Facilitation template – First lab agenda sesión 12:00-13:15 “The root causes of inequalities”

2.3.2. Second in-presence lab

Proposed agenda

Time	Title of the activity
9:00-9:05	Welcome and signature of new participants' informed consent forms
9:05-9:25	Our future visions of the twin transitions
9:25-10:25	Identifying systemic changes: workgroups per sectors
10:25-11:30	Identifying policies for change: workgroups per sectors
11:30-11:45	Coffee break
11:45-13:00	Co-creating policies for change
13:00-13:20	Plenary discussion
13:20 – 13:30	Wrap-up and next steps

Table 3: Second in-presence lab's agenda

The second in-presence meeting was conceived as a structured transition from systemic diagnosis to the co-creation of transformation pathways. Its overarching objective was to identify and prioritise the systemic changes and policy measures required to move from the current system configuration—validated in previous sessions—towards a fair and inclusive twin transition by 2050. The methodology combined back casting, socio-technical systems analysis, deliberative prioritisation and structured policy design tools.

At the beginning of the session, validated system maps for each sector (energy and housing/transport) were presented together with the project's future vision of a just twin transition, proposed as part of T3.2 findings. This vision was framed as plural and exploratory rather than normative end-states, allowing participants to engage with multiple possible transition pathways under conditions of uncertainty.

During the **9:25-10:25 session “Identifying systemic changes: workgroups per sectors”** a socio-technical roadmap exercise was proposed (De Vicente Lopez & Matti, 2016) grounded in back casting logic. Participants worked in sectoral groups and were invited to identify the structural transformations required to bridge the gap between the current system and the 2050 vision. At this stage, policy instruments were deliberately excluded and participants focused instead on systemic shifts across the other four systemic dimensions present in the system maps: environment; society and culture; economy; and science and technology. This encouraged reflection on structural transformations such as the emergence or disappearance of practices, infrastructures, power relations, behavioural norms, and stakeholder roles.

These changes were aimed to address the sectoral challenges identified in the sectoral maps. After the initial proposal, they were then mapped relationally, identifying causal dependencies and sequencing dynamics. This step clarified which changes functioned as enabling conditions for others and how interactions across system dimensions shaped the overall transition pathway.

During the **10:25-11:30 slot “Identifying policies for change: workgroups per sectors”**, the workshop moved to structured policy design by turning the proposed changes into concrete policy proposals, i.e. the mitigation measures to counteract the twin transition negative effects. Policy proposals could be of two types (i) adjustments to existing policies (including scaling, modification or removal) and (ii) new policy proposals. Four types of policy instruments were proposed to participants:

- Regulation: laws, standards, bans, requirements.
- Economic incentives: subsidies, taxes, grants, penalties.
- Public services: programmes, infrastructure investment.
- Information/education: campaigns, training, awareness.

Participants were explicitly encouraged to combine politically feasible measures —more commonly associated to the adaption of current policy proposals— with more ambitious or disruptive proposals, which could be potentially more transformative and generate wider impact in the systems of the associated sectors.

The template used for the previous two timeslots is presented in Figure 5.

Identifying systemic changes

FITTER-EU

Needed changes for achieving a just twin transition 	Environment					Vision 2050
	Society & culture					
	Economy & finance					
	Science, technology & infrastructure					
Policies to address these changes 	Adjustment in existing policies					
	New policies					

2030 2035 2040 2045

Funded by the European Union
This project has received funding under the Horizon Europe Programme, GA n°101132546

Figure 5: Facilitation template – Second lab agenda slots 9:25-11:30

During the final slot of the lab **11:45-13:00** the actual co-creation of policies was carried out. For the detailed content development of policies, the following two templates in form of policy factsheets were provided to partners.

Section	Guiding questions / What to fill in
Title of the policy (existing)	Which current policy, programme, or regulation are you adapting? Provide its official name/title if possible.
Sectoral focus	Which sector is the policy linked to (energy/housing, transport, etc.)?
Original policy objectives	What was the policy originally designed to achieve?
Prioritised challenge addressed	Which of the identified changes is the adjustment in this policy trying to address?
Limitations or gaps	Why is the current policy insufficient, ineffective, or unfair?
Proposed adaptations	What changes should be made? (e.g. broaden eligibility, adjust criteria, simplify access, strengthen enforcement).

Policy type(s)	Tick which category the adaptation concerns: ☐ Regulation (laws, standards, bans, requirements) ☐ Economic incentives (subsidies, taxes, grants, penalties) ☐ Public services (programmes, infrastructure investment) ☐ Information/education (campaigns, training, awareness)
Participatory dimension & stakeholders	Who should be involved in the adaptation process (civil society, unions, local authorities, etc.)?
Target population: only one question is allowed per answer	
Age range	☐ <18 ☐ 25–35 ☐ 35–65 ☐ >65
Living in a house with low energy efficiency:	☐ Yes ☐ No
Gender	☐ Woman ☐ Male ☐ Non-binary ☐ Other
Need of a car to perform daily activities:	☐ Yes ☐ No
Level of education	☐ No formal education ☐ Primary ☐ Secondary ☐ Further formal education
Location of residency	☐ Urban area ☐ Suburban area ☐ Rural area
Economic status	☐ Employed ☐ Unemployed ☐ Retired
Care responsibility as the main activity	☐ Yes, remunerated ☐ Yes, non-remunerated ☐ No
EU citizenship	☐ Yes ☐ No
Disability or long-term condition	☐ Yes ☐ No
Level of income	☐ Low income ☐ Medium income ☐ High income
Tenancy status	☐ Homeowner ☐ Tenant
Location of residency	☐ Rural ☐ Urban ☐ Suburban
Systemic focus	Which systemic dimensions are strengthened through the proposed adaptation (environment, society & culture, economy & finance, science/tech/infrastructure, policy/governance)?
Potential risks/barriers	What factors or actors might prevent the adaptation from working?
Drivers/enablers	What existing factors or actors can support this adaptation?
Time horizon	When should the adaptation be implemented (short, medium, long term)?
Feasibility & resources	What resources are needed to adapt the policy effectively (financial, institutional, technological)?

Table 4: Policy factsheets to adapt existing policies

Section	Guiding questions / What to fill in
Title of the policy (new proposal)	Give the policy a short, descriptive name.
Sectoral focus	Which sector is this policy addressing (energy, housing, transport,)?
Prioritised challenge addressed	Which one of the identified changes is this policy aiming to solve?
Policy description	Summarise in 2–3 sentences what the policy is about.
Policy type(s)	Tick or specify which applies: - Regulation (laws, standards, bans, requirements) - Economic incentives (subsidies, taxes, grants, penalties) - Public services (programmes, infrastructure investments) - Information/education (campaigns, training, awareness)
Participatory dimension & stakeholders	Who should be involved in design and implementation? (civil society, unions, private sector, local authorities, etc.)
Target population: only one question is allowed per answer	
Age range	☐ <18 ☐ 25–35 ☐ 35–65 ☐ >65
Living in a house with low energy efficiency:	☐ Yes ☐ No

Gender	☐ Woman ☐ Male ☐ Non-binary ☐ Other
Need of a car to perform daily activities:	☐ Yes ☐ No
Level of education	☐ No formal education ☐ Primary ☐ Secondary ☐ Further formal education
Location of residency	☐ Urban area ☐ Suburban area ☐ Rural area
Economic status	☐ Employed ☐ Unemployed ☐ Retired
Care responsibility as the main activity	☐ Yes, remunerated ☐ Yes, non-remunerated ☐ No
EU citizenship	☐ Yes ☐ No
Disability or long-term condition	☐ Yes ☐ No
Level of income	☐ Low income ☐ Medium income ☐ High income
Tenancy status	☐ Homeowner ☐ Tenant
Location of residency	☐ Rural ☐ Urban ☐ Suburban
Systemic focus	Which dimensions of the system does it act on? (environment, society & culture, economy & finance, science/tech/infrastructure, policy/governance).
Potential risks/barriers	What factors or actors could block or limit this policy? (political resistance, funding gaps, technology access, etc.)
Drivers/enablers	What factors or actors could support this policy? (existing programmes, strong social demand, available technology, etc.)
Time horizon	What is the implementation time horizon?
Feasibility & resources	What is required for implementation (financial, institutional, technological)? How feasible it is considered to be?

Table 5: Policy factsheets for new policy proposals

2.3.3. Third in-presence lab

Proposed agenda

Time	Title of the activity
9:00-9:05	Welcome and signature of new participants' informed consent forms
9:05-9:35	Presenting the co-created policy measures in the second lab
9:35-11:00	Validating the co-created policy factsheets and completing the co-creation process
11:00-11:15	Coffee break
11:15-12:15	Analysing transformative potential of policies
12:15-12:45	Scoring of policies according to their transformative potential
12:45-13:00	Wrap-up and next steps

Table 6: Third in-presence lab's agenda

The third in-presence meeting was dedicated to concluding labs' final outcome: a set of policy measures, including adaptations to existing policies and new policy proposals. The **9:35-11:00** agenda session's objective was to consolidate the co-created policy portfolio by completing the co-creation of policy factsheets. In some country case studies, FITTER-EU partners pre-filled some of the policy factsheets with desk research to complement the ideas that had emerged during the first lab, completing them with references to norms, laws and relevant public programmes. During the session, policy factsheets filled by partners were validated. The validation was structured around explicit criteria: consistency with identified

sectoral challenges; contribution to a fair and inclusive transition pathway; adequacy of equity and mitigation considerations; and clarity and operational completeness. Feedback was required to be specific and actionable, referencing concrete sections of each factsheet. Where new proposals had emerged, additional factsheets were co-created to ensure completeness of the policy portfolio. For more details on the specificities of the methodology implemented in each lab, see Chapter 3.

The final agenda session **11:15-12:15 “Analysing transformative potential of policies”** introduced a structured assessment framework to evaluate both the systemic impact and feasibility of the proposed policies (see facilitation template in Figure 6). In sectoral groups, participants analysed all policy proposals using guiding dimensions including:

- Impact criteria: directly addressing identified challenges; systemic ripple effects; capacity to reshape institutional rules; potential to influence paradigms and political priorities; equity implications; and implementation barriers.
- Feasibility criteria: political, cultural and financial obstacles; and timing required for implementation.



Analysing transformative potential of policies



Funded by
the European Union

This project has received funding under the
Horizon Europe Programme, GA n°101132546

List of policies (include names)	Impact					Scoring transformative potential	Feasibility and implementation		Scoring feasibility
	Sectoral challenge → Does it directly address a specific problem?	Systemic effects → Does it generate wider impacts? In what domains?	Rules of the game → Does it reshape institutions/regulations?	Paradigms & attitudes → Can it shift political priorities?	Equity dimension → Does it reduce inequalities in the green/digital transition?		Barriers → Are there political, cultural, or financial obstacles?	Timing → Can it be implemented now, or does it require preparation?	

Figure 6: Facilitation template: Third in-presence lab agenda slot 11:15-12:15 “Analysing transformative potential of policies”

Following collective reflection, participants were asked to individually score each policy along the two axes of analysis —impact and feasibility—. These scores were then discussed and used to position policies within a two-dimensional classification matrix distinguishing between categories. In this analysis a typology of five categories was proposed to analyse the policies (see

Table 7).



Table 7: Typology of policy proposals

2.4. Operational process to develop the methodological guidelines

Methodological guidelines were developed by SV and provided to partners in three sets of documents, delivered sequentially to organise the sessions. The first set, provided in August 2025, included the initial engagement guidelines, the Gantt chart of the overall co-creation process, and the invitation materials. Subsequently, methodological guidelines were provided in two packages of documents: the first for the first in-person meeting, and the second for the remaining meetings (second and third in-person meetings) and the online validation sessions. Each of these two packages included the detailed methodological guidelines explained in Section 2.4, with the detailed agenda, facilitation script, and templates to be used in each session, as well as the reporting template. Partners were asked to deliver two reports: one at the end of the first in-person lab and one at the end of the implementation of the whole co-creation process. Please see Annex I: List of organisations participating in the national labs.

Initial engagement guidelines

The document was provided to partners on August 13th 2025: launch the co-creation process and initiate the engagement phase. It included a set of practical information and materials, including:

- Overview of the whole co-creation process: detail of labs, content of the sessions and target participants.
- Detailed agenda for the first lab.
- Stakeholder engagement for the co-creation labs
 - Logistic details
 - Materials for the engagement: invitation letter and informed consent form.

Link for the document: [Initial guidelines first sessions 13-08.pdf](#)

Guidelines for the first in-presence lab

- First version of the guidelines (word format), PPT template for the initial presentation contemplated in the agenda and reporting template provided to partners on September 12th 2025.
- Online training session and Q&A session organised with partners on September 17th 2025 from 9:00-11:00 CEST.
- Updated version of the guidelines September 22nd. The first draft of guidelines was updated following comments received by partners during the training session. The most relevant modifications were to simplify the sequence of exercises, minimising the use of templates to keep a more focused discussion.

Link for the final version of the documents: [T4.2 - Detailed guidelines first session v7](#)

Guidelines for the second and third in-presence labs and the online validation sessions

- First version of the guidelines and facilitation templates provided to partners on October 8th PPT format).
- Training session during the in-presence project meeting in Dublin on October 22nd.
- Updated version of the guidelines, facilitation templates and the reporting template for the final national reports: November 5th. The modification mainly included the elimination of a session dedicated to matching hazards included in the T4.1 survey with the mitigation measures. After a coordination meeting with TU Dublin and AITEC, it was decided that this was not required for the FITTER-EU platform. Therefore, it was considered more appropriate to leave the possibility for mitigation measures — policy proposals — co-created within the labs to address higher-level challenges — at sectoral level, as identified in all national case studies during the first lab — instead of at the level of specific policy interventions, as in the case of the T4.1 hazards.

Link for the final version of the guidelines: [Guidelines 2nd set v6 26-11-25.pptx](#)

2.5. Limitations in the methodology

This section details a set of methodological limitations identified after the implementation of the six laboratories, as reported by partners in their national reports. It is worth noting that the co-creation process in the labs represent a first step to start conceptualising a just transition in the energy, housing and transport sectors in the six country case studies of the project. The result of the co-creation process is a set of policy proposals in the six country case studies, which could be then improved with further research on the specific technical and legislative considerations.

Engagement challenges

Sustaining consistent and inclusive stakeholder engagement throughout the co-creation process proved to be one of the main challenges. Despite substantial outreach efforts, ensuring the continuous participation of organisations representing vulnerable groups was difficult. As participation was not financially compensated, several civil society organisations indicated that they lacked the resources and staff time to remain engaged across multiple sessions. This challenge was particularly visible in Spain and Ireland, where attendance declined in subsequent in-person meetings due to the time commitment required and the high level of expertise demanded.

Furthermore, in several cases, intermediary organisations participated in the labs on behalf of DGs, rather than direct representatives of those groups themselves. While this approach enabled the collection of diverse perspectives, it did not necessarily guarantee that the full range of intersectional inequalities was authentically represented. The German case, for instance, revealed that certain vulnerable groups—such as refugees—were insufficiently addressed during the workshops and required additional bilateral follow-up discussions to capture their specific structural challenges in accessing housing and energy support.

At the same time, contextual differences influenced engagement dynamics. In Portugal (Azores), participation remained stable due to the close-knit nature of the local community and stakeholders' perception that their contributions were visible and valued. This contrasts with larger urban contexts such as Rome or Madrid, where highly engaged professionals faced significant time constraints.

Time restrictions

The ambitious methodological architecture required significant time to implement fully. In practice, time constraints limited the depth of analysis that could be achieved within single-day sessions. Although the intention was to iteratively unpack complex system dynamics, limited time often shifted the focus towards generating mitigation proposals rather than deepening systemic diagnosis.

In Germany, the number and scientific detail of the methods—combined with participants' diverse backgrounds—made transitions between exercises cognitively demanding. Completing multiple policy factsheets in detail required legal and technical expertise that was not always present, and time was insufficient to reach consensus on focus points. Similarly, in Ireland and Spain, facilitators noted that simplifying or streamlining certain steps could have freed up time for deeper policy co-creation.

Fatigue towards the end of sessions was also reported (Ireland), suggesting that the cognitive intensity of the methodology may have affected the quality of later-stage outputs.

Lack of flexibility and contextual adaptation of the methodology

The methodology was deliberately standardised and maintained a national-level of analysis to ensure comparability across countries and enable integration into the FITTER-EU platform. However, this standardisation limited flexibility and, in some cases, constrained the capacity to adapt the process to specific institutional and territorial realities.

This limitation was particularly relevant in countries where regions have more autonomy, such as Italy, Spain and Germany. In the latter, for instance, the strong federal system—where legislative powers and financing responsibilities are distributed across national, regional (Länder) and municipal levels—did not fully align with the national-level focus of the labs. Participants found it difficult to formulate recommendations on issues primarily regulated at regional or municipal level, such as social housing allocation or refugee accommodation. This revealed a structural tension between the methodology's harmonised design and multi-level governance complexity.

Cognitive and methodological complexity

The density and scientific detail of the methodological framework were perceived as demanding, particularly for participants with heterogeneous backgrounds. In Germany, Italy and Portugal, facilitators reported that moving rapidly between structured exercises required continuous adaptation to prevent overload. While the structure was appreciated (Ireland), greater modularity and prioritisation—clarifying which steps were essential and which optional—could have reduced cognitive pressure and improved sustainability of engagement.

Furthermore, policy proposal development required substantial technical expertise, including knowledge of legal frameworks, fiscal instruments and administrative competences. Co-creation processes addressing structural reforms may require complementary legal and economic expertise.

Cross-sectoral interdependencies

Although the methodology separated participants into sectoral groups (e.g. energy, housing, transport), national reports—particularly from Ireland—highlighted that these sectors are deeply interconnected. Subgroups frequently found themselves discussing overlapping policies, especially when developing transformative proposals. In Hungary, for instance, participants decided to elaborate a cross-sectoral proposal with most individual policies being proposed as transversal to the energy and transport sectors.

Power structures within the labs

Implicit power asymmetries may have influenced discussions inside the lab. FITTER-EU partners and institutional representatives from the public administration —'policy-makers' often possessed greater technical knowledge than disadvantaged groups represented in the labs, which could shape how problems and solutions were framed.

In addition, the methodological framing introduced by FITTER researchers inevitably influenced the way participants conceptualised problems. While this ensured comparability across countries, it may also have limited alternative framings or grassroots narratives.

Operationalising proposals and creating impact

The actual impact of the policy proposals will depend on the usage and dissemination of the FITTER-EU platform of the FITTER-EU platform because these policy proposals will be included inside the platform. Although the creation of a policy portfolio in each case study focused on identifying leverage points and mitigation pathways, the outputs remain conceptual and deliberative. Their feasibility, scalability and fiscal implications require further technical validation through modelling, legal analysis and negotiation with competent authorities. For instance, in Spain some of the participants criticised the value of their participation and questioned the potential of the workshops to reach real impact in public policy design.

3. The implemented co-creation labs

3.1. Overview of the FITTER-EU lab sessions and participants

The table below presents a consolidated overview of participation in the FITTER-EU labs across the six countries, detailing the implemented lab sessions and total number of participants per session. In total, 26 lab meetings were carried out, involving 237 external participants. Each country addressed two sectors in their national labs, with energy being a cross-cutting sector for all case studies. The other two analysed sectors were divided evenly among the countries: housing was addressed in Germany, Ireland and Portugal, while transport was covered in Italy, Hungary and Spain.

Two countries (Portugal and Germany) strictly followed the methodology—that foresaw three in-person and two online meetings— with the last validation session being optional. Variations to the methodology entailed the following:

- The first Hungarian first lab was organised directly with members of DGs instead of intermediary organisations. SFC liased and collaborated with the CSO Igazgyöngy Alapítvány, which was their main point of contact. The reason for this deviation was that the first lab had already been organised prior to the provision of T4.2 methodological guidelines and the Hungarian partners considered this as a most suitable approach to their national context: being that most of the stakeholders and intermediaries were fully aligned to governmental positions, the process would have been at risk of resulting bias. In the following sessions of the Hungarian lab, the sessions were organised with members of intermediary organisations representing DGs and policy makers, similar to all of the other case studies. The online workshops were conducted as internal sessions within the partner’s team (see section 3.2 on deviations in the methodology for more details).
- Italian and Spanish partners implemented four meetings and did not organise the final validation session, which was specified to be optional in the guidelines.
- Spain organised the second and third in-person labs on the same day as a full working day, with morning and afternoon sessions, rather than holding them on separate half-day dates.
- Irish partners organised only the three in-presence meetings as part of their national co-creation path. Even if Irish country case leaders had planned the first online validation lab, they needed to cancel it because of the low number of confirmed participants.

The total number of participants was considerably higher in the first meeting, with 99 external stakeholders, compared to the total number of attendants to the second and third labs —47 and 43, respectively. This is primarily explained by the highest level of attendance to the first Hungarian Lab as reported above.

As can be seen in the table, the number of external participants was lower than the number originally requested in the initial engagement guidelines (15–20 participants per national lab). This highlights the considerable challenges faced regarding stakeholder engagement, which were experienced by all partners (see Section 2.5 for more details on limitations in the methodology and co-creation process).

Country	Sectors addressed	Number of lab	Date	Venue	Facilitators	Participants per lab session	Total participants from intermediary organisations
DE	Energy Housing	1st in-presence lab	09.10.2025	EineWeltHaus München	3	7	29
		1st online lab	30.10.2025	Online	2	4	

Country	Sectors addressed	Number of lab	Date	Venue	Facilitators	Participants per lab session	Total participants from intermediary organisations
		2nd in-presence lab	06.11.2025	EineWeltHaus München	3	3	
		3rd in-presence lab	20.11.2025		3	9	
		2nd online lab	11.12.2025	Online	3	6	
HU	Energy Transport	1st in-presence lab	25.09.2025	Told, Igazgyöngy Foundation HQ	2	47*	32
		1st in-presence lab	09.10.2025				
		1st online lab	29.10.2025	Online	1	5	
		2nd in-presence lab	05.11.2025	Rumbach Space, Budapest	2	12	
		3rd in-presence lab	02.12.2025	Rumbach Space, Budapest	2	9	
		2nd online lab	04.12.2025	Online	1	3	
IE	Energy Housing	1st in-presence lab	10.10.2025	TU Dublin (Dublin)	4	10	23
		1st online lab	N/A	N/A	N/A	N/A	
		2nd in-presence lab	14.11.2025	TU Dublin (Dublin)	6	8	
		3rd in-presence lab	28.11.2025	TU Dublin (Dublin)	6	5	
		2nd online lab	N/A	N/A	N/A	N/A	
IT	Energy Transport	1st in-presence lab	20.10.2025	Wire Working (Roma)	3	15	36
		1st online lab	20.10.2025	Online	3	9	
		2nd in-presence lab	14.11.2025	Wire Working (Roma)	3	6	
		3rd in-presence lab	05.12.2025		3	6	
		2nd online lab	N/A	N/A	N/A	N/A	
ES	Energy Transport	1st in-presence lab	02.10.2025	Media and Communication Faculty (Avenida Complutense, 3). Madrid	2	11	39
		1st online lab	20.10.2025	online	2	8	
		2nd in-presence lab	11.12.2025 (10 a.m. to 2 p.m.)	Media and Communication Faculty (Avenida Complutense, 3). Madrid	2	10	
		3rd in-presence lab	11.12.2025 (3 p.m. to 6 p.m.)		2	10	
		2nd online lab	N/A	N/A	N/A	N/A	
PT	Energy Housing	1st in-presence lab	07.11.2025	Angra de Heroísmo, Terceira Island, Portugal	2	9	34
		1st online lab	14.11.2025	Online	1	7	

Country	Sectors addressed	Number of lab	Date	Venue	Facilitators	Participants per lab session	Total participants from intermediary organisations
		2nd in-presence lab	24.11.2025	Angra de Heroísmo, Terceira Island, Portugal	1	8	
		2nd online lab	05.12.2025	Online	1	6	
		3rd in-presence lab	11.12.2025	Angra de Heroísmo, Terceira Island, Portugal	1	4	
		(*) 44 individuals from DGs and 3 from intermediary organisations representing DGs					

Table 8: Overall sessions in the six FITTER-EU labs

Country-level participation trends are presented in Figure 7, with the exception of the first session of the Hungarian lab —HU reported from the second in-presence lab—, to avoid a distortion in the graph due to the outlier value. On average, participation per meeting was approximately eight participants, although trajectories varied across countries. Spain maintained a constant group of participants throughout the lab process. Portugal, Italy and Ireland experienced a decrease in participation after the first meeting, keeping a core group during the whole co-creation lab. Germany recorded higher participation in the first and final meetings. In Hungary, following the initial large-scale engagement of disadvantaged individuals, a stable group of intermediary organisations continued to participate in subsequent sessions. Considering the total number of external participants in each FITTER-EU lab, five out of the six FITTER-EU countries had between 29 to 39 participants throughout all the organised sessions. The exception was Ireland with a total number of 23 participants, which is explained by the organisation of only three out of the five sessions.

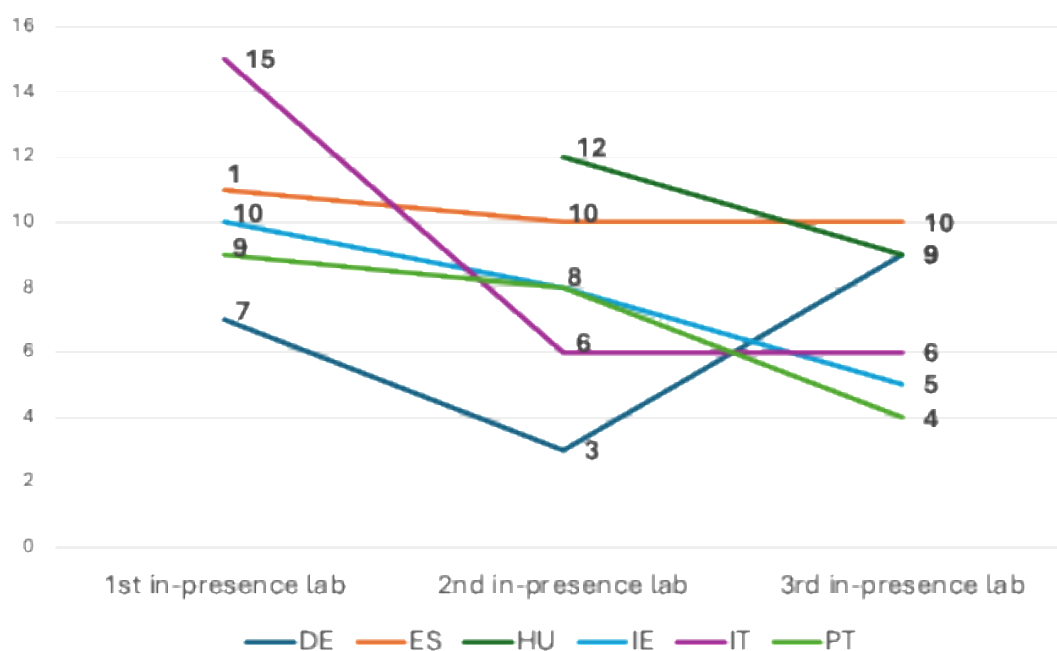


Figure 7: Participation in FITTER-EU national labs (number of external participants)

Inclusiveness was an important feature of FITTER-EU engagement strategies and co-creation methodologies—refer to D2.5 presenting the stakeholder engagement of the project and D2.6 with the dedicated stakeholder engagement strategy for vulnerable groups—, that was consistently sought for by organising partners and with overall positive results:

- In **Italy**, overall gender participation was strongly skewed towards women, with only three men participating throughout the process—one in the first in-person meeting, one in the first online meeting, and one in the second in-person meeting. Most participants were experts in the transport sector. During the first lab, almost two-thirds of participants were from the transport sector, while the remaining third were from the energy sector. In the last two in-person sessions, participation consisted exclusively of transport experts. In terms of group dynamics, interactions were generally constructive, and participants engaged actively, allowing one another to speak and responding to each other's contributions.
- In **Germany**, overall gender participation was balanced between men and women. An exception occurred during the second in-person workshop, where only three organisations participated and all attendees were men, resulting in a temporary gender imbalance. Additionally, in the first and fifth labs, the energy subgroup was composed exclusively of men. In terms of group dynamics, while overall interaction was constructive and participants engaged actively—allowing each other to speak and responding to one another's inputs—two older male participants (one in-person and one online) required moderation to prevent disproportionate speaking time. From a geographical perspective, participation in in-person sessions was mainly composed of local and regional actors, as national-level experts could not be successfully engaged. The discussions were therefore strongly oriented towards urban realities, particularly in relation to housing ownership and tenancy as key dimensions of inequality. Direct representation of structurally disadvantaged groups was limited. Although some individuals working closely with vulnerable populations participated, recruitment of representatives from these groups proved challenging, partly due to time constraints and the voluntary nature of the exchange. Regarding thematic expertise, housing-related discussions were more fluid than those on energy, and participants often highlighted the difficulty of separating the two sectors, given the alignment of consumer vulnerabilities across energy and housing contexts.
- In **Hungary**, the first in-person lab was characterised by direct participation of disadvantaged individuals. Of the 47 participants, 37 were women, resulting in a strong female majority. This imbalance was attributed to socio-economic dynamics in the local context, where men are more frequently engaged in employment while women assume childcare responsibilities, making them more available to participate. Several women attended with underage children, although children did not participate in discussions. All participants came from the village of Told in Eastern Hungary, although many had previously relocated there from nearby towns or more remote areas due to personal circumstances such as marriage or bereavement. Participants faced significant socio-economic challenges, including poverty and intersectional vulnerabilities related to education, employment and health. Their mobility capacities were limited, and their engagement was facilitated through ongoing mentoring relationships with the Igazgyöngy Foundation. In subsequent sessions, participation shifted towards non-governmental organisations (NGO) representatives, local government actors and one union representative, alongside some party-affiliated experts. Gender balance in these later sessions was approximately equal, with a slight majority of women. Expertise spanned diverse areas including child support, green transition policies and workers' rights
- In **Spain**, the participant group consisted of ten individuals, with a balanced gender distribution (six men and four women). Participants were primarily affiliated with research institutions and social

organisations, including researchers and social workers with direct professional experience working with vulnerable population groups affected by the green and digital transitions. The diversity of professional backgrounds contributed to rich discussions that integrated technical knowledge, social expertise and lived-experience perspectives, particularly regarding disability, energy poverty and accessibility in transport. The group demonstrated strong awareness of equity considerations and structural inequalities, which informed both the analysis of systemic challenges and the formulation of policy proposals. No imbalances in participation dynamics were observed.

- In **Portugal**, most participants were women, with only one man participating in the sessions. During the lab sessions, participants spontaneously organised in two groups based on shared interests, institutional affinities and hierarchical roles in their organisations. Thus, one of the groups included several individuals holding leadership positions roles —the presidents or senior representatives of the organisations—, while in the other prevailed participants with a political and more junior role. These groups were cross-sectoral in composition, with each bringing together organisations active across both the energy and housing sectors. Consequently, discussions were not structured along sector-specific lines but rather developed through integrated and cross-cutting exchanges that reflected the interdependencies between energy and housing vulnerabilities.
- Finally, in **Ireland**, the participant group was composed of ten individuals representing a diverse mix of civil society organisations, public authorities, and research-oriented initiatives. Gender representation was relatively balanced, with six women and four men participating. Women were particularly represented in civil society and advocacy roles, including organisations focused on gender equality, socio-economic disadvantage and support for older people, while men were more present in research, digital skills provision and central government policy roles. No significant gender-based imbalances in speaking time or interaction dynamics were reported. From an institutional perspective, the group combined three policy-making actors (a regional elected representative, a civil servant from the Department of Climate, Energy and the Environment, and a public service research and policy manager) with civil society and research organisations operating at national and local levels. Geographically, participation spanned local, regional and national levels, with a predominance of nationally operating organisations.

Organisational profile in each lab

This section presents the organisational profile of the FITTER-EU labs, reporting the characteristics of the organisations participating rather than those of individual participants. The number of organisations participating in each lab differs from the number of individual participants reported in Table 8, as in some cases more than one participant from the same organisation attended the lab sessions.

In total, 83 organisations and one independent expert (Germany) participated in the FITTER-EU labs —see the full list of organisations in Annex II: List of organisations participating in the national labs—. A quarter of the organisations (25%) participating in the FITTER-EU labs were German (20 organisations plus one expert), followed by 19% of Spanish organisations (16), 18% of Italian organisations (15) and 17% of Irish organisations (14). The countries with less organisations involved in the labs were Hungary (10 organisations) and Portugal (8 organisations).

Overall, more than half of the organisations involved in the labs (55%, 46) had an exclusively national scope, while 8% (10) had a regional scope and 16% (20) a local scope. The remaining organisations had a mixed scope, primarily combining regional and national levels. Table 9 presents the geographical scope of organisations in each country case study. Organisations with a mixed focus on national and regional/local levels were excluded from the table.

Half of the countries — Hungary, Italy and Ireland — involved organisations predominantly with a national focus, while one — Spain — was balanced between the two levels, with the regional organisations based in Madrid. In the case of Germany, although national organisations were still in the majority, there was a higher number of regional organisations compared to the other countries. The German partners were based in Munich and aimed to organise a workshop with regional impact, addressing the housing crisis in the city; therefore, they involved a wide number of relevant organisations in the local context. A similar situation occurred in the Portuguese lab, as the workshop was organised in the Azores islands, thereby involving a majority of local stakeholders.

Contry	Local and regional	National
DE	38%	43%
ES	50%	50%
HU	30%	60%
IE	21%	71%
IT	27%	67%
PT	63%	38%

Table 9: Geographical scope of organisations participating in the national FITTER-EU labs

Regarding the sectoral focus, 43% of organisations (36 out of 83) did not report a specific sectoral orientation. In Spain and Hungary no organisations with a specific sectoral focus were reported. In Portugal, participation was equally divided between organisations without a sectoral focus and those focused on housing. In Germany, all participating organisations reported a sectoral focus (nine in energy, nine in housing and three in both sectors). In Ireland, five organisations reported no sectoral focus, while the remaining nine were distributed across energy, housing and transport. In Italy, two organisations reported no sectoral focus and the remaining thirteen were distributed across sectors, with only two organisations explicitly focused on transport.

Figure 8 shows the distribution of represented organisations whose primary function relates to policy formulation or implementation—including both civil servants and policy experts, as detailed in Chapter 2. In total, 56% of participating organisations can be classified within this category. Germany presents the highest proportion (67%), followed by Italy (60%), while Spain shows the lowest percentage (19%).

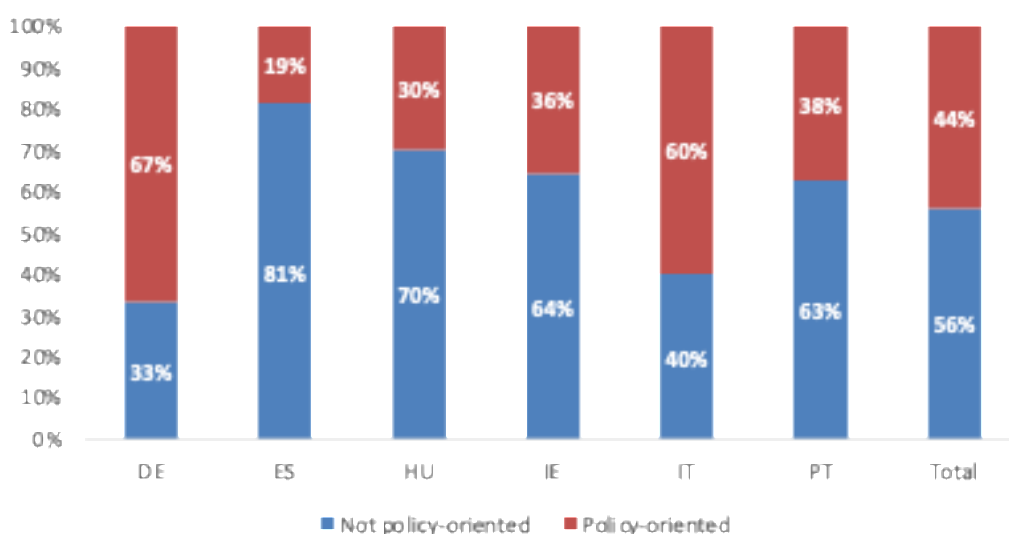


Figure 8: Participating policy makers and policy experts in the lab (% of organisations)

Finally, Figure 9 presents the distribution of DG-focused organisations by country: 71% of participating organisations reported an explicit focus on DGs. This constitutes the majority profile in all countries except Germany, where 70% of organisations did not report a specific DG focus. In the remaining countries, at least 67% of participating organisations focus on DGs, reaching 100% in Spain.

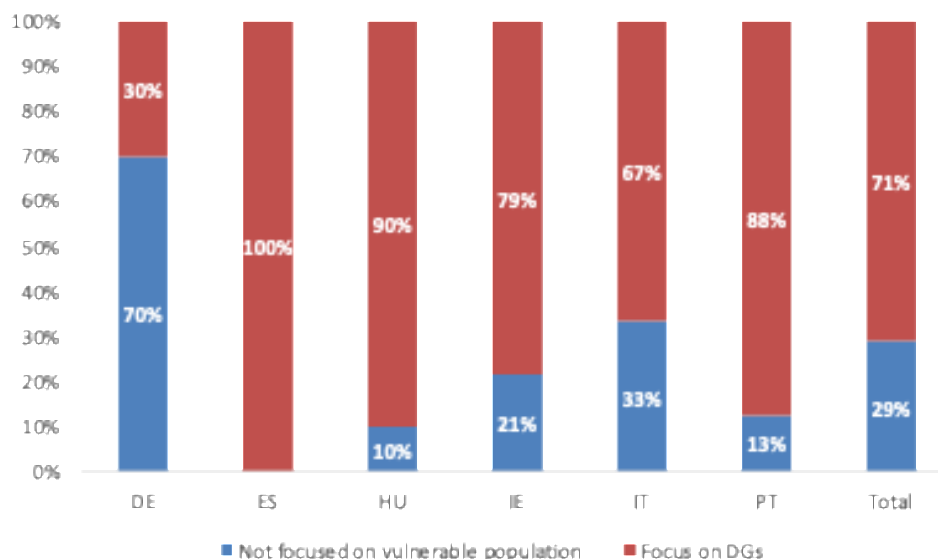


Figure 9: Participating organisations with a focus on DGs (% of organisations)

3.2. Deviations from the methodology

This section presents the most relevant deviations from the methodological guidelines —see Chapter 2— implemented in each FITTER-EU lab and explain some of the participation patterns presented in previous chapters. Across the six countries, the implementation of the co-creation methodology shows both a high level of overall adherence to the common framework and a set of context-driven adaptations concerning sequencing, sectoral organisation and validation dynamics. Germany, Hungary and Portugal implemented all the five sessions foreseen (three in-presence labs and two online validation sessions), whereas Italy and Spain conducted four sessions, cancelling the second online validation. It is worth noting that this session was defined as optional in the methodological guidelines to be conducted if the validation of policy factsheets could not be finalised during the third in-presence lab. Finally, Ireland implemented only three in-person labs, cancelling both online validations: even though FITTER-EU Irish partners had organised the first online validation session, it had to be cancelled because of the limited number of participants.

There were some modifications regarding the proposed sequencing of the labs¹: Spain merged the second and third labs into a full-day in-presence session and validated policy factsheets there; Portugal anticipated the second online session to take place between the second and third labs and added a post-session written validation phase. Hungary diverged more substantially because, before the T4.2 guidelines were provided, the FITTER-EU Hungarian partners had liaised with an NGO with which they had previously collaborated in other project participatory activities and arranged to organise the co-creation workshops with them and their beneficiaries —vulnerable population—. The first Hungarian lab was therefore organised with

¹ The sequence of lab sessions established in the methodological guidelines was: (i) the first in-presence lab, (iii) first online session to validate the system maps, (ii) second in-presence labs, (iv) third in-presence lab and (v) second online validations session (optional).

individuals from DGs rather than with intermediary organisations, and it was held as a two-day event. After this initial workshop, the Hungarian partners were asked to follow the T4.2 methodological guidelines, and the remaining sessions were organised as foreseen in the methodology, inviting other CSOs and policymakers. In this process, the online validation sessions were transformed into internal working meetings within the national FITTER-EU team, primarily to align the outputs of the first workshop with the T4.2 guidelines.

Regarding sectoral focus and working group configurations inside each lab session, Germany, Ireland, Italy and Spain largely followed the methodology by organising participants into two sector-oriented subgroups, at least in the initial labs. However, small participant numbers frequently led to adjustments: during the second in-presence labs, Germany and Italy reduced one lab to a single sector-focused group —housing and transport, respectively—; Ireland’s and Italy’s third labs were conducted in plenary due to limited attendance. Portugal departed from the sector-based structure entirely, allowing participants to self-organise into cross-sectoral groups based on institutional affiliation and hierarchical role. Hungary combined approaches: subgroups —with no sector-specific orientation— were used only in the first lab with vulnerable participants, while subsequent sessions with external stakeholders were conducted in plenary with a cross-sectoral, citizen-focused perspective.

In terms of agenda management and co-creation methods, time constraints and participant availability were recurrent drivers of adaptation. Ireland and Italy shortened or merged agenda slots, which reduced the number of policy factsheets produced or modified scoring procedures. Spain and Germany adjusted discussion formats to allow deeper qualitative exchanges before scoring transformative potential. Preparatory work on policy factsheets varied significantly: Spain and Italy prepared in advance draft proposals ahead of later labs; Germany prepared initial topics and ideas for selected factsheets in advance based on existing relevant literature (see below); Hungary finalised and complemented system maps and policy factsheets internally; while Portugal and Ireland reported no methodological deviations in this respect.

3.2.1. Germany

Number and sequence of sessions:

- All five sessions planned in the methodology implemented: three in-presence labs and two online validation sessions.
- The analysis of transformative potential was conducted in a dedicated second online session rather than integrating it into the third in-presence lab.

Sectoral focus inside the labs and working groups:

- With the exception of the second in-presence lab, all sessions were organised into two sector-oriented subgroups: one focusing on energy and one on housing. Due to the relatively small number of participants overall, only two subgroups were formed per session.
- Participants were invited based on their expertise in climate, energy, or housing policy, or as representatives of DGs.
- The second in-presence lab was thematically narrowed to one sector (housing) due to the limited number of participants (three).

Agenda and co-creation methods:

- First online session:

- A Mentimeter questionnaire (three questions) was introduced at the beginning of the session to stimulate interaction and gather initial reflections on tenancy, digitalisation, and sustainability in the building and energy sectors.
- Instead of addressing sectors sequentially in plenary, participants were divided into two breakout groups (energy and housing) to enable more in-depth discussions.
- The discussion questions provided in the common methodology were adapted to better reflect the German regulatory and policy context.
- Due to limited staff capacity, two WECF facilitators combined moderation and note-taking tasks.
- Second in-presence lab:
 - Owing to the reduced number of participants (3), the session focused exclusively on the housing sector.
 - Similarly to the Italian case, the template “Identifying systemic changes and policies” was not used —see agenda slots 9:25-11:30 in Section 2.3.2—, since systemic changes had already been discussed in the previous lab. Instead, a positioning exercise (“I agree – I do not agree”) was conducted to assess participants’ views on proposed future visions and necessary changes in the housing sector.
 - WECF had prepared two specific topics for the policy factsheets in advance to guide the discussion (modernisations levy and funding programmes for efficient buildings). The aim was to include knowledge from existing publications that had been shared during the first lab by participants².
- Third in-presence lab and final online validation session:
 - The third lab concentrated on adjusting existing policies and co-creating new policies through the completion of Policy Factsheets.
 - A short discussion on potential advocacy activities for 2026 was included, responding to participants’ requests expressed during the first in-presence lab.
 - A dedicated online validation to merge the analysis of the transformative potential of co-created policies was organised. This allowed additional experts — including some who had not participated in earlier workshops — to contribute an external perspective. Participants received the policy factsheets in advance. As they were well prepared, only a short introduction to each recommendation was necessary, enabling focused discussion on transformative potential, risks, and feasibility.

3.2.1. Hungary

Number and sequence of sessions:

- All five sessions planned in the methodology implemented: three in-presence labs and two online validation sessions.
- FITTER-EU Hungarian partners started the preparation of the co-creation process on May 2025, before the provision of T4.2 official methodological guidelines elaborated by SV in mid-September 2025. As a result, some discrepancies between the initial agreements with partners and the final guidelines were unavoidable.
- To organise the workshops, Hungarian partners contacted Igazgyöngy Alapítvány, a Hungarian non-governmental organisation (NGO) established in 1999 that works with a comprehensive

² The references shared during the first German lab were: (Bundesamt für Wirtschaft und Ausfuhrkontrolle, BAFA, 2025); (Bundesverband Erneuerbare Energien, BEE, 2025); (Kaiser & Schaffert, 2025); (Lange & Thesfeld, 2025) and (Ortego & Steitz, 2024)

settlement-development model targeting marginalised Roma communities and families living in deep poverty one of the most disadvantaged rural regions of Eastern Hungary. They originally planned to work directly with people affected by vulnerabilities (members of DGs and not intermediary organisations). As a result, the first in-presence lab was organised according to this original programme, collecting the needs of DGs and the programme was simplified to collect their feedback in a more open format —not following all the structured steps and templates in the methodological guidelines—.

- The second and third in-presence labs were aligned with T4.2 guidelines, working with intermediary organisations representing DGs —members of Igazgyöngy Alapítvány — and other CSOs, as well as policy makers.
- The first and second online validation sessions were transformed into internal working sessions within the organising team (SFC). These internal meetings served to align earlier results with the FITTER-EU methodological requirements and to finalise system maps and policy factsheets. In particular, during the first session, SFC partners prepared the systems maps as contemplated in the methodological guidelines, based on the inputs collected during the first in-presence lab with members of DGs. During the second internal working session, SFC partners conducted preparatory work on the co-creation factsheets, to be finalised during the third in-person lab with the external participants — intermediary organisations representing DGs and policymakers.

Sectoral focus inside the labs and working groups:

- During the first lab, two subgroups were formed per session, each with one facilitator and one note-taker. Discussions were guided by structured question sets covering perceptions, effects and adaptation, drawing on prior FITTER-EU tasks (T3.1 and T3.3).
- In the second and third in-presence labs with external stakeholders, no sector-oriented subgroups were formed. Although organisations could broadly be associated with energy (5 in the second lab; 3 in the third lab) and transport/housing (5 in the second lab; 3 in the third lab), participants typically worked across sectors with a citizen-focused approach (e.g. municipalities, anti-poverty organisations, sustainable water management organisations, green transition advocates, rule-of-law organisations). Given the cross-sectoral mandates of the participating organisations and the limited number of participants, discussions were conducted in plenary to ensure an integrated perspective on energy poverty and mobility challenges.

Agenda and co-creation methods:

- First in-presence lab with vulnerable participants (persons directly affected by twin transition negative effects):
 - The programme was split into two events (green and digital focus) to accommodate participants' needs, while maintaining an intersectional perspective on the twin transition.
 - To align with the T4.2 methodology, system maps were developed by the SFC team and Igazgyöngy managers after the meeting, based on participants' inputs. These maps were not directly presented to vulnerable participants, as they were considered too abstract and complex. Instead, they were later validated with stakeholders involved in subsequent workshops.
- First online session:
 - Converted into an internal working session inside the FITTER-EU Hungarian team to align with T4.2 guidelines.
 - The organising team finalised system maps based on earlier discussions and feedback from Igazgyöngy Alapítvány.

- Second in-presence lab: Validation exercises originally foreseen for the online session were implemented during this in-person meeting with external stakeholders.
- Third in-presence lab: Some validation components foreseen in the methodology were partially implemented due to time constraints.
- Second online session: Conducted as an internal working meeting. FITTER-EU partners complemented the co-creation of policy factsheets on the basis of the discussions held during the third in-presence lab with external stakeholders.

3.2.2. Ireland

Number and sequence of sessions:

- Three meetings implemented of the five planned in the methodology: three in-presence sessions. The two online validation sessions were cancelled.
- First online validation session was cancelled 3 days before the planned event, due to very small number of people registered (2).
- The second online validation session was not held. Instead, the validation of the factsheets on adjustments to existing policies co-created in the second in-person session took place during the third in-person session. For this purpose, the chairperson prepared a set of questions for each factsheet and brought them to the group for detailed discussion.
- The third in-person Lab was shortened from the recommended 4 hours to 3 hours and 15 minutes (-45 minutes).

Sectoral focus inside the labs and working groups:

- During the first and second in-person labs the sector-oriented exercises were implemented in two subgroups, one focusing on energy and the other on housing, as foreseen in the methodology.
- During the third in-presence lab no subgroups were formed due to the small number of external participants (5 in total) and participants worked collaboratively. Since all of them had already taken part in previous in-person Open Lab sessions, participants who had previously worked in the energy subgroup validated the energy policy factsheets, while those who had worked on housing validated the housing policy factsheets. The second part of the session, dedicated to the co-creation of new transformative policies, all participants (external and internal from the FITTER-EU Irish partner) worked together collaboratively.

Agenda and co-creation methods:

- Second in-presence lab: due to time constraints the session dedicated to the co-creation of policy measures was shortened and exclusively focused on the adjustment of existing policies. This reduction affected the output: each subgroup produced 3 factsheets, instead of the 10 recommended in the guidelines.
- Third in-presence lab: the first part was dedicated to the validation of the policy factsheets co-created during the first lab and to the co-creation of new policies, taking more time than planned. The final session focused on the analysis of the transformative potential of policies was shortened, merging the two agenda slots planned in the methodology, which meant that the scoring component couldn't be implemented.

3.2.3. Italy

Number and sequence of sessions:

- Four meetings implemented of the five planned in the methodology: three in-presence sessions and first online validation session. The second online validation session was cancelled.
- Only the last validation session was cancelled because the policy proposals — mitigation measures — were validated as part of the third lab.

Sectoral focus inside the labs and working groups:

- During the first in-presence lab, participants worked in two sector-oriented groups, as foreseen in the methodological guidelines: two thirds of the participants were in the transport group, while one third were in the energy group.
- During the second and third in-presence labs, there was only one participant from the original energy group. Therefore, the second lab was exclusively focused on the co-creation of transport policies.
- Before the third in-presence lab, FITTER-EU partners carried out preparatory work aimed at concluding a draft version of the policy proposals based on the inputs collected from the first two labs. Due to the limited number of participants, all activities in the third lab were implemented in plenary, covering the two prioritised sectors — energy and transport.

Agenda and co-creation methods:

- First lab: no deviations.
- Second in-presence lab: the FITTER-EU facilitation group decided to shorten and merge the two initial agenda sessions (9:25–10:25 ‘Identifying systemic changes: workgroups per sector’ and 10:25–11:30 ‘Identifying policies for change: workgroups per sector’; see section 2.3.2), because during the first lab the required and proposed changes to reach a just transition had already been identified. Thus, it was possible to dedicate more time to the co-creation of policies. The group worked on the co-creation of five policy factsheets, covering approximately half of the agreed proposals during the first time slot.
- As preparatory work for the third co-creation lab, the Italian FITTER-EU partners finalised a draft version of the proposals to be validated during the third lab by: (i) concluding the policy factsheets of the proposed transport policies agreed upon in the second lab, and (ii) proposing a set of policies and filling in the policy factsheets for the energy sector.
- Third in-presence lab: in order to maintain a thematic focus, it was decided to firstly address the transport sector and then the energy sector, analysing the policy factsheets one by one and covering all agenda items: (i) validation of the content of the policy factsheets and analysis for their transformative potential; and (ii) scoring the two dimensions of transformative potential. This last exercise, unlike the guidelines — which foresaw individual scoring followed by collective discussion — was conducted collectively in plenary from the outset due to time constraints. The final agenda slot (12:15–12:45 ‘Scoring of policies according to their transformative potential’; see section 2.3.3), specifically the part dedicated to categorising the policies, was not implemented because of time constraints; however, this exercise was carried out ex post by FITTER-EU partners, who assigned the categories based on the scores agreed during the lab.

3.2.4. Spain

Number and sequence of sessions:

- Four meetings implemented of the five planned in the methodology: three in-presence sessions and first online validation session. The second online validation session was cancelled.
- The content of the second and third in-person labs was integrated into one full-day session, instead of two half-day sessions as originally proposed in the methodology.
- During this integrated session, the validation of the co-created policy factsheets was carried out. Therefore, the second online validation session, initially foreseen, was deemed unnecessary by the Spanish FITTER-EU partners and was cancelled.

Sectoral focus within the labs and working groups:

- All sessions were implemented through two sector-oriented working groups and plenary discussions, as foreseen in the methodology.

Agenda and co-creation methods:

- First in-person lab: no modifications.
- The Spanish FITTER-EU partners decided to pre-elaborate a first draft of the policy proposals in advance —presenting the ideas but not filling the policy factsheets—, after collecting inputs from the first lab. This draft was presented during the integrated session of the second and third labs.
- Second in-person lab (morning session): no major deviations from the proposed methodology were implemented. More time than initially planned was dedicated to collective discussion. Participants expressed the need to clarify how existing policies (e.g. Plan MOVES, National Energy Poverty Strategy) are experienced in practice by vulnerable groups. This adjustment enriched the identification of policy gaps and limitations but reduced the time available for generating additional new policy proposals.
- Third in-person lab (afternoon session):
 - The agenda was slightly adapted to allow for a more in-depth qualitative discussion of the transformative potential of the policies before proceeding to scoring. This adjustment aimed to ensure a shared understanding of key concepts such as systemic change, equity, and feasibility, given the complexity of the issues addressed.
 - Time management was the main constraint. Due to the participants' extensive professional experience, discussions were particularly detailed and could not be significantly shortened. While this limited the number of new proposals generated, it strengthened the qualitative depth and coherence of the outputs.

3.2.5. Portugal

Number and sequence of sessions:

- All five sessions planned in the methodology implemented: three in-presence labs and two online validation sessions.
- Instead of implementing the second online session at the end of the co-creation process, it was carried out between the second and third in-presence labs.
- A post-session validation phase was planned, involving the circulation of draft reports to participants for review and final feedback.

Sectoral focus inside the labs and working groups:

- Instead of forming sector-oriented groups for the development of the methodology, participants organised themselves freely according to their institutional affiliations and hierarchical roles within their organisations, with one group bringing together those in more senior and leadership

positions. The way of operating in throughout the whole co-creation process was therefore not in sector-oriented groups as indicated but in two groups each with a cross-sectoral orientation.

Agenda and co-creation methods:

- No deviations

4. Twin transition challenges identified in the labs

This chapter reports on the prioritised sectoral challenges identified during the first FITTER-EU in-presence labs, in particular within ‘The systemic dimensions of challenges’ session part of the agenda of the first lab —see section 2.3.1—. During this session, after individually identifying challenges affecting vulnerable profiles — conducted through the guided brainstorming exercise with personas — lab participants were guided to consider the most pressing negative effects of the twin transition in each sector. This served as the starting point for building the system maps and proposing mitigation measures, whose aim was to address these challenges.

4.1. Energy sector

Across all six countries, participants identified affordability constraints and energy poverty as central structural barriers to a fair energy transition. In Hungary, Spain, Portugal and Italy, energy poverty was described as severe and multidimensional, particularly where low income intersects with age, disability, unemployment, precarious work, rural or peripheral residence, and single-person households. In Germany and Ireland, affordability pressures were strongly linked to rising rents and energy costs, which reduce disposable income and limit the capacity to invest in renewable energy, energy-efficient appliances, or sustainable mobility choices. In all contexts, participants highlighted that subsidy schemes for solar panels, heat pumps or retrofitting require upfront investment, thereby structurally excluding low-income households. In Spain and Italy, limited digital literacy and complex application procedures further restrict access to bonuses and incentives.

A closely related cross-country issue concerns the landlord–tenant divide and housing insecurity, particularly salient in Germany, Ireland and Portugal, and present to a lesser extent in Italy. Lab discussions emphasised that tenants lack agency to retrofit dwellings or join energy communities, while homeowners disproportionately benefit from incentives. In Germany, modernisation levies were reported to increase rents and exacerbate social segregation, contributing to clustering by income and in some cases prompting suburban displacement. This dynamic was linked to increased commuting distances and fossil fuel dependency. German participants also raised the issue of “remanence” (under-occupation of oversized dwellings) and criticised housing policy for following a predominantly market-oriented logic that neglects systemic inefficiencies. In Ireland and Portugal, reduced decision-making autonomy and lack of tenure security were highlighted as structural barriers.

All country labs pointed to policy design and governance gaps. A recurring concern was the mismatch between ambitious climate targets and the socio-economic realities of vulnerable groups. Policies were described as fragmented, technocratic, or overly focused on technological solutions without sufficient attention to affordability, income constraints or social equity. In Italy, institutional fragmentation across national, regional and local levels, uneven administrative capacity and bureaucratic complexity were identified as major implementation barriers. In Hungary, participants stressed regulatory inconsistency, frequent feed-in restrictions, and the influence of partisan political interests over energy policy decisions, including preferential treatment of certain investments and limited decentralisation. In Spain, a strong policy–reality gap was discussed, including ineffective communication, non-transparent tariffs and declining institutional trust. Irish participants highlighted the absence of a human-rights-based approach and the limitations of one-size-fits-all policy design that does not sufficiently account for regional variation. Portuguese discussions emphasised that existing measures are limited, fragmented and insufficiently aligned with the real needs of financially fragile households.

Market structure and price asymmetries emerged prominently in Hungary and Italy, and indirectly in Germany and Spain. Italian participants described energy as being treated primarily as a market commodity rather than a common good, with large operators retaining most profits, energy sold to the grid poorly remunerated, and purchased electricity remaining expensive. The proliferation of commercial offers was seen as generating confusion and distrust. Hungarian labs highlighted state control of the grid, distorted market incentives, and reduced investor confidence. German discussions criticised the limited social components in energy legislation (e.g. GEG), while Spanish participants referred to opaque tariffs and service conditions.

Significant infrastructural and technological constraints were identified across Hungary, Ireland, Italy, Germany and Portugal. These include insufficient grid capacity, weak local grids (particularly in small municipalities), absence of smart-grid integration, widespread non-smart meters (Italy), and limited storage infrastructure. Hungary and Ireland stressed challenges related to renewable intermittency and limited economies of scale (e.g. district heating). Several countries reported shortages of qualified technical professionals and insufficient investment in innovation targeting energy poverty reduction. In Germany, limited mobility infrastructure and insufficient digitalisation were linked to suboptimal urban planning and relocation pressures.

Environmental dimensions were discussed in connection with these structural constraints. Italy's dependence on fossil fuel imports was associated with geopolitical vulnerability and slow emissions reduction. Hungary's limited renewable integration contributes to continued pollution. Germany's emphasis on new construction over renovation was described as generating material waste and higher energy consumption. Ireland and Portugal raised concerns about the environmental consequences of limited retrofitting capacity and slow decarbonisation.

Finally, several country-specific elements emerged. German labs connected the housing crisis to risks of social polarisation and democratic strain. Hungarian participants emphasised the geopolitical dimension of energy policy and the overriding of environmental considerations by partisan or economic interests. Spanish labs foregrounded digital exclusion and techno-ableism affecting older people and persons with disabilities. Portuguese participants stressed demographic ageing, low levels of engagement and limited organisational support. Italian participants underlined institutional fragmentation, highlighted regulatory capacity constraints (e.g. ARERA, GSE), territorial disparities and the commodification of energy, while Irish labs focused on tenure insecurity, time poverty and insufficient support for collective energy initiatives, as well as limited skills within communities, and restrictions on commercial participation in energy communities.

4.1.1. Germany

Sectoral challenges prioritised: High and rising rents

Key problems

- Tenancy / housing insecurity:
 - High rents reduce tenants' ability to invest in energy efficiency or renewable energy.
 - The energy transition is increasingly perceived as a project of elites.
 - Renting households cannot participate meaningfully in energy communities or self-generation (e.g., balcony modules have minimal impact due to long amortisation times).
 - Modernisation levies and the consequent burden on tenants increase social inequality and creates social divide and urban segregation connected to the clustering of people with similar financial possibilities in neighbourhoods.

- Rising rents leave less disposable income and it can lead to a movement out of overly crowded cities into sub-urban and rural areas, changing working patterns, increasing the mobility between place of work and of living.
- High individual mobility due to unaffordable housing often increases dependence on fossil fuels for commuting.
- Policy gaps:
 - Housing policy follows a neoliberal market logic, ignoring systemic issues like remanence (tenants remaining in oversised apartments).
 - Housing crisis (lack in available, decent and affordable housing) is turning into a threat to democracy, since mostly the needs of the upper economic part of society are addressed through policies (lack of social components in EEG-and GEG law).
 - Political responses tend to treat symptoms, not underlying structural causes.
- Environmental challenges:
 - Focus on new construction over renovation leads to higher energy consumption and material waste.
 - Tenants' limited spending power reduces adoption of environmentally sustainable behaviours (e.g. renewable electricity choices, energy-efficient appliances or environmentally friendly food or mobility choices).
- Technical and infrastructure constraints:
 - Lack of mobility infrastructure prevents ideal urban planning (e.g., 15-min cities).
 - Insufficient digitalisation and grid capacity, especially for those relocating to suburban/rural areas.

4.1.2. Hungary

Sectoral challenges prioritised:

- Renewable energy sources also have environmental impacts, their production is fluctuating and difficult to store, which poses a major challenge to the capacity of the electricity grid and requires significant investment.
- Energy policy is closely intertwined with the struggle between different interest groups and geopolitical forces, with continuous economic growth and consumerism pitted against environmental constraints.
- Energy poverty is severe, the housing stock is energy inefficient, and energy prices are fixed by the state.

Key problems

- Energy poverty
 - Poorer households are excluded from using renewable energy.
 - State policy sustains energy poverty instead of reducing it.
 - Political polarisation hindering the effective management of energy poverty: party sympathy determines attitudes towards energy poverty.
- Policy gaps
 - State support for solar investments encourages development, but the system penalises production fluctuations, feed-in restrictions are frequent and regulation lacks consistency.
 - State control of the energy grid distorts the market and hinders innovation.
 - Energy policy decisions are driven by partisan political logic.

- State energy policy illegally favours certain investments (e.g. battery factories) while lacking decentralisation.
- 'Green' policies are not enforced due to geopolitical interests.
- Economic rationality is overridden by partisan interests maintaining monopolies.
- Environmental negative effects: limited use of renewable energy sources causes pollution and hampers the achievement of climate neutrality.
- Limited in infrastructure and technology
 - Innovation and investments do not target technologies that reduce energy poverty.
 - Due to shortcomings in education, research and technology, smart-grid integration is absent.
- Distortions in the energy market reduce investment willingness and increase price fluctuations.

4.1.3. Ireland

Sectoral challenges prioritised: tenancy, access to finance.

Key problems

- Tenancy
 - As tenants have no agency to retrofit, they are left further behind, widening divide between homeowners and tenants.
- Access to finance
 - Unavailability to afford upfront costs to retrofit.
 - Increasing energy costs.
- Inadequate sectoral policies:
 - Policies are targeted at individuals, but the transition policies are collective and do not take a human rights approach.
 - Policies do not enable security of tenure; targeted at those on middle to high income who can advocate for themselves.
 - Policy design following a one size fits all approach without considering local conditions into account: policies should be targeted by regions/areas.
 - Energy communities need to be incentivised but lack in skills, time poor, commercial partners not allowed.
- Environmental challenges related to the limitations in retrofitting: increase of greenhouse emissions.
- Technical and technological constraints to expand the energy transition:
 - Limited supply of workers and technological development; limited grid capacity; increasing energy costs.
 - Not enabling economies of scale with regard to the switch to renewable energy sources e.g. district heating.

4.1.4. Italy

Sectoral challenges prioritised:

- Absence of a coherent and equitable framework for a just energy transition.
- Energy is not perceived as an essential common good.

Key problems

- Policy gaps:
 - Although a legal system, well articulated policies and financial instruments for the energy transition exist, the implementation of policies is not homogenous throughout the national territory and is dependent on the capacity of the local authorities.
 - Fragmented governance and bureaucratic complexity:
 - Institutional fragmentation across national, regional and local levels.
 - Administrative capacity varies significantly between territories.
 - High bureaucracy creating implementation delays that increase long-term social and economic costs.
 - Limited management capacity of regulation entities (ARERA, GSE).
 - Limited benefits of energy policies
 - Limited economic accessibility in policies: Incentives often benefit higher-income households.
 - Limitation in the implementation of energy communities: energy selling to the grid produce limited benefits.
- Market concentration and price asymmetry:
 - Energy is treated primarily as a market commodity rather than a common good.
 - Large operators retain most profits.
 - Energy sold to the grid is poorly remunerated; purchased energy remains expensive.
 - Proliferation of commercial offers generates confusion and distrust.
- Energy poverty and territorial disparities:
 - Higher vulnerability in rural and peripheral areas.
 - Cognitive and digital barriers (limited digital literacy) limit access to support mechanisms (bonus and energy communities).
 - Negative narratives link transition policies to rising costs, weakening public consensus.
- Environmental challenges
 - Dependence on fossil fuel imports that creates high exposure to geopolitical risks and limited national renewable production capacity.
 - Slow progress in emissions reduction.
- Technical and infrastructural constraints:
 - Widespread presence of non-smart meters.
 - Weak local grids, particularly in small municipalities.
 - Shortage of qualified technical professionals.
 - Infrastructural limitations hinder renewable energy communities.

4.1.5. Spain

Sectoral challenges prioritised:

- Public policies do not sufficiently target vulnerable groups.
- Digital devices and applications are not affordable or accessible.
- Energy tariffs and services are not transparent.
- Solar subsidies are not relevant for vulnerable households.

Key problems:

- Energy poverty: Energy poverty is most severe where low income overlaps with age, disability, unemployment, loneliness, and precarious work. Single-person households are particularly exposed.
- Inaccessible subsidy design: Grants require upfront payments (e.g. solar panels, heat pumps), excluding low-income households.
- Digital exclusion and techno-ableism:
 - Digital innovation perpetuates ableism and ageism. Limited access to elderlies, low-educated people and people with disabilities.
 - Not affordable to many people.
- Policy–reality gap:
 - Strong mismatch between ambitious environmental targets and daily survival needs of vulnerable households.
 - Public discourse on green innovation feels disconnected from lived realities.
 - Limited trust in institutions due to bureaucratic complexity and slow decision-making.
 - Communication of policies is not effective: language not clear, complex offers of tariffs and available reductions not effectively communicated.

4.1.6. Portugal

Central sectoral challenges prioritised: Financial / Economic; aging population; accessibility in terms of finances and organisational support; low level of interest.

Key problems:

- Lack of financial stability, which pushes energy sector priorities—such as investing in energy efficiency improvements—into the background.
 - The income of the average household strongly influences the ability to install energy-efficient solutions at home and to pay energy bills, which often include a significant share of renewable energy that tends to be more expensive.
 - Limited access to renewable energy sources.
 - Financial difficulties related to installing green energy solutions and accessing renewable energy.
 - Reduced decision-making autonomy for tenants due to dependence on landlords or neighbours in rental housing.
- Insufficient investment in public infrastructure and modernisation, including the infrastructure required for energy storage and transmission.
- Absence of transition policies that reflect real needs.
 - In the energy sector, existing measures are limited, fragmented, or overly focused on technological solutions, with insufficient attention to affordability, income constraints, and tenant–landlord dynamics, preventing vulnerable households from benefiting in practice.
 - Limited accessibility in terms of financial support (e.g. incentive programmes) and organisational support to implement twin transition policies.
- Ageing population: older people often have limited knowledge of recent technological advancements, a situation exacerbated by the absence of policies targeting different age and social groups.
- Low level of interest and engagement.

4.2. Housing sector

Across all three countries —Germany, Ireland and Portugal—, the most consistent issue concerns unequal access to housing and homeownership, closely linked to financial insecurity, labour market position and social vulnerability. Homeownership is widely associated with income stability, creditworthiness and long-term employment security, structurally excluding lower-income households, younger people and those in precarious work. Participants highlighted that sustainability and retrofitting schemes are largely designed for homeowners, leaving tenants with limited access to benefits while continuing to bear high energy costs in inefficient dwellings. In Germany and Ireland, this dynamic reinforces a divide between property owners—who can access subsidies and asset appreciation—and tenants, who face rising costs without corresponding support. In Portugal, restricted access to affordable and social housing was emphasised, particularly affecting families with children, migrants facing restrictive eligibility criteria, and individuals experiencing addiction, mental health challenges or job loss. Across contexts, inefficient housing stock increases energy expenditure, reduces disposable income and heightens exposure to energy poverty.

A second major area of convergence concerns the interaction between housing affordability and spatial displacement, particularly pronounced in Ireland and Portugal and indirectly reflected in Germany. High rental and ownership costs push households towards peripheral or rural areas, resulting in longer commuting distances, increased travel time and greater dependence on private vehicles. Irish participants described uneven and insufficient public transport provision, especially outside Dublin, limited rural connectivity, inadequate service timings (including after-hours services), and safety concerns limiting cycling uptake. Car dependency was associated with additional economic burdens, including vehicle purchase, fuel costs, expensive electric vehicles requiring subsidies, congestion-related public expenditure, and financial strain on older people reliant on taxis. Portuguese discussions similarly emphasised peripheralisation, insufficient reimbursement mechanisms for commuting costs, and limited public transport services adapted to older adults and persons with special needs. In Germany, social and spatial discrimination were highlighted as pushing disadvantaged groups out of well-served neighbourhoods, contributing to socio-spatial segregation and unequal access to climate-friendly housing and infrastructure.

Policy and governance shortcomings were consistently identified across all countries. In Germany, modernisation levies were described as increasing rents and disproportionately benefiting homeowners, while tenancy laws were considered to contribute to financial insecurity and, in some cases, discriminatory outcomes. Frequent legal and funding changes create uncertainty and undermine equitable planning. In Ireland, the absence of a minimum Building Energy Rating (BER) requirement for rental properties leaves tenants exposed to inefficient housing conditions, and insufficient investment in public and active transport exacerbates spatial inequalities. In Portugal, participants stressed underfunding, weak implementation, mismanagement and lack of coordination across housing policies, alongside restrictive eligibility criteria that exclude vulnerable groups from support programmes. Across all contexts, a perceived mismatch between policy frameworks and lived socio-economic realities was emphasised, particularly regarding affordability, retrofitting obligations and access to social housing.

Social vulnerability and discrimination were particularly explicit in Germany and Portugal. German labs identified financial insecurity and discriminatory practices—including those linked to religion or social background—as restricting access to well-served neighbourhoods, with unequal housing access potentially contributing to social polarisation and the strengthening of exclusionary political rhetoric. In Portugal, private and family instability, including addiction-related challenges, was described as affecting housing security. Migrants were reported to face additional barriers due to limited eligibility for support schemes.

Technological and informational barriers were also discussed. In Germany, misinformation regarding infrastructural reforms (such as heating regulations) was described as limiting public understanding and adoption. Smart home technologies were perceived as primarily designed for efficiency objectives rather than accessibility, posing challenges for individuals with low digital literacy. Portuguese participants additionally raised concerns about the potential long-term impact of artificial intelligence on employment structures, noting that shifts in labour markets could undermine income stability and, consequently, housing affordability.

Environmental implications were closely linked to these structural factors. Inefficient housing stock across countries hinders progress towards sustainability targets, increases emissions and elevates household costs. In Ireland, potential fines for failing to meet climate targets were seen as ultimately affecting the public, while transport-related tailpipe emissions generate negative health impacts.

Divergences emerged in the specific structural and political dimensions emphasised in each country. In Germany, discussions focused strongly on social and spatial discrimination, including references to religious discrimination, and on the role of modernisation levies and tenancy laws in increasing insecurity. Political and legal uncertainty, frequent regulatory changes and public misinformation around heating reforms were identified as exacerbating inequities. Furthermore, a specific link between housing inequality and right-wing populist rhetoric was asserted. Smart home technologies were also described as insufficiently accessible for low digital literacy groups. In Ireland, the housing–transport nexus was particularly pronounced, emphasising the car dependency as structural outcome of housing crisis. The insufficient investment in public transport and the contribution of housing costs to labour mobility constraints and emigration were central concerns. In Portugal, participants stressed restricted access to social housing for migrants and other vulnerable groups due to eligibility criteria, emphasising individuals with addictions and mental health challenges. Other aspects mentioned were private and family instability affecting housing security, and emerging risks linked to labour market transformations, including the potential impact of artificial intelligence on income stability and affordability.

4.2.1. Germany

Sectoral challenges prioritised: Financial insecurity, discrimination, and unjust distribution

Key problems:

- Unequal access to housing and homeownership:
 - Home ownership is closely tied to financial security, age, creditworthiness, job stability, and sometimes discrimination (including religion).
 - Tenants and lower-income households face energy and transport poverty in inefficient buildings.
 - Social and spatial discrimination:
 - Financial insecurity and discrimination push certain groups out of well-serviced neighbourhoods, increasing social segregation.
 - Access to climate-friendly housing is uneven, and DGs often live in energy-inefficient buildings.
 - Right-wing rhetoric is fuelled by unequal access to housing and resources.
 - Limited autonomy to adopt renewable energy or participate in sustainable energy programmes.
- Policy gaps:
 - Modernisation levies and funding programmes benefit homeowners, offering little relief to tenants.

- Tenancy laws contribute to financial insecurity and discrimination.
- Political and legal uncertainty (frequent changes to laws/funding) exacerbates inequities.
- Technological and informational barriers:
 - Public misinformation about infrastructural updates (e.g., heating law) limits adoption.
 - Smart home technologies are designed mainly for energy efficiency, not accessibility or usability for low-tech literacy groups.

4.2.2. Ireland

Sectoral challenges prioritised: House ownership, transport needs

Key problems:

- House ownership and limited access to housing
 - Housing policies for retrofitting are only accessible to homeowners.
 - Inefficient housing is expensive and reduces disposable income.
 - High housing costs affect labour mobility and regional balance.
 - Housing costs are contributing to worker emigration.
- Transport needs linked to limited housing alternatives
 - Spatial displacement and increased commuting demand: High housing costs and limited alternatives require people to live further from employment centres, increasing commuting distances.
 - Insufficient and uneven transport provision
 - Limited public transport options, particularly in rural areas, reinforce car dependency.
 - Transport infrastructure and investment are unevenly distributed and insufficient
 - Service timings are inadequate (e.g. after hours).
 - Economic and social impacts of car dependency
 - Long-distance commuting forces car purchase and use
 - Electric vehicles remain expensive and require subsidies.
 - Seniors relying on taxis face high costs and reduced disposable income.
 - Congestion generates public costs.
 - Time and accessibility constraints
 - Longer travel times reduce free time and affect quality of life.
 - Safety concerns limit uptake of alternatives such as cycling.
- Inadequate sectoral policies
 - Policies are only available to homeowners.
 - No minimum Building Energy Ratio (BER) rating for rental properties.
 - Insufficient investment in public and active transport.
 - Uneven transport infrastructure development across regions —concentrated in Dublin.
- Environmental challenges related to housing and transport
 - Housing stock remains energy inefficient and makes it difficult to reach sustainability targets.
 - Fines for not meeting targets would be borne by the public.
 - Tailpipe emissions are significant in the transport sector producing negative health impacts.

4.2.3. Portugal

Sectoral challenges prioritised: lower income level; access to affordable housing; sustainability-related costs; risks of Artificial Intelligence (AI) impacting sectoral structures, social dynamics, and employability.

Key problems:

- Financial difficulties, including challenges related to housing costs and overall living expenses.
 - Lack of employability and its impact on housing stability.
 - Financial pressures faced by families with children.
 - Sustainability-related costs, including expenses linked to retrofitting and energy efficiency requirements.
- Restricted access to affordable housing.
 - Limited accessibility of social housing for lower-income families, especially those with children. Other vulnerable groups affected by economic hardship include people with addictions, mental health challenges, or those who have lost their jobs. For people with a migrant background, existing support programmes and policies may be limited or entirely unavailable.
 - Limited access to housing support programmes and policies for migrants due to restrictive eligibility criteria.
 - Insufficient availability of social housing for DGs, including people facing addiction, mental health issues, or similar challenges.
- Travel and transport burdens resulting from housing affordability constraints, whereby high ownership or rental costs push households to peripheral areas, leading to longer commuting distances, insufficient reimbursement mechanisms, and limited public transport services adapted to people with special needs and older adults.
- Lack of home ownership and reliance on shared or insecure housing arrangements.
- Private and family instability affecting housing security (e.g. addiction-related problems).
- Perceived mismatch between existing policies and the real needs of the general population (e.g. housing affordability, retrofitting, etc.).
 - Policies are described as underfunded and poorly implemented, with limited availability of affordable and social housing and eligibility criteria that exclude vulnerable groups. Overall, there is a lack of sufficiently targeted, coordinated, and well-resourced policies capable of addressing structural barriers.
 - Mismanagement leading to inadequate allocation of resources and underfunding of the population's actual needs.
- Risk of AI impacting sectoral structures, social dynamics, and employability: concerns that the further expansion of AI could alter the employment landscape, potentially affecting income stability and, consequently, housing affordability.

4.3. Transport sector

Across the countries examined in the transport sector — Hungary, Italy and Spain — the co-creation process identified a foundational structural tension between public and private transport models. This dichotomy underpins many of the inequalities observed: a car-centric development pathway has progressively displaced investment and strategic prioritisation of public transport. Two interrelated divides emerge. First, the imbalance between public and private transport investment reinforces car dependency as the dominant mobility paradigm. Second, this imbalance generates differentiated access to mobility, with those unable to rely on private vehicles disproportionately affected.

Limitations in public transport were consistently identified as the primary source of accessibility problems. These limitations extend beyond mobility in a narrow sense: public transport was framed — particularly in Italy — as a mechanism for reducing territorial inequality and enabling access to employment, education, healthcare and other essential services. Weak or underdeveloped networks therefore constrain not only movement but also social and economic opportunity. Across countries, rural and peripheral areas were described as lacking adequate public transport coverage, with insufficient infrastructure density and services misaligned with commuting patterns. These gaps reinforce territorial inequality between urban, suburban and rural areas and increase reliance on private vehicles as a compensatory strategy.

Several cross-cutting governance and infrastructural shortcomings contribute to these public transport deficiencies. Participants highlighted limited long-term planning, insufficient monitoring of service provision, uneven quality standards and a perceived under-prioritisation of public transport relative to road and automotive investments. In Italy, bureaucratic fragmentation, political instability and limited local programming capacity were identified as obstacles to coherent transition planning, alongside concerns regarding transparency in the allocation of transition funds and budget reallocations affecting resources intended to address transport poverty. In Spain, limited institutional responsiveness was emphasised: policies were described as well-intended but insufficiently targeted to reach disadvantaged groups, who are rarely incorporated into technological innovation processes. In Hungary, local governments were described as having restricted influence over industrial and infrastructural decisions, limiting their capacity to rebalance investment priorities.

Accessibility and social inclusion concerns intersect strongly with these public transport gaps. Elderly people, persons with disabilities and rural residents face structural barriers where infrastructure is not adapted to accessibility needs. Italian labs placed particular emphasis on safety and discrimination within mobility spaces, including harassment affecting women, LGBTQIA+ individuals and migrants. Digitalisation introduces additional exclusionary dynamics: limited digital literacy among users and insufficient operator capacity to design accessible digital services were reported in Italy and Spain. Hungarian discussions added a broader social dimension, noting that public transport could function as a communal space facilitating access to services and social interaction, yet current system design does not enable this role.

Within this context of public transport underdevelopment, the second structural component concerns the reinforcement of a car-centric system. Participants across countries stressed that policy frameworks and financial incentives tend to favour private vehicle ownership, particularly through electric vehicle (EV) strategies. EV subsidies were widely perceived as regressive in distributional terms: high upfront costs, advance-payment models and limited accessible financing mechanisms (such as social leasing or second-hand markets) mean that benefits accrue primarily to higher-income households and corporations. Consequently, low-income households remain dependent on older combustion-engine vehicles, deepening socio-economic divides. Electrification strategies were therefore questioned for reducing tailpipe emissions without addressing the structural causes of car dependency or territorial exclusion.

Environmental and industrial contradictions further complicate this car-centred pathway. In Hungary, battery factory production associated with the expansion of the EV industry was linked to local air, soil and water pollution and high water use, raising tensions between environmental objectives, community health and workers' interests. Weak regulatory enforcement and monitoring were also reported. In Italy and Spain, infrastructural constraints — including insufficient EV charging networks and limited vehicle range — hinder widespread adoption, while the continued dominance of fossil fuels limits progress towards climate neutrality. Across contexts, participants stressed that without substantial reinvestment in public and active transport modes (cycling, walking, micromobility), electrification alone cannot deliver an equitable or sustainable transition.

Divergences in emphasis nonetheless emerged. Hungary placed strong focus on industrial policy favouring the automotive sector, environmental harms linked to battery production and tensions between labour — worker's interests— and environmental objectives. Italy highlighted governance fragmentation, limited administrative capacity, transparency deficits in fund allocation, and pronounced safety and discrimination issues in mobility environments, alongside the strategic role of public transport in equalising access to services. Spain framed transport poverty as a structural barrier to opportunity, emphasising rural vulnerability, inaccessible physical infrastructure for persons with reduced mobility and the gap between policy intention and effective outreach to disadvantaged groups.

4.3.1. Hungary

Sectoral challenges prioritised:

- Green energy sources, such as battery factories, cause significant environmental pollution.
- The focus on (electric) cars perpetuates car dependency, increases income inequality, and hinders the development of community and alternative forms of transport.
- Public transport is not given sufficient priority, and there is a lack of organisational, infrastructural, and educational developments. Limited promotion of alternatives that promote active transport modes (cycling, walking) and micromobility.

Key problems

- Policy gaps
 - National policy prioritises the interests of the automotive industry over sustainability considerations.
 - Local governments have limited or no influence over industrial production, and their ability to enforce environmental regulations is restricted.
 - State-subsidised tax incentives for electric cars disproportionately benefit higher-income households and large corporations.
 - Transport infrastructure investments mainly support electric vehicles while neglecting the development of public transport.
 - Lack of regulatory enforcement and insufficient monitoring.
- Environmental effects
 - Due to the dominant share of fossil fuels and limited penetration of electric vehicles, climate neutrality and environmental protection are not ensured.
 - Local air, soil and water pollution, high water use.
- Limitations associated to electric vehicles
 - Reinforce the unsustainable nature of urban transport: reduce emissions but do not eliminate their causes.
 - Increase social inequalities and segregation, deepening the urban-rural divide.
- Social conflicts associated with the twin transition: tension between environmental and workers' interests, health and environmental justice.
- Accessibility problem: public transport is not only a means of travel but also a communal space where people spend time and could access various services; however, currently is poorly suited for this purpose.

4.3.2. Italy

Sectoral challenges prioritised: Accessibility — ensuring equitable access to mobility, green technologies and services. It implies access to transport services, reducing territorial inequalities between urban, suburban and rural areas, increasing the opportunities of the population.

Key problems

- Territorial inequality in mobility access:
 - Peripheral and rural areas lack adequate public transport networks, creating the need to rely on private transport.
 - Public transport schedules often fail to reflect commuting patterns.
 - Insufficient long-term planning and monitoring of service providers.
- Policy gaps
 - Bureaucracy and political instability that obstruct continuity in transition policies.
 - Limited programming capacity of the local administrations to effectively respond to territorial needs.
 - Budget reallocations threaten funds intended for energy and transport poverty.
 - Limited transparency in the allocation of transition funds.
- Infrastructural weaknesses
 - Limited density and efficiency of transport networks.
 - Inadequate EV charging infrastructure.
 - Perceived low quality of public transport services.
- Affordability barriers to sustainable mobility:
 - High cost of electric vehicles.
 - Limited administrative capacity to use EU funds effectively.
 - Insufficient development of social leasing or second-hand EV markets.
- Digital exclusion and service design gaps:
 - Limited digital literacy among users.
 - Transport operators lack capacity to design accessible digital services.
 - Uneven infrastructure distribution exacerbates exclusion.
- Safety and social exclusion:
 - Women, LGBTQIA+ people and migrants face bias, harassment and exclusion in mobility spaces.
 - Disability and low income intensify access barriers.
 - Lack of awareness and prevention of violence in public transport environments.

4.3.3. Spain

Sectoral challenges prioritised:

- Transport should enable access to opportunities for DGs.
- Rural public transport networks are insufficient.
- Electric vehicle subsidies are not relevant for DGs.

Key problems

- Transport poverty among vulnerable groups:
 - Elderly people and persons with disabilities face structural mobility barriers.
 - Rural residence significantly increases transport vulnerability.

- Public transport infrastructure is not adapted to accessibility needs.
- Affordability gap in clean mobility:
 - Low-income households rely on old combustion-engine vehicles.
 - Electric vehicle subsidies are inaccessible due to high upfront costs.
 - Advance-payment schemes exclude vulnerable households.
- Underdeveloped electric infrastructure:
 - Insufficient public charging stations.
 - Limited travel range of electric vehicles.
- Urban environment as barrier:
 - Physical infrastructure remains inaccessible for people with reduced mobility.
 - Vulnerability remains invisible in policy design.
- Low institutional responsiveness:
 - Policies are well-intended but do not reach those most in need.
 - DGs are rarely considered in technological innovation processes.

5. Results of the co-creation process

This chapter presents the outcome of the co-creation process in the six FITTER-EU countries: each lab delivered a policy portfolio comprising two types of policy measures: (i) adjustments to existing policies and (ii) new policy proposals, covering two sectors: energy and housing—in Ireland, Germany, and Portugal—energy and transport—in Italy, Spain, and Hungary. The proposed policy measures were framed in factsheets based on a common template (see the template in Chapter 2.3.2, Table 4 and Table 5), covering several dimensions, such as:

- The sectoral challenges that the policy aims to address
- A description of the proposal, specifying elements to be adjusted in existing policies versus new proposals
- The target population
- The stakeholders to be involved in policy design, implementation and evaluation
- Consideration of the systemic effects of the policy
- Potential risks and barriers, as well as drivers or enablers
- The time horizon, feasibility and required resources

The chapter is structured as follows: it first presents an overview analysis of all co-created policy factsheets in quantitative terms, followed by the overall description of the factsheets' content in each analysed sector and policy type (adjustment versus new proposal).

Annex III: The co-created policy portfolios presents the detailed policy portfolios for each country, consisting of the complete set of policy proposals co-created within each lab. The policy factsheets included in this annex are summarised versions of the template used in the labs (presented in Chapter 2). Each summarised factsheet presents the sectoral challenges the policy aims to address, as well as the concrete proposal. In addition, each factsheet includes the results of the analysis of the policies' transformative potential (see the agenda session of the third lab, 'Analysing Transformative Potential of Policies', in Section 2.3.3). In this exercise, each policy proposal was assigned a score across two dimensions—impact and feasibility. Based on these criteria, a policy typology was established (see Table 7: Typology of policy proposals in Chapter 2).

It is important to note that this latter exercise presented several limitations in the co-creation process, mainly due to time constraints, as most partners prioritised the development of the policy factsheets. As described in Section 2.5 on methodological deviations, most partners simplified the proposed analytical steps—with the exception of Germany and Portugal—and many concluded the analysis in an abbreviated manner. Moreover, in one of the countries—Ireland—it was not possible to implement this session during the labs; only general reflections were provided. As a consequence, some policy factsheets do not include these elements. In general, methodological deviations also account for the different level of detail featuring the policy factsheets descriptions.

Due to these limitations, the information presented in this chapter emphasises the policy proposals as such, but doesn't cover the results of the "transformativeness" analysis and the resulting policy categories. The consortium is currently assessing how to further develop and deepen this analysis potentially within WP6, which is dedicated to validation.

5.1. Overall analysis of the co-created policies

The six policy portfolios co-created within the FITTER-EU labs comprise a total of 82 policy proposals. Figure 10 illustrates their distribution across countries, revealing a significant concentration in two cases: Italy and Spain account for approximately half of all proposals, representing 27% (22 policies) and 22% (18 policies), respectively. Hungary and Germany each account for 15% of the total (12 policies per country), followed by Ireland (12%, 10 policies) and Portugal (9%, 8 policies).

One possible explanation for the higher number of proposals in Italy and Spain is that the FITTER-EU national partners adapted the methodology (see section 3.2 reporting methodological deviations) and undertook preparatory work prior to the third lab. This included drafting an initial version of the policy portfolios by compiling ideas generated during the first diagnostic session. Such preparatory work may have facilitated the development and consolidation of a larger number of proposals during the co-creation process.

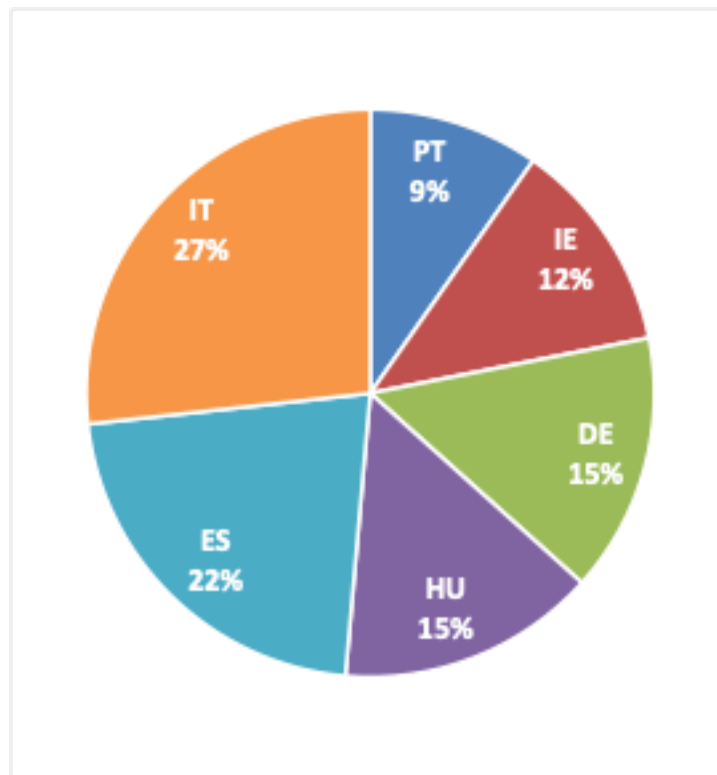


Figure 10: Distribution of policy proposals within the countries

Clear differences also emerge when considering the type of policies co-created, distinguishing between adjustments to existing policies and new policy proposals.

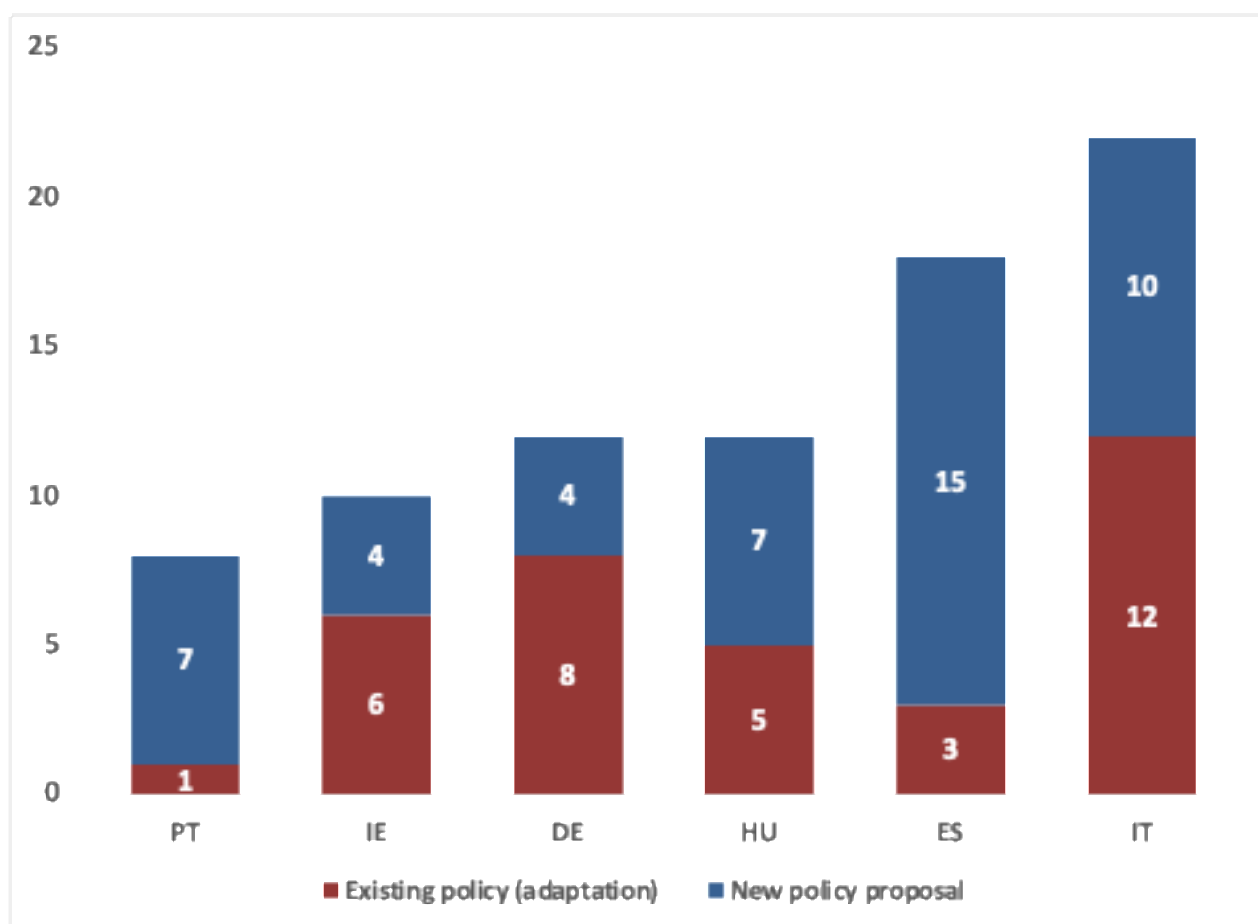


Figure 11: Distribution of Policy Proposals by Country, by Type (Adjustments to Existing Policies and New Proposals)

Germany and Ireland display a stronger focus on adjustments to existing policies. As detailed in Section 3.2, this orientation is closely linked to their respective co-creation methodologies. In Germany, the process was initiated based on proposals derived from existing literature, which framed the discussion around refining and adapting established policy approaches. In Ireland, the co-creation process prioritised the development and refinement of existing policies, with comparatively less time allocated to the design of entirely new proposals. Moreover, new policies in Ireland were developed with a cross-sectoral perspective, addressing both sectors simultaneously rather than generating sector-specific proposals in energy or housing.

Italy presents a slightly higher proposals adjusting existing policies than new policy proposals. In contrast, Hungary shows a greater emphasis on new policy proposals, which represent 58% of its portfolio (7 out of 12 policies). This trend is even more pronounced in Portugal and Spain, where new policy proposals constitute the majority of the portfolios: 88% in Portugal (7 out of 8 policies) and 83% in Spain (15 out of 18 policies).

Types of policy proposals and sectoral focus

The following table presents the distribution of policy proposals according to the types of instruments included, distinguishing between adaptations of existing policies and newly proposed measures. It is worth noting that each policy could be classified to belonging to one or more of the four categories:

- Regulation: laws, standards, bans, requirements

- Economic incentives: subsidies, taxes, grants, penalties
- Information or education: campaigns, training, awareness
- Public services: programmes, infrastructure investments

To enhance interpretability and reduce fragmentation caused by numerous specific instrument combinations categorising each policy, the original categories were analytically regrouped into seven conceptually coherent clusters. The recategorisation follows two principles: (i) distinguishing single-instrument interventions from multi-instrument mixes, and (ii) differentiating incremental from structurally integrated policy designs.

Type of policy proposal		Adaptation of existing policy proposals	New policy proposals	Total
Single-instrument interventions	(1) Economic incentives (ECON)	5	1	6
	(2) Regulation (REG)	6	0	6
	(3) Information, educational campaigns (INFO)	0	4	4
	(4) Public services (SERV)	0	6	6
Structurally integrated (multi-instrument) policy interventions	(5) Economic-regulatory combination ¹	13	7	20
	(6) Combinations including an informational component ²	6	9	15
	(7) Comprehensive systemic package ³	5	20	25
Total		34	47	81

Figure 12: Types of policy proposals included in the FITTER-EU national portfolios

¹ Policy proposals combining economic incentives and regulatory instruments (excluding informational components). Categories included: (i) ECON/REG, (ii) REG/SERV, (i) ECON/SERV, (iv) ECON/REG/SERV

² All combined policies including an informational/educational component. Categories included: (i) ECON/INFO, (ii) INFO/REG, (iii) INFO/SERV, (iii) INFO/REG/SERV, and (i) ECON/INFO/SERV

³ Policy proposals including the 4 types of policy instruments: ECON/REG/INFO/SERV.

The table reveals a clear structural distinction between policy adaptation and new policy proposals.

On the one hand, policy adaptations tend to focus on regulatory and economic instruments. This applies both to single-instrument interventions (ECON, REG) and to traditional economic–regulatory combinations—see row (5) of the table, where 13 of 20 proposals are adaptations of existing policies—. Notably, pure regulatory proposals—see row (2) of the table— appear exclusively as adaptations. Probably the reason for this is the limited legal expertise of the lab participants and time limitations in the co-creation process to propose a detailed regulatory adjustment in the sectoral policies. The same pattern emerges for proposals based on pure economic incentives: 5 out of 6 correspond to adaptations of existing policies.

On the other hand, new policy proposals are strongly associated with service provision and informational instruments—see rows (3) and (4) of the table—. Most significantly, the comprehensive four-instrument package is overwhelmingly associated with new proposals (20 out of 25 cases). This indicates that when participants envision structural/systemic change, they do so through integrated policy mixes combining

economic incentives, regulation, information, and services. A possible explanation for this could be the adoption of the systemic approach proposed by the methodology, which encourages the development of more integrated policy proposals with multiple elements addressing the different dimensions underlying the identified sectoral challenges.

Overall, policy proposals co-created on the six FITTER-EU labs suggest the coexistence of two distinct logics: (i) an incremental logic, centred on regulatory and economic adjustment within established policy architectures, (ii) a transformative logic, characterised by multi-instrument integration and the expansion of state capacity through service provision and informational components.

Transformative thinking present in the co-creation of new policy proposals inside the FITTER-labs manifested as instrument diversification and policy mix complexity.

The following table presents the sectoral distribution of policies proposed per country. Results are disaggregated per country because they depend on the co-creation methodologies implemented in each context — see section 3.2 on deviations in the methodology.

Sectors	DE	ES	HU	IE	IT	PT	Total
Energy	5	8	1	2	12	4	31
Housing	7	0	0	1	0	2	10
Transport	0	10	1	0	9	0	20
Energy & Housing	0	0	1	6	1	2	10
Transport & Energy	0	0	1	0	0	0	1
Energy & Housing & Transport	0	0	0	1	0	0	1
Others	0	0	8	0	0	0	8
Total	12	18	12	10	22	8	81

Table 10: Sectoral distribution of policies per country

Half of the Labs (Germany, Spain, and Italy) co-created policies focusing exclusively on the sectors they were addressing.

The strong interdependence between Housing and Energy policies was evident in two of the countries addressing these sectors. In Ireland, as shown in the Annex III: The co-created policy portfolios, proposals to adjust existing policies were sector-specific, whereas proposals for new policies were cross-cutting, addressing both the Housing and Energy sectors simultaneously. Portugal followed a similar approach but, for both types of proposals — adjustments and new co-created proposals were either sector-specific or cross-cutting (Housing–Energy), depending on the case.

Finally, in Hungary, as specified in the Methodological Deviations section 2.5, no sector-specific policies were co-created. Instead, the focus was entirely on cross-cutting policies. Of the twelve co-created policies, four belonged to one or more of the three sectors addressed in the co-creation process. The remaining

eight pertained to other sectors, including Agriculture (1), Education (1), Digital Inclusion or Digitalisation of Public Administration (2), and the general Green Transition (4).

5.2. Crosscutting and Lab's specific emerging approaches to just twin transitions

Across the six national labs, the co-created policy portfolios reveal a similar understanding of the twin green and digital transition as a deeply distributive and institutional transformation, rather than a primarily technological one. Although the countries differ significantly in political systems, welfare traditions and administrative capacities, several shared orientations emerge.

First, there is a strong and recurring concern that existing transition frameworks risk reproducing or even exacerbating social inequalities. Stakeholders observed that current subsidy schemes, market incentives and regulatory designs often disproportionately benefit higher-income households, property owners or established market actors. As a result, many proposals aim to introduce social differentiation, progressive cost allocation and targeted prioritisation of vulnerable groups.

Second, the transition is consistently reframed from a narrow climate or technological perspective toward a broader governance and institutional lens. Questions of electricity market design (Italy, Germany), landlord-tenant regulation (Germany, Ireland), local benefit-sharing (Hungary), participatory governance (Hungary, Ireland, Spain), administrative simplification (Italy, Ireland, Portugal), and the integration of energy poverty into public health frameworks (Spain) are treated as central to achieving both environmental effectiveness and social justice.

Third, many proposals reflect an approach to policy design based on proactive inclusion and structural accessibility. Instead of assuming that households or communities will navigate complex administrative systems, several portfolios call for simplified procedures, local support structures and targeted outreach (Italy, Ireland, Spain, Portugal), automatic enrolment mechanisms (Spain), direct supplier payments to remove pre-financing barriers (Portugal), or for rebalancing structural power asymmetries within markets (Germany, Hungary, Spain).

At the same time, broad features seem to emerge for the main approaches followed by each lab. In some countries, the designed policy measures concentrate on redistribution within established liberalised market systems (Germany). In others, the Lab seemed to prioritize expanding accessibility and eligibility within existing schemes (Ireland, Portugal). The Italian Lab centred attention on correcting structural inefficiencies in market design and implementation capacity, while in Hungary policy proposals extended toward broader institutional and constitutional redesign embedding sustainability principles across governance structures. Finally, the Spanish lab combined redistributive tariff design, fiscal stabilisation instruments and rights-based framing of energy access.

In what follows, specific considerations per sector are provided, distinguishing between adjustments to existing policies and entirely new proposals.

5.2.1. Sectoral overview: energy

Three shared thematic priorities can be identified across all participating countries:

- Correcting economic incentives for energy transition creating regressive effects;
- Expanding citizen and community participation;
- Addressing price volatility and fairness in electricity markets.

However, the depth of intervention varies. Some Labs focused on recalibrating existing frameworks (Ireland, Germany, Portugal), while others move toward structural market redesign or governance reform (Italy, Hungary, Spain).

Adjustment to existing energy policies

A first cluster of adjustments addresses energy poverty and targeted support. In multiple countries (Ireland, Germany, Spain), proposals seek to broaden eligibility criteria beyond narrow income thresholds by incorporating housing efficiency, energy expenditure burdens or specific vulnerability indicators. In Spain, particular emphasis is placed on automatic enrolment in social tariffs and reducing bureaucratic barriers to accessing the energy poverty strategy. In Portugal, adjustments to photovoltaic subsidy schemes aim to eliminate pre-financing requirements through upfront payments or direct transfers —by the State— to certified suppliers. Across cases, the objective is to remove liquidity and administrative barriers that disproportionately exclude low-income households. In Ireland, particular attention is given to including groups historically excluded from schemes due to housing form or administrative barriers. In Germany, redistributive differentiation is introduced through socially graded mechanisms within taxation, i.e. reducing the electricity tax for end consumers of green electricity or for households affected by a negative economic situation —graded by standardised socioeconomic indicators—.

A second group of adjustments concerns energy taxation and cost allocation. In Germany, proposals differentiate taxation more clearly between fossil and renewable energy, aiming to correct regressive effects. In Ireland and Spain, reforms emphasise refining allocation mechanisms and introducing more progressive tariff structures. Across countries, the common logic is to prevent low-income households from bearing disproportionate transition costs and to align price signals with distributive fairness.

Administrative simplification constitutes a third recurring theme. In Italy and Ireland, complex procedures are identified as key barriers to accessing retrofit and renewable schemes. Similar concerns arise in Spain and Portugal, where beneficiaries face difficulties navigating application processes or advancing payments. Proposed adjustments therefore include local advisory structures, clearer guidance, standardised procedures, enhanced coordination between levels of government and “automatic” cross-referencing of social data, thus reducing the bureaucracy to access the social bonus when a person is identified as belonging to a vulnerable group.

A fourth cluster addresses price stability and consumer protection. In Italy and Germany, particular attention is paid to the volatility of electricity prices linked to fossil fuel markets. While Germany’s approach tends to focus on redistributive correction within existing structures, Italy places stronger emphasis on addressing structural price formation mechanisms. In Spain, proposals include the creation of a public stabilisation fund financed through windfall profit taxation to automatically offset bills during price spikes. In Hungary, local compensation mechanisms for renewable installations are introduced to ensure territorial fairness and social acceptance.

New energy policy proposals

In Italy, structural electricity market reform is proposed to reduce the transmission of fossil fuel price volatility into electricity pricing. In Germany, the proposals aim to strengthen public oversight and accountability in the governance of energy infrastructure through increased public ownership. They envisage the federal government acquiring a significant stake in transmission grid operators via a state-backed investment vehicle, with revenues from grid operations reinvested in network expansion.

In Hungary, new governance-oriented instruments introduce institutional safeguards for sustainability, including enhanced oversight and structured participatory mechanisms. Ireland complements this governance focus with strengthened community energy frameworks and local participatory structures. Spain adds progressive and automatic energy tariffs differentiated by income and household composition, as well as the institutional recognition of energy poverty within public health systems. Portugal proposes territorially anchored renewable neighbourhood programmes and integrated energy-agriculture initiatives such as agrivoltaics.

Another prominent theme across countries (Germany, Italy, Ireland, Spain, Portugal) is the strengthening of community and small-scale energy production, through support from the public institutions —capacity building, training and others such as support desks—, priority grid access, simplified administrative procedures and financial instruments that reduce upfront barriers. In Spain and Portugal, particular attention is paid to ensuring that such community models explicitly include vulnerable households.

Whereas adjustments often aim to improve inclusion within existing markets, new proposals tend to reconfigure governance, embed participatory mechanisms, deploy redistributive fiscal instruments and question the structural distribution of power within energy systems. Hungary stands out for linking energy governance with broader institutional and intergenerational accountability reforms, while Spain advances automaticity and progressive tariff design as structural correctives.

5.2.2. Sectoral overview: housing

Across countries, there is strong concern that energy-efficiency and retrofit policies may unintentionally exacerbate housing inequality, particularly in rental markets or among liquidity-constrained households.

Two shared priorities emerge:

- Redistributing renovation costs more equitably;
- Targeting low-income and underserved households more effectively.

However, the intensity and scope of intervention vary. Germany places particularly strong emphasis on landlord-tenant regulation, Ireland on targeted eligibility to access the incentives for house retrofitting, Hungary integrates housing renovation into broader social infrastructure concerns, while Spain and Portugal emphasise comprehensive support for vulnerable households and neighbourhood-level inclusion.

Adjustments to existing housing policies

A central adjustment across several countries introduces socially differentiated subsidy schemes, with higher support rates or full funding for lower-income households. In Portugal, concerns about upfront co-financing requirements mirror similar critiques in other countries, prompting calls for eliminating advance payments and strengthening public support structures. In Germany, proposals explicitly link public support to restrictions on rent increases, seeking to prevent renovation-induced displacement. Ireland emphasises area-based prioritisation using deprivation indicators and calls for stronger enforcement of minimum energy standards in rental housing.

Reforms of landlord-tenant cost allocation mechanisms are particularly prominent in Germany, where proposals limit the share of renovation costs transferable to tenants and strengthen automatic safeguards. Ireland similarly proposes clearer minimum standards and stronger compliance mechanisms. These measures reflect a shared concern with “green gentrification,” though Germany’s proposals intervene more directly in rent regulation.

Administrative accessibility again emerges as a key theme. Retrofit schemes are criticised for favouring households with administrative literacy and financial capacity. Adjustments therefore propose simplified procedures, proactive targeting and strengthened local implementation support.

In Hungary, housing-related adjustments integrate broader infrastructural considerations, such as water network rehabilitation. Portugal similarly expands the housing discussion toward accessible public infrastructure and inclusive public spaces, linking energy efficiency with universal accessibility.

New housing policy proposals

The most transformative proposals (Ireland, Germany) involve fully funded deep retrofits for energy-poor households, eliminating upfront contributions and extending safeguards to rental properties. These measures frame housing decarbonisation as a tool for structurally reducing energy poverty rather than merely improving efficiency.

In some contexts (Germany), solidarity-based funding mechanisms are proposed to finance renovation through progressive contribution structures. In others (Hungary), housing is embedded within broader welfare and environmental governance frameworks. Portugal introduces integrated neighbourhood programmes combining training, social inclusion and infrastructure investment, aiming to reposition social housing as a transitional and enabling solution rather than a site of permanent marginalisation.

Compared to the energy sector, new housing proposals focus less on market redesign and more on deepening redistribution, strengthening social protection and embedding housing policy within broader social inclusion strategies.

5.2.3. Sectoral overview: transport

Transport policies display a clear and increasingly convergent orientation around transport poverty, territorial inequality, structural car dependency and the distributive effects of mobility incentives. Across countries, mobility is framed not only as a decarbonisation challenge but as a condition for access to work, care, education and public services.

A shared concern emerges that existing incentive structures — particularly subsidies for private electric vehicles — tend to privilege higher-income groups, urban cores or VAT-registered actors, while neglecting low-income households, caregivers, rural residents and shift workers. In both Italy and Spain, strong critiques are directed at electric vehicle schemes that primarily subsidise private purchases (e.g. Ecobonus; MOVES), with limited redistributive impact and weak alignment with traffic reduction objectives. This reinforces the broader cross-country shift from vehicle electrification alone toward rebalancing investment in collective and accessible public transport.

At the same time, differences in emphasis remain. Hungary prioritises territorial inclusion and rural accessibility. Italy combines social differentiation in concessions with governance reform and intermodality in freight. Spain advances a more explicit fiscal and regulatory strategy to structurally discourage private car use and finance universal public transport.

Adjustments to existing transport policies

Italy proposes expanding the transport bonus beyond narrow age-based or existing socio-economic eligibility criteria to target a wider range of population groups. These include people over 70, young people up to the age of 26, low-income individuals, caregivers (including those caring for elderly or disabled relatives), residents in peripheral areas, and workers with rigid or night shifts —currently, only upper secondary school students are included. This reflects a more nuanced understanding of changing family

structures, demographic ageing and hidden economic vulnerability. Transport access is thus reframed as a support mechanism for unpaid care and social reproduction.

Italy also introduces adjustments at the governance level of Local Public Transport, calling for increased and protected funding (including earmarking revenues from fuel excise duties), prioritised investment in peripheral and internal areas, 24-hour service expansion (modelled on metropolitan systems such as Vienna), improved accessibility standards (including revision of wheelchair weight limits), and transparency in fund allocation. These proposals respond to territorial disparities between north and south, and between urban and rural areas, aligning with concerns raised in Hungary and Portugal regarding spatial inequality.

In Spain, adjustments to the MOVES programme³ shift the focus from subsidising private electric vehicles toward strengthening zero-emission public transport fleets. Stakeholders highlight regional inequalities in fund allocation and pinpointed at the regressive effects of first-come-first-served schemes, the taxability of subsidies, and the limited accessibility of electric vehicle ownership for vulnerable groups, including persons with disabilities. The proposed reorientation prioritises collective mobility and universal adaptability rather than private vehicle ownership.

Across Italy and Spain, therefore, a common corrective logic emerges; rebalancing incentives away from private car electrification toward public transport accessibility, while introducing more socially differentiated eligibility criteria.

Digitalisation and information access also constitute an adjustment domain in Italy. Proposals address fragmentation of mobility apps, lack of interoperability between electric charging providers, insufficient data governance and digital exclusion among elderly populations. Suggested measures include a unified public mobility platform (single login), mandatory interoperability standards, digital literacy support points and periodic social impact assessments. Transparency reforms for Limited Traffic Zones (ZTLs) and urban traffic rules — through unified portals and digital signage — similarly aim to reduce unequal informational burdens.

Finally, Italy proposes tightening freight regulation in urban areas, designating specific logistics zones and limiting tourist vehicle circulation in historic centres, linking emission reduction with spatial justice and public space quality.

Overall, adjustments across countries share three orientations: expanding social eligibility, correcting regressive private-vehicle incentives —i.e. which benefit higher income households—, and strengthening the accessibility and governance of public transport systems.

New transport policies

In the Hungarian lab, integrated mobility planning of public transport, cycling and reduced car dependency through spatial redesign principles were prioritised. Ireland linked rural transport infrastructure to social inclusion and service accessibility.

In Italy a National Plan for Intermodal Freight Transport was proposed to promote modal shift from road to rail through intermodal hubs, infrastructure investment, incentives, and alignment with EU logistics corridors. This proposal directly addresses congestion and emissions from freight transport and introduces a transnational dimension, including protections for cross-border workers. Complementary proposals include awareness campaigns targeting policymakers and users to shift cultural norms around public transport, and measures to ensure safety in public transport for women, LGBTQIA+ people and persons with disabilities, embedding mobility within a rights and inclusion framework.

³ Subsidy program in Spain to promote sustainable mobility, offering grants for the purchase of electric/plug-in hybrid cars.

Some of the elements of the Spanish proposal for transport include:

- A Universal Mobility and Transport Programme embedding accessibility principles in education and public campaigns
- Legislative guarantees of universal access across urban and intercity transport modes;
- Progressive taxation of private vehicles (higher rates for more polluting cars or multiple-car households);
- Taxes on single-occupancy vehicle use;
- Corporate parking taxes in urban areas;
- Urban tolls for private vehicle access;
- Exclusive public transport lanes on congested roads;
- Progressive pricing of public transport monthly passes linked to income;
- A ban on new parking development in city centres.

Together, these measures use fiscal, regulatory and spatial instruments to discourage private car dependency and generate funding for accessible public transport. Compared to other countries, Spain most explicitly deploys taxation and urban access restrictions as redistributive tools to reshape mobility behaviour.

Across Italy and Spain in particular, transport policy is reframed not simply as electrification of vehicles, but as redistribution of space, incentives and fiscal burdens. Freight intermodality (Italy), progressive taxation and congestion pricing (Spain), and strengthened public transport governance (Italy) collectively illustrate a structural shift from individualised mobility solutions toward collective and publicly regulated systems.

Compared to energy and housing, transport proposals are more explicitly spatial and behavioural in orientation. They combine social eligibility reform, fiscal disincentives for private car use, intermodal infrastructure investment, digital integration and rights-based accessibility. The convergence across countries suggests that mobility is increasingly understood as a domain where decarbonisation, territorial equity and social inclusion must be addressed simultaneously through systemic redesign rather than incremental incentive correction.

List of references

- Bundesamt für Wirtschaft und Ausfuhrkontrolle, BAFA. (2025). *Energie. Bundesförderung für energieeffiziente Gebäude, Förderprogramm im Überblick*. Retrieved from https://www.bafa.de/DE/Energie/Effiziente_Gebaeude/Foerderprogramm_im_Ueberblick/foerderprogramm_im_ueberblick
- Bundesverband Erneuerbare Energien, BEE. (2025, 13.08). *BEE-Stellungnahme zum Entwurf eines Dritten Gesetzes zur Änderung des Energiesteuer- und Stromsteuergesetzes*. Retrieved from 2025-08-13_BEE-STN_zur_Aenderung_des_EnergieStG_und_StromStG.pdf
- De Vicente Lopez, J., & Matti, C. (2016). *Visual toolbox for system innovation. A resource book for practitioners to map, analyse and facilitate sustainability transitions*. Transitions Hub Series. Brussels: EIT Climate-KIC.
- European Union Institute for Security Studies, S. G. (2015). *Climate change as a 'wicked problem'*. European Union Institute for Security Studies.
- Galgóczi, B. (2023). *Towards a European socio-ecological contract*. International Journal of Labour Research, 12(1-2), 60-77.
- Joint Research Centre. (2025, February 7). *Unlocking the full potential of Behavioural Insights for policy*. Retrieved from EU Policy Lab: https://policy-lab.ec.europa.eu/news/unlocking-full-potential-behavioural-insights-policy-2025-02-07_en
- Kaiser, F., & Schaffert, A. (2025). *Reformvorschlag für eine soziale Ausgestaltung der energetischen Sanierungsförderung*. Zukunft KlimaSozial. Institut für Klimasozialpolitik.
- Lange, V., & Thesfeld, V. (2025). *Novelle des Energiewirtschaftsrechts. Referentenentwurf zur Stärkung des Verbraucherschutzes im Energiebereich, zur Änderung weiterer energierechtlicher Vorschriften sowie zur rechtsförmlichen Bereinigung des Energiewirtschaftsrechts vom 10.07.2025*. Bündnis Bürgerenergie.
- Matti, C., Martín Corvillo, J., Vivas Lalinde, I., Juan Agulló, B., Stamate, E., Avella, G., & Bauer, A. (2020). *Challenge-led system mapping. A knowledge management approach*. Transitions Hub series. Brussels: EIT Climate-KIC.
- Ortego, A., & Steitz, J. (2024). Kapitalengpässe lösen, Netzkosten reduzieren – Effekte staatlicher Beteiligungen auf den Stromübertragungsnetzausbau (S. 1–21). Dezernat Zukunft.
- TIP Resource Lab. (2022). *Pentagonal map for system analysis (individual version)*. Retrieved from TIP Resource Lab: <https://tipresourcelab.net/resource/pentagonal-map-for-system-analysis-individual-version>.

Annex I: List of organisations participating in the national labs

Below, the links to the national lab reports elaborated by partners are included:

First national lab reports

- [Germany](#)
- [Hungary](#)
- [Ireland](#)
- [Italy](#)
- [Spain](#)
- [Portugal](#)

Second national lab reports

- [Germany](#)
- [Hungary](#)
- [Ireland](#)
- [Italy](#)
- [Spain](#)
- [Portugal](#)

Annex II: List of organisations participating in the national labs

Following, the list of organisations present in each lab are presented.

Germany

- Münchner Initiative Nachhaltigkeit
- Münchner Wohnen
- Together4UkraineMünchen
- Stromsparcheck (Caritas)
- Mieterverein München
- Green City
- BayFOR
- Technical University Munich
- Umweltinstitut München e.V.
- Münchner Flüchtlingsrat
- Architects4Future
- Institut Wohnen und Umwelt
- Klima Allianz Deutschland
- Öko-Institut
- Fraunhofer ISI
- Prognos AG
- Bayernwerk

Hungary

- Igazgyöngy Alapítvány
- Civil Bázis
- Civil Kollégium Alapítvány
- Három Királyfi Három Királynő Mozgalom
- Zöld Platform
- II. Kerületi ÖnkormányzatMKKP
- VI. kerületi ÖnkormányzatMomentum Mozgalom
- Víz Koalíció
- IV. kerületi Önkormányzat (LMP)
- Vegyipari Dolgozók Szakszervezete (VDSZ)

Ireland

- Crosscare Migrant Project
- Fine Gael
- SmartD8
- Department of Climate, Energy and the Environment
- Sage Advocacy
- National Organisation of the Unemployed
- Feminist Communities for Climate Justice
- Pobal
- An Cosán
- Codema
- St Vincent de Paul

- The Irish Council for Social Housing
- Age & Opportunity
- Ecovision

Italy

- Fondazione ENI Enrico Mattei FEEM
- CGIL
- CESPII
- Ministero Lavoro e politiche sociali
- Università Trento
- Forum Terzo Settore
- Open Polis
- UGL
- Regione Toscana
- CILD: Coalizione Italiana Libertà e Diritti Civili
- Italia Solare
- Lucha y Siesta
- Comune di Roma
- Consiglio Nazionale delle Ricerche (CNR) /Fondazione IFEL
- Croce Rossa Italiana Comitato Locale

Spain

- Ana Valdivia Foundation
- Almar Foundation
- CEU San Pablo University
- Spanish Paralympic Committee
- UCM
- Leon Rugby club
- I am able
- Red Cross
- Manuel María Vicuña Foundation
- Provivienda

Portugal

- Misericórdia Altares – Mercy Charity Institution
- Misericórdia da Praia – Mercy Charity Institution
- Lar D Pedro V – social solidarity institution
- CCIAM – Chamber of Commerce and Industry of Angra do Heroísmo
- CMPV (City House – Praia da Vitória)
- CMAH (City House – Angra do Heroísmo)
- DR Obras Públicas – Regional Directorate for Public Works
- CAD – Disability Support Centre

Annex III: The co-created policy portfolios

6. German policy portfolio

6.1. Energy

6.1.1. Adjustments in existing policies

Policy code: DE_EN_AD_1 Country: Germany Sector: Energy	
Policy Title: Energy Industry Act, Energiewirtschaftsgesetz (EnWG) – Amendment 2025* <i>*This Act originally came into force in 1935 and was liberalised in 2005. Hence, it is an Act that is revised and amended throughout different policy cycles. In 2025, a new amendment was discussed in the political and public discourse. This proposal consists of additions to this discourse, hence, to add more to the draft of the amendment.</i>	
Policy Type	Regulation
Sectoral Challenge Addressed	Focus on amendment and energy sharing – citizen participation, reduction of bureaucracy
Original Policy Objective	The main objective is to make the grid-based supply of electricity, gas, and hydrogen to the population increasingly secure, affordable, consumer-oriented, efficient, environmentally friendly, and greenhouse gas-neutral, using renewable energies. Liberalization of the energy market, consumer protection, and expansion of renewables
Existing Gaps	Insufficient consideration of energy sharing to date – hence amendment and proposed changes from civil society
Proposed Adjustment	• Support: Establishment of a uniform, central internet platform for data exchange (Section 20b EnWG-E), which could significantly facilitate the practical implementation of local supply models, especially for network operators. • However, citizen energy must be explicitly named as a possible operator, see Lange, V. & Thesfeld, V. (2025). • Active participation of households and SMEs in the energy market; additional amendment of socially graded subsidies (e.g., subsidies for eG memberships). • Gender and social aspects with regard to participation in citizen energy projects (e.g., promotion of memberships, tax deductibility of membership fees, or flat-rate allowances for voluntary work on committees) • Social aspects with regard to consumers – binding social targets should be introduced that oblige energy suppliers to support low-energy households and to take a differentiated approach to electricity cut-offs. The term “protected customers” also falls under

Policy code: DE_EN_AD_1	
Country: Germany	
Sector: Energy	
	this category (including hospitals and social institutions). Duty to provide information for low-energy households, e.g. by requiring local authorities or municipalities to provide a brochure.
Transformative Category	Quick win
Transformative scoring	3,5
Feasibility scoring	3

Policy code: DE_EN_AD_2	
Country: Germany	
Sector: Energy	
Policy Title: Citizen participation laws (Bürgerbeteiligungsgesetze) Standardisation of and liability in participation laws, from federal/state level to national level	
Policy Type	Regulation (laws, standards, bans, requirements) Economic incentives (subsidies, taxes, grants, penalties)
Sectoral Challenge Addressed	Lack of citizens' (financial and democratic) participation in renewable energy plants → lack of acceptance
Original Policy Objective	Participation for citizens in renewable energy projects, e.g. wind power projects; rise in acceptance rates
Existing Gaps	To date, there is no uniform liability across federal states for plant operators to make a financial contribution to the municipalities on whose grounds the plant is located; these are only regulated in selected federal states. At the same time, adequate funding for participation in municipalities is required. Finally, participation often remains limited to local authorities, not citizens.
Proposed Adjustment	• Standardisation of legislation and of liable financial contribution • Education and information for all citizens in a municipality • Advantage: local value creation, participation and acceptance (maybe: Creation of jobs in rural areas)
Transformative Category	Quick win
Transformative scoring	4,5
Feasibility scoring	3

Policy code: DE_EN_AD_3	
Country: Germany	
Sector: Energy	
Policy Title: Electricity Tax Act	
Policy Type	Regulation (laws, standards, bans, requirements) Economic incentives (subsidies, taxes, grants, penalties)
Sectoral Challenge Addressed	Financial vulnerability: Redistribution issues, financing inequalities

Policy code: DE_EN_AD_3 Country: Germany Sector: Energy	
Original Policy Objective	Environmental factor: Higher electricity tax should lead to lower electricity consumption and a transition to renewable energy. <i>See Bundesverband für Erneuerbare Energien (2025).</i>
Existing Gaps	• Benefits for companies, but not for end consumers • No distinction between different energy sources or fossil fuels vs. renewables • Uncertainty as to the extent to which the benefits will affect end consumers
Proposed Adjustment	• Reduction of electricity tax for end consumers of green electricity, reduction according to social grading • Higher pricing for fossil fuel companies, including the gas industry • Distinction between climate-damaging and climate-friendly companies
Transformative Category	Short-term stabilizer
Transformative scoring	2,2
Feasibility scoring	3

6.1.2. New policies

Policy code: Country Sector	DE_EN_N_1 Germany Energy
Policy title: Abolishing the standard energy supply/the Default system	
Policy Type	Regulation Information/education
Sectoral Challenge Addressed	Financial vulnerability; Energy illiteracy – citizens are not sufficiently informed about their energy rights and energy supply. Green electricity and renewable energy for all – it should not remain an elite project.
Description of the Proposal	Transparency and educational work on energy sources/energy providers. Conscious decision-making for consumers ‘Similar to waste separation’ – clear educational concept on sustainable behaviour • In Germany, sometimes when tenants move to a new apartment, they have to take over the standard energy supply or electricity supply when they sign a contract. Sometimes, the provider is not the most sustainable, neither the cheapest one. To change this and allow tenants to decide upon their own energy consumption and

Policy code:	DE_EN_N_1
Country	Germany
Sector	Energy
Policy title: Abolishing the standard energy supply/the Default system	
	provider, owners of apartments could be taken into account to provide the right information. Or cities/municipalities.
Transformative Category	Transformative bet/quick win
Transformative scoring	3,9
Feasibility scoring	2,5

Policy code:	DE_EN_N_2
Country	Germany
Sector	Energy
Policy title: State participation in the transmission grid	
Policy Type	Regulation (laws, standards, bans, requirements) Public services (programmes, infrastructure investment)
Sectoral Challenge Addressed	• Lack of democratic control and participation • Structurally rising electricity prices/grid fees due to grid expansion and profit orientation • Risk of power outages: Grid stabilization through financing of expansion and thus capacity for renewable energy, simplified coordination in planning and expansion
Description of the Proposal	• The federal government pools equity capital via KfW (German state-owned investment and development bank), or a new state-owned energy infrastructure company (EIG) and acquires a 50% stake (financed by loans) in transmission grid operators. • Income from operations can thus be reinvested in grid expansion.
Transformative Category	Not scored or classified by participants
Transformative scoring	
Feasibility scoring	

6.2. Housing

6.2.1. Adjustments in existing policies

Policy code	DE_H_AD_2
Country	Germany
Sector	Housing
Policy Title: Federal funding for energy-efficient buildings (BEG)	
Policy Type	Regulation (laws, standards, bans, requirements) Economic incentives (subsidies, taxes, grants, penalties)
Sectoral Challenge Addressed	Financial vulnerability: Ideally a policy change would tackle an unequal distribution of resources and financial challenges. The premise is that

Policy code	DE_H_AD_2
Country	Germany
Sector	Housing
	this can be addressed through the restructuring of the BEG, particularly concerning the sub-programs Individual Measures (BEG EM) and Residential Buildings (BEG WG).
Original Policy Objective	The Federal Funding for Efficient Buildings, abbreviated as BEG, consolidates prior funding initiatives aimed at enhancing energy efficiency and promoting renewable energy within the building sector. It supports various initiatives, including the implementation of new heating systems, the optimization of existing heating systems, improvements to the building envelope, and the adoption of advanced system technologies. See BAFA (2025).
Existing Gaps	It concentrates on investors, particularly homeowners. Tax deductibility advantages households within the upper income bracket, and there is also a lack of transparency concerning this matter.
Proposed Adjustment	• Segmentation into smaller, targeted funding programs for specific groups • Socially stratified funding for comprehensive renovation, partial renovation, and heating system replacement; refers to the demands from Zukunft Klimasozial (An institute working on social-just climate solutions), see Schaffert, A. & Kaiser, F. 2025. • For households with an equivalised annual net income below €30,000 (which corresponds to roughly 60 percent of the total population), the subsidy amount is augmented by 10 percent. If the equivalised annual income falls below €15,000 (accounting for approximately 25 percent of households), the increase is 30 percent. • Concentrate on equivalence-weighted income to ensure that both pensioners and families are automatically prioritised. • Commercial landlords are granted enhanced subsidies when apartments are designated as social housing or transferred to non-profit housing. This measure can contribute to mitigating rent increases. • Low- and middle-income households, along with private landlords who charge rents below the local rent index [or the local comparative rent], receive a WPB subsidy. ○ Prioritisation of support for landlords who remain below the average • Local implementation, e.g. in Munich: Local support funding is called KFG and has special clauses for worst performing, material life cycle analysis, neighbourhood redevelopment.
Transformative Category	Transformative bet
Transformative scoring	3,75
Feasibility scoring	3,5

Policy code	DE_H_AD_3
Country	Germany
Sector	Housing
Policy Title: German Civil Code – Modernisation levy (Modernisierungsumlage BGB, §559)	
Policy Type	Regulation (laws, standards, bans, requirements)
Sectoral Challenge Addressed	Financial vulnerability: Landlords pass on the costs of purely cosmetic renovations to tenants, and the costs of energy-efficiency upgrades are disproportionately borne by tenants, while this is an investment in the property of owners and tenants are rarely included into any decisions.
Original Policy Objective	The idea was to provide owners with financial incentives for modernization measures. It was intended to act as a renovation accelerator by allowing them to pass on costs to tenants (introduced into the German Civil Code in 2001, based on the 1975 Rent Control Act).
Existing Gaps	Modernization levies have repeatedly led to sharp rent increases in the past. Households in the lower and, to some extent, the middle income brackets are already at or above their financial limit. Further increases in housing costs threaten to overwhelm many financially. Increased property values at the expense of tenants exacerbate financial inequality/opportunities.
Proposed Adjustment	Proposals are based on KlimaSozial publication see above, but discussed and advanced in 3. Lab) Increased use of subsidies Under the one-third model, the modernization surcharge is reduced from 8 percent to 3 percent, thus lowering the financial burden on tenants. In return, subsidy rates increase. Landlords can retain the subsidy and no longer have to deduct it from modernization costs as before. This reduces the financial burden on landlords, creates an incentive to apply for subsidies, and simultaneously curbs rent increases. Use of subsidies as a prerequisite for the modernization levy The passing on of modernization costs from landlords to tenants should be linked to the receipt of subsidies. Since the portion of the renovation costs financed through subsidies cannot be passed on to the rent, the modernization surcharge is reduced. This could limit rent increases. This opportunity is already possible when changing heating system (where a 10% modernization levy can be chosen by landlords but then they need to deduce state subsidies from the costs), but not to other, deep renovation subsidies (energy-efficient buildings and individual measures such as windows etc.). Modernization measures can only be passed on to tenants if the GEG minimum standards are met. Currently, there is no requirement that the statutory minimum standard stipulated in the Building Energy Act (in particular Annex 7 to Section 48 GEG) must be met through modernization. This could be amended to stipulate that a rent increase after modernization is only permissible if the modernization ensures compliance with the regulatory minimum standard. <i>This was contested by participants in our</i>

Policy code	DE_H_AD_3
Country	Germany
Sector	Housing
	<i>labs, as some think the standards are too high, they are influenced by the building lobby and lead to less renovations taking place due to high financial barriers.</i> Lump sum and deduction of maintenance costs in the modernization levy For example, a guideline stipulating that 50 percent of the total costs of an energy-efficient renovation may not be passed on via the modernization levy if the last building envelope renovation was 20 years ago or more. Automated regulations should be found to prevent tenants from having to take legal action against their own landlords. Restriction of the modernization levy to energy-efficient renovation measures
Transformative Category	Quick win
Transformative scoring	4,25
Feasibility scoring	3,75

Policy code	DE_H_AD_5
Country	Germany
Sector	Housing
Policy Title: Tax relief, e.g., through adjustment of income tax (EstG)	
Policy Type	Economic incentives (subsidies, taxes, grants, penalties)
Sectoral Challenge Addressed	Financial vulnerability; Create incentives for measures that promote environmental and social actions of landlords
Original Policy Objective	Tax payment
Existing Gaps	The current mechanism is only profitable for owner-occupiers, who can deduct modernization costs from their income tax (Income Tax Act (EstG) § 35c), up to a maximum of 20% of the expenses (maximum 40,000 euros), spread over three years. Tenants therefore do not benefit as landlords are not incentivised.
Proposed Adjustment	Owners who rent out their properties should be incentivised in the same way, but this should be based on certain criteria, e.g. • Maximum building/apartment ownership or turnover limit • Cooperatives • If landlords do not use the modernization surcharge—i.e., they do not burden their tenants financially with the costs • If landlords rent below the local rent index or local comparable rent
Transformative Category	short-term stabiliser
Transformative scoring	3
Feasibility scoring	3,75

Policy code	DE_H_AD_1
Country	Germany
Sector	Housing
Policy Title: CO2 levy - CO2 Cost Allocation Act (CO2KostAufG) Based on the incentive system of the Fuel Emissions Trading Act (translating ETS 1 into national law)	
Policy Type	Regulation (laws, standards, bans, requirements)
Sectoral Challenge Addressed	A “fair” distribution of the consumption of CO2 under the incentive system of the Fuel Emissions Trading Act.
Original Policy Objective	Penalizing the use of fossil fuels and incentivizing the use of renewable energies → Long term goal = reducing CO2 emissions The purpose of this law is to divide the costs of carbon dioxide between landlords and tenants according to their areas of responsibility and influence on the carbon dioxide emissions of a building.
Existing Gaps	Tenants often have no influence on the existing infrastructure, but often pay more. When renting single-family homes or apartments with floor gas heating, the mechanism requires tenants to reclaim the money from landlords—which in turn creates obstacles.
Proposed Adjustment	• Change the way costs are distributed, e.g. landlords pay everything as they are responsible for the infrastructure • Set higher thresholds for landlords than currently planned, so that landlords change their heating systems. • Adjust regulations depending on the type of heating. • Introduce a hardship clause, for tenants that cannot pay extra costs or have special needs for high energy consumption due to chronic sickness or disabilities.
Transformative Category	Short-term stabiliser
Transformative scoring	2,75
Feasibility scoring	4

Policy code	DE_H_AD_4
Country	Germany
Sector	Housing
Policy Title: German Civil Code; Costs of heat supply as operating costs, regulatory authority (BGB § 556c)	
Policy Type	Regulation (laws, standards, bans, requirements)
Sectoral Challenge Addressed	• To avoid increased heating costs for tenants, protection especially when no higher efficiency is reached as fossil fuel based district heating implies. • Due to the possibility for designing price adjustment clauses within the contracts the prices are very volatile, which at some point was justified as the district heating market needs to take high investments and therefore needs planning security, but they are

Policy code	DE_H_AD_4
Country	Germany
Sector	Housing
	not at all controlled at the moment and carry a high risk of energy poverty
Original Policy Objective	This paragraph was established to protect tenant against rising heating costs from their landlords.
Existing Gaps	District heating infrastructure that in the future could be more environmentally friendly heat supply (currently 1/3 is still fossil fuel powered), is not being implemented because gas is significantly cheaper and an upgrade would needed to be paid by the landlords under §556c and they are unwilling/unable to bear the costs alone. à The upfront costs are affordable, but the running costs are very high/ à Further, oftentimes the district heating grid is monopolised by one local energy suppliers.
Proposed Adjustment	- State subsidies for district heating connections if they are powered by 100% renewable energy sources E.g. integrating into the BEW funding (abbreviation for Federal funding for efficient heating networks) à Subsidy programme specifically for landlords with regard to district heating connections (but running costs would not be covered). - Value added tax (VAT) exemption on district heating energy sources would cover running costs for end consumers and might reduce the price volatility.
Transformative Category	Transformative bet/ quick win
Transformative scoring	3,5
Feasibility scoring	3,75

6.2.2. New policies

Policy code	DE_H_N_1
Country	Germany
Sector	Housing
Policy title: Educational work on the topic of CO2 levies, etc.	
Policy Type	Information/education (campaigns, training, awareness)
Sectoral Challenge Addressed	Financial vulnerability: Few tenants know what the CO2 surcharge is or what costs they will/may face in the coming years. Even homeowners often make decisions without knowing about the tax burdens or including them specifically in their cost calculations.
Description of the Proposal	Like a Tenant ABC—but specifically on the topics of energy consumption, rights and obligations, opportunities for participation, etc.

Policy code	DE_H_N_1
Country	Germany
Sector	Housing
Policy title: Educational work on the topic of CO2 levies, etc.	
	→ Provided in part by consumer advice centers, but only reaches homeowners, not tenants. → Involvement of social and religious institutions at the local level/social multipliers, youth organisations. → Connection with the planned one-stop shops (see EU Energy Performance of Building Directive), which should also provide targeted information for landlords.
Transformative Category	Non-starter
Transformative scoring	1,75
Feasibility scoring	3

Policy code	DE_H_N_2
Country	Germany
Sector	Housing
Policy title: Social and financial responsibility – a joint task	
Policy Type	Economic incentives (subsidies, taxes, grants, penalties) Public services (programmes, infrastructure investment) Information/education (campaigns, training, awareness)
Sectoral Challenge Addressed	There are people who are heavily burdened by the system and often suffer multiple forms of discrimination (economic difficulties, health problems, precarious tenancy situations), while there are people with a lot of capital and assets who need to be held more accountable.
Description of the Proposal	Strengthening justice and solidarity: there could be an extra tax on wealthy individuals and high earners, which could be used for energy-efficient renovation of German buildings and renewable energy supply.
Transformative Category	Non-starter
Transformative scoring	2,5
Feasibility scoring	2,25

7. Hungarian policy portfolio

7.1. Adjustments in existing policies

Policy code	H_AD_1
Country	Hungary
Sector	Energy efficiency and Housing
Policy Title: Social Housing and Home Refurbishment Programme (inc. specific rural funding) – State funding from municipal social housing and Social Component for Low-Income Households (This refers to in part this national regulation: https://net.jogtar.hu/jogszabaly?docid=a2400389.kor but also, municipal level social housing projects in general)	
Policy Type	Regulation Economic incentives Public services
Sectoral Challenge Addressed	• Energy poverty is severe; the housing stock has low energy efficiency, while energy prices are fixed by the state. • Large numbers of low-income households cannot participate due to the requirement of own contribution. (50% funding intensity) • Municipal rental housing is heavily depreciated, resulting in poor living conditions and long-term asset loss for local authorities. • Outdated and leaking water networks cause structural damage and inefficiency, yet are not integrated into refurbishment schemes. • Without intervention, inequalities in the green and digital transitions will deepen.
Original Policy Objective	The original programme aimed to improve the energy efficiency of the national housing stock, modernise residential buildings and encourage home renovation through state-supported subsidies.
Existing Gaps	• The current system excludes the poorest citizens because they cannot pay the mandatory self-contribution. • Municipalities lack financial state support to maintain or upgrade their housing stock. • Water network rehabilitation is not linked to building refurbishment, although both problems appear together in deprived areas. • The programme does not target social inequalities or provide mechanisms to ensure that the most vulnerable beneficiaries are prioritised.
Proposed Adjustment	• Creation of a dedicated social component within the programme, in which the state or municipality covers the self-contribution for low-income households.

Policy code	H_AD_1
Country	Hungary
Sector	Energy efficiency and Housing
	• Introduction of a yearly benchmark, such as a fixed number of beneficiary households, to ensure predictable long-term funding and strengthen financing channels. • Central government remains the source of the subsidy, but local municipalities receive the authority to make final allocation decisions, as they are more familiar with local needs and socio-economic realities. • Integration of water network rehabilitation into the refurbishment programme in areas with high infrastructure decay. • Establishment of a normative support scheme that helps municipalities reduce long-term asset depreciation in their rental housing stock. • Use of ESCO-type models where appropriate to reduce upfront financial burdens for residents. • The reform does not change the overall goals of the programme but alters the decision-making route so that support reaches vulnerable groups more effectively.
Transformative Category	Quick wins
Transformative scoring	4
Feasibility scoring	3

Policy code	H_AD_2
Country	Hungary
Sector	Transport - Energy
Policy Title: EV Incentive and Subsidy Programme – Adjusted to Support Used EVs (Gov. decree: 243/2019. (X. 22.) and call for proposals: RRF-REP-10.10.1-24	
Policy Type	Regulation Economic incentives Public services
Sectoral Challenge Addressed	• Due to the still dominant share of fossil fuels and the limited penetration of electric cars, climate neutrality and environmental protection are not ensured. • New EVs are affordable only for higher-income groups. • Current subsidy design excludes most households, since only corporations are allowed to participate and only new cars can be bought, and does not significantly reduce car dependency. • Support for new EVs does not align with long-term sustainability or social fairness. • High environmental footprint associated with battery logistics and reverse transport.
Original Policy Objective	The programme originally aimed to promote the uptake of new electric vehicles, reduce transport emissions, and stimulate the domestic EV market.

Policy code	H_AD_2
Country	Hungary
Sector	Transport - Energy
Existing Gaps	- Subsidies currently apply mainly to new EVs and corporate fleets, excluding the majority of citizens as many people can only afford to buy used cars, or cannot acquire a company car. - Support for new car purchases increases total car numbers and does not reduce congestion or emissions meaningfully. - State resources spent on new EV buyers leave little funding for public transport development. - Used EV buyers have no support despite lower cost and higher social impact potential. - Environmental externalities of battery transport and disposal are insufficiently addressed.
Proposed Adjustment	- Shift the programme entirely from new EV purchases to supporting used EV acquisitions. - Remove subsidies for new EVs and redirect all funds to a used-vehicle swap scheme, similar to the original incentive, a significant subsidy would be available to buy used EVs. Additionally, if a person can exchange an older combustion engine car, together with the subsidy they could get a used EV for free. - Make used EVs accessible to private individuals, especially lower- and middle-income groups. - Link the reform to public transport investment, using freed resources to strengthen alternatives to car dependency. - Address battery disposal by prioritising EVs produced domestically, where recycling can also happen locally. - Consider solutions for handling the inflow of used EVs and their potential re-export.
Transformative Category	Quick wins
Transformative scoring	2
Feasibility scoring	3

Policy code	H_AD_3
Country	Hungary
Sector	Digital inclusion - this is / would be a foundational reform that allows children in deep poverty to learn digital skills and get access to virtually unlimited educational material online. Without such empowerment their chances of breaking out of poverty would be significantly lower and they would only have very restricted information about the state of the world including environmental issues, governance etc.
Policy Title: Digital Devices Programme – Social Prioritisation and Inclusive Allocation (call for proposals: RRF-1.2.1-2021-2021-00001	
Policy Type	Economic incentives Public services Information/education
Sectoral Challenge Addressed	The current programme does not differentiate between households based on socioeconomic status.

Policy code	H_AD_3
Country	Hungary
Sector	Digital inclusion - this is / would be a foundational reform that allows children in deep poverty to learn digital skills and get access to virtually unlimited educational material online. Without such empowerment their chances of breaking out of poverty would be significantly lower and they would only have very restricted information about the state of the world including environmental issues, governance etc.
	• High-income families receive the same devices as low-income families, reducing the availability of laptops for those who cannot afford any device at all. • The programme is institution-based, meaning laptops are tied to schools rather than individuals. • Students in higher education (high school and university) are not included despite measurable need. • The digital divide persists across generations, particularly for disadvantaged children and university students.
Original Policy Objective	The original programme aimed to increase digital access for students by providing laptops to public schools, improving educational outcomes and reducing the digital divide.
Existing Gaps	• The current programme does not differentiate between households based on socioeconomic status. • High-income families receive the same devices as low-income families, reducing the availability of laptops for those who cannot afford any device at all. • The programme is institution-based, meaning laptops are tied to schools rather than individuals. • Students in higher education (high school and university) are not included despite measurable need. • The digital divide persists across generations, particularly for disadvantaged children and university students.
Proposed Adjustment	• Introduce a social prioritisation model in which lower-income and disadvantaged households are placed at the front of the allocation queue. • Maintain universal eligibility so that high-income families are not fully excluded, but receive devices only if stock remains after prioritisation. • Shift from institution-based to person-based allocation, allowing students to keep the device as they move through the education system. • Extend eligibility to university students, particularly first-generation and low-income learners. • Allow students to keep the device at the end of its lifecycle rather than returning it, reducing electronic waste and administrative burden. • Consider providing two devices across a student's educational path, in line with realistic device lifespan (e.g., two devices across 10 years). • Include optional refurbishment or buy-back schemes only if they do not penalise low-income students.

Policy code	H_AD_3
Country	Hungary
Sector	Digital inclusion - this is / would be a foundational reform that allows children in deep poverty to learn digital skills and get access to virtually unlimited educational material online. Without such empowerment their chances of breaking out of poverty would be significantly lower and they would only have very restricted information about the state of the world including environmental issues, governance etc.
	Encourage better device care by clarifying long-term ownership expectations.
Transformative Category	Structural shifts
Transformative scoring	3
Feasibility scoring	5

Policy code	H_AD_4
Country	Hungary
Sector	Energy
Policy Title: Wind Turbine Installation Programme – Community Compensation Mechanism (government decree: 253/1997. (XII. 20.))	
Policy Type	Regulation Economic incentives Public services
Sectoral Challenge Addressed	- Due to the still dominant share of fossil fuels and the limited penetration of electric cars, climate neutrality and environmental protection are not ensured. - Renewable energy sources also have environmental impacts; their production is fluctuating and difficult to store, which poses major challenges for the capacity of the electricity grid and requires substantial investment. - Local communities often bear environmental or visual burdens from wind farms without receiving direct benefits. - Opposition from municipalities slows down installations and reduces acceptability. - There is currently no consistent compensation model that recognises local externalities. - Infrastructure issues such as maintenance, storage capacity and ecological impacts complicate deployment.
Original Policy Objective	The original programme aimed to increase the share of renewable energy production by enabling the development of wind turbines and related infrastructure at suitable locations.
Existing Gaps	- Communities near wind farms do not receive guaranteed compensation or share in economic benefits. - Local governments have limited influence over wind farm siting or benefit distribution, reducing trust in investors. - Maintenance responsibilities and long-term operational risks are insufficiently clarified.

Policy code	H_AD_4
Country	Hungary
Sector	Energy
	- Storage challenges and ecological concerns require more nuanced planning. - The current framework does not integrate community ownership or usage rights in a stable or legally defined form.
Proposed Adjustment	- Introduce a mandatory community compensation model in which municipalities receive a fixed percentage of the profit or energy value generated by wind turbines. - Provide the local community with either a share of ownership or a legally protected usage right to a portion of the produced energy, without requiring them to bear proportional maintenance risk. - Ensure that compensation is tied to the actual externalities experienced by the municipality, such as noise, visual impact or land-use restrictions. - Avoid forcing municipalities into formal ownership positions that may carry financial liabilities; instead, strengthen their entitlement to benefits. - Allow the model to apply not only to wind turbines but also to geothermal or other renewable energy facilities where local impact is significant. - Maintain constructive ambiguity for investors while ensuring fairness for communities, thereby increasing acceptance and accelerating deployment.
Transformative Category	Non-starter
Transformative scoring	2
Feasibility scoring	4

Policy code	H_AD_5
Country	Hungary
Sector	Digital public administration
Policy Title: Digital Citizenship App (DÁP) https://net.jogtar.hu/jogszabaly?docid=a2300103.tv – Strengthened Data Security and Privacy Framework	
Policy Type	Regulation Public services Information/education
Sectoral Challenge Addressed	- Storing all personal information in one place creates severe cybersecurity vulnerabilities. - A single breach can expose complete medical, administrative or personal histories. - Citizens' trust in government digital services is fragile and essential for adoption. - Human actors remain the greatest cybersecurity risk in any large-scale system.

Policy code	H_AD_5
Country	Hungary
Sector	Digital public administration
Original Policy Objective	The original DÁP intends to centralise access to public services and personal administrative data through a single digital interface, simplifying interactions between citizens and the state.
Existing Gaps	• Current design centralises sensitive data excessively, increasing the severity of potential attacks. • There are insufficient safeguards to separate, compartmentalise or pseudonymise data. • Public communication does not adequately address risks or reassure citizens. • High dependency on a constantly evolving cybersecurity landscape makes static security designs obsolete. • No clear accountability structure defines who is responsible for breaches and citizen restitution.
Proposed Adjustment	• Introduce a decentralised or layered data storage architecture to reduce the vulnerability of a single breach. • Ensure that data used by the application is accessible when needed but stored with isolation principles, preventing attackers from viewing full datasets at once. • Strengthen authentication and encryption protocols, including mandatory multi-factor verification for sensitive operations. • Create an independent cybersecurity supervisory body responsible for auditing the DÁP continuously. • Develop transparent communication channels to inform citizens about how their data is protected, how breaches would be handled and what rights they retain. • Incorporate adaptive security measures so that the system evolves in response to new threats. • Provide clear accountability and restitution mechanisms to maintain public trust.
Transformative Category	Non-starter
Transformative scoring	3
Feasibility scoring	3

7.2. New policies

Policy code	H_N_1
Country	Hungary
Sector	Education
Policy title: Planetary Boundaries Education Framework – Integrated School and Adult Learning System	
Policy Type	Information/education
Sectoral Challenge Addressed	• Energy policy is closely intertwined with power struggles among interest groups and with geopolitical conflicts; the pursuit of continuous economic growth and consumerism stands in contrast to environmental limits. • Renewable energy sources also have environmental impacts; their production is fluctuating and difficult to store, which poses major challenges for the capacity of the electricity grid and requires substantial investment. • Insufficient understanding of planetary boundaries, climate science and sustainability among both children and adults. • Lack of consistent, science-based environmental literacy across the education system. • Social and behavioural changes needed for long-term ecological resilience are not adequately supported by current curricula. • Adult populations have limited access to accessible and engaging sustainability education.
Description of the Proposal	The policy introduces a comprehensive education framework built on the scientific concept of planetary boundaries, aligned with IPCC-based sustainability knowledge. It embeds age-appropriate environmental literacy throughout primary and secondary education and provides accessible adult education modules supported by municipalities. The programme includes incentives, community events and public communication campaigns to ensure long-term cultural integration of sustainability principles.
Transformative Category	Structural shifts
Transformative scoring	5
Feasibility scoring	3

Policy code	H_N_2
Country	Hungary
Sector	General green transition
Policy title: Environmental, Social and Governance (ESG)-Based Company Valuation Authority and Transparency Framework	
Policy Type	Regulation Economic incentives Public services Information/education

Policy code	H_N_2
Country	Hungary
Sector	General green transition
Policy title: Environmental, Social and Governance (ESG)-Based Company Valuation Authority and Transparency Framework	
Sectoral Challenge Addressed	• Energy policy is closely intertwined with power struggles among interest groups and with geopolitical conflicts; the pursuit of continuous economic growth and consumerism stands in contrast to environmental limits. • Current company valuation systems rely primarily on financial indicators and self-reported ESG data, which allows greenwashing and reduces accountability. • Corporate self-assessment enables manipulation, selective disclosure and influence over political decision-making. • The previous Green National Champions programme lacked transparency and clear criteria, making corruption risks significant. • Local communities and environmental impacts have limited influence on corporate evaluation and regulatory outcomes.
Description of the Proposal	The proposal establishes an independent ESG-Based Company Valuation Authority that measures corporate value by integrating financial, environmental and social impacts into a single, mandatory assessment system. The authority replaces and absorbs the Green National Champions programme, using a transparent, audited methodology that prevents greenwashing and ensures equal treatment of small, medium and large enterprises. It includes the power to verify ESG reports, impose sanctions, and communicate results publicly, ensuring accountability and reducing the influence of corporate lobbying.
Transformative Category	Structural shifts
Transformative scoring	5
Feasibility scoring	3

Policy code	H_N_3
Country	Hungary
Sector	General green transition
Policy title: Environmental and Social Welfare Minimum of people as a benchmark for All Development and Investment Projects	
Policy Type	Regulation Economic incentives Public services Information/education
Sectoral Challenge Addressed	• Energy policy is closely intertwined with power struggles among interest groups and with geopolitical conflicts; the pursuit of continuous economic growth and consumerism stands in contrast to environmental limits. • Current investment and development decisions often prioritise economic gain over residents' fundamental wellbeing.

Policy code	H_N_3
Country	Hungary
Sector	General green transition
Policy title: Environmental and Social Welfare Minimum of people as a benchmark for All Development and Investment Projects	
	- There is no unified welfare threshold that defines the minimum acceptable level of environmental and social conditions (such as noise, water quality, air quality, sleep, energy access). - Communities exposed to industrial projects or infrastructure face declining living standards without adequate mitigation. - Existing ESG frameworks do not consistently incorporate welfare metrics affecting everyday life.
Description of the Proposal	The policy introduces a legally defined Environmental Welfare Minimum that establishes measurable thresholds for key aspects of human wellbeing, including clean water, air quality, noise levels, sleep quality, local environmental impacts and energy access. Any planned investment or industrial development must demonstrate compliance with these minimum thresholds. If a project threatens to reduce residents' wellbeing below the defined minimum, this must be reflected in the project's feasibility, cost structure or required mitigation measures. The welfare minimum becomes a core component of ESG evaluation and long-term planning.
Transformative Category	Structural shifts
Transformative scoring	4
Feasibility scoring	3

Policy code	H_N_4
Country	Hungary
Sector	Agriculture and land use
Policy title: Territory-Specific Sustainable Water Management and Retention Framework	
Policy Type	Regulation Economic incentives Public services Information/education
Sectoral Challenge Addressed	- Green energy sources, such as battery factories, are associated with significant environmental pollution. - Hungary faces increasing water scarcity in some regions and flooding in others, while outdated water networks cause major losses. - Industrial water use can reduce the availability of water for households, lowering environmental and welfare standards. - Centralised, uniform regulations do not account for regional differences in hydrology, soil conditions or climate risks. - Public awareness about water conservation remains low, and behavioural change is limited.
Description of the Proposal	The policy introduces a territory-specific, expert-led water management and retention system that ensures sustainable household water access and protects ecological balance. It mandates local hydrological

Policy code	H_N_4
Country	Hungary
Sector	Agriculture and land use
Policy title: Territory-Specific Sustainable Water Management and Retention Framework	
	assessments to guide decisions on agricultural water use, industrial consumption and community-level retention strategies. The framework establishes targeted financial support for the renovation of old, leaking water networks and promotes integrated water-saving practices. Public campaigns raise awareness about water scarcity, responsible consumption and long-term conservation needs.
Transformative Category	Structural shifts
Transformative scoring	5
Feasibility scoring	3

Policy code	H_N_5
Country	Hungary
Sector	Transport
Policy title: Priority of Alternative and Public Transportation in Urban and Regional Mobility Systems	
Policy Type	Economic incentives Public services Information/education
Sectoral Challenge Addressed	There is a lack of prioritisation and development for public transport, cycling, walking, and micromobility. • Car dependency limits mobility options, increases emissions and worsens congestion. • Existing transport structures often privilege private car use over sustainable alternatives. • Infrastructure for public transport, cycling and micromobility remains insufficient in many regions. • The distribution of daily activities requires long distances, reducing accessibility for low-income and car-free households. • Settlements are not always designed to support short-distance living or 15-minute city principles.
Description of the Proposal	The policy prioritises public transportation, micromobility and other sustainable alternatives over private car use. It supports the development of integrated mobility systems where daily needs can be met locally, and where public and active transport options are accessible, reliable and attractive. The framework encourages the creation of 15-minute cities, promotes sustainable settlement size and strengthens the role of public transport as the backbone of mobility planning. Investment, regulation and urban design all aim to reduce car dependency and enhance equitable, environmentally friendly mobility. See the current strategy here.
Transformative Category	Transformative bets
Transformative scoring	4
Feasibility scoring	3

Policy code		H_N_6
Country		Hungary
Sector		Green transition in general
Policy title: Enhanced Ombudsperson for Future Generations with Expanded Autonomy and Long-Term Planning Mandate		
Policy Type	Regulation Public services Information/education	
Sectoral Challenge Addressed	• Energy policy is closely intertwined with power struggles among interest groups and with geopolitical conflicts; the pursuit of continuous economic growth and consumerism stands in contrast to environmental limits. • Short political cycles often undermine long-term sustainability goals. • Existing institutions responsible for protecting long-term public interests lack autonomy, authority and enforcement power. • Sustainability considerations are frequently overridden by short-term economic or political priorities. • There is no strong institutional guardian ensuring that long-term ecological and social impacts are systematically integrated into policymaking.	
Description of the Proposal	The policy strengthens the institution of the Future Generations' Commissioner by expanding its legal powers, autonomy and role in national decision-making. The Commissioner becomes responsible for overseeing long-term planning, reviewing legislation for sustainability impact and ensuring that policies do not compromise the rights, wellbeing or environmental security of future generations. The office gains ombudsman-level independence, with guaranteed resources and the authority to request information, issue legally binding opinions and monitor governmental compliance with intergenerational obligations.	
Transformative Category	Structural shifts	
Transformative scoring	4	
Feasibility scoring	2	

Policy code		H_N_7
Country		Hungary
Sector		Green transition in general
Policy title: Institutionalised Citizens' Assembly System for Participatory and Legitimate Decision-Making		
Policy Type	Regulation Public services Information/education	
Sectoral Challenge Addressed	• Current policymaking processes often lack meaningful public participation, reducing legitimacy and weakening trust in institutions. • Citizens have limited channels to influence decisions directly, especially on complex or controversial matters.	

Policy code	H_N_7
Country	Hungary
Sector	Green transition in general
Policy title: Institutionalised Citizens' Assembly System for Participatory and Legitimate Decision-Making	
	• Representative systems alone do not always capture the diversity of public perspectives, leading to disengagement and political polarisation. • There is no permanent institutional mechanism ensuring participatory deliberation beyond ad hoc consultations
Description of the Proposal	The proposal establishes a permanent, institutionalised Citizens' Assembly composed of a statistically representative sample of the population. The Assembly participates regularly in policy development, evaluates legislative proposals and provides binding or semi-binding recommendations. Participants are compensated for their time, trained in deliberative methods and supported by independent facilitators. The model enhances legitimacy, transparency and democratic learning, enabling both concrete decisions and long-term civic capacity building
Transformative Category	Structural shifts
Transformative scoring	4
Feasibility scoring	2

8. Irish policy portfolio

8.1. Energy

8.1.1. Adjustments in existing policies

Policy code	IE_EN_AD_1
Country	Ireland
Sector	Energy/Housing (co-created in Energy subgroup)
Policy Title: Warmer Homes Scheme	
Policy Type	Economic incentives
Sectoral Challenge Addressed	Policies are inaccessible to many groups in the population.
Original Policy Objective	To improve the energy efficiency, warmth, and comfort of homes owned by low-income households, while reducing energy poverty and supporting national climate goals.
Existing Gaps	Selection criteria are not including caravans: leaves Travellers vulnerable to energy poverty
Proposed Adjustment	Broaden eligibility to bring the Traveller Community into the scheme, as people living in Caravans/Mobile Homes are not benefitting as these do not fit the criteria for what counts as a “home”. This policy adaptation should be co-designed with Travellers
Transformative Category	
Transformative scoring	
Feasibility scoring	

Policy code	IE_EN_AD_2
Country	Ireland
Sector	Energy
Policy Title: Community Energy Grant	
Policy Type	Economic incentives Public services Information/education
Sectoral Challenge Addressed	Access to finance
Original Policy Objective	To support energy efficiency upgrades and renewable energy projects in community buildings, homes, and businesses.
Existing Gaps	The policy faces two main limitations: insufficient support and financing during the project application and implementation phases, and lower-than-expected uptake and effectiveness within the residential sector. These limitations stem from a mismatch between policy design assumptions and the actual administrative, technical, and financial capacity of residential and community-level applicants. They are further

Policy code	IE_EN_AD_2
Country	Ireland
Sector	Energy
	compounded by implementation constraints related to stakeholder capacity and the availability of support mechanisms. From a policy design perspective, complex eligibility criteria and project requirements tend to favour larger, well-resourced applicants with established technical expertise. The emphasis on project ambition, scale, and demonstrable community benefit (often delivered through coordinated project structures) can disadvantage smaller or simpler residential projects, reducing accessibility and participation. From an implementation perspective, barriers arise from administrative complexity, limited pre-application and implementation support, and capacity constraints among intermediaries and project coordinators responsible for project delivery. These factors increase the administrative burden of participation and limit the scheme's effectiveness in reaching less-resourced residential and community applicants, highlighting the need for more streamlined processes and strengthened support throughout the project lifecycle.
Proposed Adjustment	Provide additional support for communities to manage the project implementation phase. Use deprivation index data to prioritise projects. Create more energy support local offices for grant application support
Transformative Category	
Transformative scoring	
Feasibility scoring	

Policy code	IE_EN_AD_3
Country	Ireland
Sector	Energy
Policy Title: Fuel Allowance	
Policy Type	Economic incentives Public services
Sectoral Challenge Addressed	Energy poverty due to increasing heating and electricity costs; growing pains of adapting infrastructure; ageing housing stock
Original Policy Objective	Payment designed to help households with the cost of heating their homes during the winter months. It is paid to eligible people who are dependent on long-term social welfare payments, and only one person per household can receive it.
Existing Gaps	Good that it is managed by DSP (Department of Social Protection) but eligibility must be based on existing evidence of energy poverty as it exists today rather than just keep adding new groups. Many of the groups already included will continue to qualify, but the framework needs to be more coherent and evidence based. Currently, eligibility is largely based on monetary poverty or welfare status, rather than on direct evidence of energy poverty. While this approach ensures

Policy code	IE_EN_AD_3
Country	Ireland
Sector	Energy
	administrative simplicity and targets many vulnerable households, it does not explicitly account for other key drivers of energy poverty, such as poor energy efficiency, high energy needs, or housing deprivation.
Proposed Adjustment	Broaden eligibility. While it was broadened in Budget 2026, the baseline approach should recognise that energy poverty is not defined solely by low income, but also by poor energy efficiency and deprivation. Over time, eligibility has been expanded by adding new beneficiary groups through successive budgets. While this has improved coverage, it has resulted in an incremental eligibility framework that remains only indirectly linked to actual energy poverty outcomes. Some households experiencing high energy costs due to inefficient housing or specific energy needs may still be excluded if they do not meet the income or welfare-based criteria. The proposed adaptation does not suggest replacing the existing income-based criteria, but rather complementing them with a more explicit, evidence-based definition of energy poverty. This would involve broadening eligibility to consider additional factors, such as: 1) Poor energy efficiency of the dwelling (e.g. low BER ratings or absence of key efficiency measures); 2) High energy expenditure relative to income, indicating energy cost vulnerability; 3) Indicators of deprivation or housing inadequacy that increase heating needs
Transformative Category	
Transformative scoring	
Feasibility scoring	

8.2. Housing

8.2.1. Adjustments in existing policies

Policy code	IE_HO_AD_1
Country	Ireland
Sector	Housing
Policy Title:	Minimum BER in the Rental Sector This is a proposal to introduce minimum BER standards for private rental properties. This is part of current Irish policy commitments, although it has not yet been fully enacted into law. The commitment to introduce minimum BER standards for private rental properties is included in the Government’s <u>Housing for All strategy</u>, which envisages introducing such standards “where feasible” from 2025. In the Irish Parliament, <u>specific draft legislation was discussed in 2025</u> to make minimum BER requirements mandatory for private rented accommodation, for example, proposals to require a minimum BER of D2 by end of 2026 and C1 by end of 2028 were mentioned.
Policy Type	Regulation

Policy code	IE_HO_AD_1
Country	Ireland
Sector	Housing
Sectoral Challenge Addressed	House ownership. Policies are inaccessible to tenants. Tenancy status limits agency
Original Policy Objective	Bring reductions to CO2 emissions in housing in the rental sector
Existing Gaps	It is a proposal, not implemented in legislation yet. An enormous cohort of people are trapped in poor quality/damp/ energy inefficient rated housing stock
Proposed Adjustment	Bringing proposal into legislation as committed in Housing for All, with clear compliance timelines (e.g., phased minimum BER thresholds such as D2 by 2026 and C1 by 2028). Strengthen enforcement, making sure RTB (Residential Tenancies Board) makes compliance checks. Bring both landlords and tenants into the compliance framework.
Transformative Category	
Transformative scoring	
Feasibility scoring	

Policy code	IE_HO_AD_2
Country	Ireland
Sector	Housing & Energy (co-created in the Housing sub-group)
Policy Title: Warmer Homes Scheme	
Policy Type	Regulation
Sectoral Challenge Addressed	Policies are inaccessible to many
Original Policy Objective	To improve the energy efficiency, warmth, and comfort of homes owned by low-income households, while reducing energy poverty and supporting national climate goals.
Existing Gaps	The funding criteria. It excludes those who can't afford & don't qualify
Proposed Adjustment	Broaden eligibility
Transformative Category	
Transformative scoring	
Feasibility scoring	

Policy code	IE_HO_AD_3
Country	Ireland
Sector	Housing & Energy
Policy Title: All homes to be B2 BER rated.	
Policy Type	Economic incentives (more accessible grants) Education/awareness
Sectoral Challenge Addressed	Rising greenhouse gas emissions from the residential sector due to the prevalence of older, energy-inefficient housing, combined with unequal access to retrofit opportunities. While retrofit schemes are available, under-served households (e.g. low-income households, renters,

Policy code	IE_HO_AD_3
Country	Ireland
Sector	Housing & Energy
	households in poorly performing housing stock, and communities with low administrative capacity) face structural barriers, such as high upfront costs, limited access to finance, information gaps, and complex application processes, which limit their ability to participate.
Original Policy Objective	There is a target in place of 500,000 homes rated B2 by 2030 with the objectives to: Reduce of CO2 emissions; reduce energy poverty; increased comfort in homes; lower national energy demand; build a retrofit industry capable of delivering at scale
Existing Gaps	Current retrofit schemes are largely demand-led and require households to self-identify, navigate complex application processes, and often co-finance works upfront. This favours higher-income, better-informed homeowners and limits uptake among under-served communities. Educational and informational gaps, as well as administrative and financial capacity constraints, further reduce participation by vulnerable households.
Proposed Adjustment	Entry point to schemes should be changed in order to prioritise under-served communities. Current schemes include: National Home Energy Upgrade Scheme (One-Stop Shop Service); Better Energy Homes Scheme; Better Energy Warmer Homes Scheme (Fully Funded Upgrades); Community Energy Grants; Home Energy Upgrade Loan Scheme These schemes are largely demand-led and rely on household initiative, financial capacity, and administrative literacy, which favours better-resourced homeowners and limits participation by under-served communities, despite their higher exposure to energy poverty and poor housing conditions.
Transformative Category	
Transformative scoring	
Feasibility scoring	

8.3. Energy and housing

8.3.1. New policies

Policy code	IE_ENHO_N_1
Country	Ireland
Sector	Housing & Energy
Policy title: Free Deep Retrofit for Energy-Poor Households	
Policy Type	Regulation Economic incentives Public services Information/education

Policy code	IE_ENHO_N_1
Country	Ireland
Sector	Housing & Energy
Policy title: Free Deep Retrofit for Energy-Poor Households	
Sectoral Challenge Addressed	Policies are inaccessible to many groups of the population; Access to finance: Individuals cannot retrofit – they cannot afford upfront costs; Large stock of old houses means we will not reach targets; Energy poverty
Description of the Proposal	Expand the Better Energy Warmer Homes Scheme to cover full B2 BER retrofits, not just basic insulation. The proposal would expand the existing Better Energy Warmer Homes Scheme to provide fully funded deep retrofits up to a minimum B2 BER standard for households experiencing energy poverty. Eligibility would be based on energy poverty and low-income criteria, building on the current means-tested approach of the Warmer Homes Scheme. The expanded scheme would: 1) Cover the full cost of retrofit works, removing all upfront financial contributions from eligible households; 2) Include comprehensive retrofit measures required to achieve a B2 BER rating, such as insulation, heating system upgrades, ventilation, and renewable energy technologies where appropriate; 3) Apply to both owner-occupied and rental properties housing energy-poor households, subject to safeguards to prevent rent increases or displacement. Eligibility thresholds would align with existing social protection and energy poverty indicators (e.g. receipt of qualifying welfare payments, income below defined thresholds, or identification through area-based energy poverty programmes), ensuring continuity with current administrative systems while significantly increasing the depth of intervention.
Transformative Category	Transformative bet
Transformative scoring	5
Feasibility scoring	1

Policy code	IE_ENHO_N_2
Country	Ireland
Sector	Housing & Energy
Policy title: A new framework for climate governance	
Policy Type	Information/education * This is an initiative focused on governance primarily, although it should include information/education
Sectoral Challenge Addressed	Policies are targeted at individuals but the problems we face are collective; policy does not take a human rights approach; targeted at those on middle to high income who can advocate for themselves; policies should be targeted by regions/areas
Description of the Proposal	Four dimensions: 1) Embed Climate Justice in Policy Design (Integrate equity indicators in Climate Action Plan actions and housing retrofit programs; Mandate Just Transition principles in housing and energy policies); 2) Strengthen Participatory Mechanisms, integrating citizens into decision-making in a meaningful way (Expand National Dialogue on

Policy code	IE_ENHO_N_2
Country	Ireland
Sector	Housing & Energy
Policy title: A new framework for climate governance	
	Climate Action to include: Local Climate Assemblies, Community-led housing retrofit forums; Digital platforms for inclusive engagement; mini-publics); 3) Link climate and community development (link to SICAP- Social Inclusion and Community Activation Programme) because, in order to make this feasible, plans should be implemented at the local level, prioritising energy-poor households, renters, and rural communities in consultations, and working from the bottom-up
Transformative Category	Structural shift
Transformative scoring	4
Feasibility scoring	2

Policy code	IE_ENHO_N_3
Country	Ireland
Sector	Housing & Energy
Policy title: Community Energy Cooperatives/ District Heating projects	
Policy Type	Economic incentives Public services Information/education campaigns
Sectoral Challenge Addressed	Policies are inaccessible to many groups of the population; Access to finance; Individuals cannot retrofit – they cannot afford upfront costs; Large stock of old houses means we will not reach targets; Energy poverty. Energy communities need strong incentives to participate, but many lack the necessary skills, have limited time, and are restricted from partnering with commercial actors.
Description of the Proposal	This policy will support vulnerable and disadvantaged communities to develop and own renewable energy projects- including solar, wind, and district heating- through targeted grants and dedicated technical assistance. By enabling community ownership, the initiative will generate local income, strengthen energy resilience, and empower areas experiencing socio-economic disadvantage. The programme should be closely aligned with Local Community Plans to ensure projects reflect local priorities and build long-term capacity. In order to prioritise disadvantaged/energy poor communities, these will need to be made clearly visible. While tenders can be organised centrally to ensure consistency and efficiency, decision-making and project management should remain at the local level to maximise community ownership and accountability. Maintaining full transparency in budget allocation and project selection will be essential to ensure fairness, trust, and public legitimacy.
Transformative Category	Quick win
Transformative scoring	4
Feasibility scoring	4

Policy code	IE_ENHO_N_4
Country	Ireland
Sector	Housing & Energy
Policy title: Strengthening Rural Transport Infrastructure and Connectivity Across Ireland	
Policy Type	Public services and infrastructure
Sectoral Challenge Addressed	Energy poverty/transport poverty
Description of the Proposal	This policy strengthens rural transport infrastructure as a tool to reduce energy and housing poverty. By improving connectivity, reducing car dependency, and enabling access to essential services and retrofit programmes, it supports vulnerable rural households facing high energy costs and poor housing conditions. The approach rebalances Ireland's transport investment away from a Dublin-centric model and ensures that rural communities can participate fully in the energy transition.
Transformative Category	Short-term stabiliser
Transformative scoring	3
Feasibility scoring	4

9. Italian policy portfolio

9.1. Energy

9.1.1. Adjustments in existing policies

Policy code:	IT_EN_AD_1
Country	Italy
Sector	Energy Energy (Renewable Energy Communities – RECs/CERs – and local entities that promote/manage them)
Policy Title: Technical-Administrative Support Programme for the Management of Renewable Energy Communities (RECs/CER) - Foundation: Legislative Decree 199/2021 = the legal foundation (definitions, rights, roles, perimeter principles). It implements the Directive (EU) 2018/2001 (RED II) – Articles 2 and 22 (Renewable Energy Communities). - Implementing measure: Ministerial Decree 23 Jan 2024 = the operational incentive framework (tariffs, access steps, and Recovery and Resilience National Plans).	
Policy Type	Public services Information/education
Sectoral Challenge Addressed	Heavy bureaucracy, unclear multi-level governance and limited administrative capacity within municipalities and cooperatives, leading to delays, procedural errors and low citizen uptake.
Original Policy Objective	Make the existing RECs/CERs regulatory framework operable by strengthening local management capacity and simplifying procedures for municipalities and cooperatives so to promote the set up of new RECs/CERs.
Existing Gaps	The current framework provides incentives and regulatory clarity but lacks structured local support, standardised operational tools and sufficient staffing/resources at municipal level. Small and inner-area municipalities struggle to manage procedures and citizen engagement effectively. Here the practical bottlenecks in procedures, local capacity, grid/DSO (Distribution System Operators) steps, timelines emerge.
Proposed Adjustment	Institutionalise a national accompaniment programme with territorial helpdesks and dedicated 'REC/community manager' roles. Provide standardised operational kits (bylaws, sharing rules, DSO/installer agreements, checklists, error-proof forms) and digital platforms for document management and tracking. Ensure stable funding for local administrative capacity and introduce simplified procedural guidelines applied consistently across regions.
Transformative Category	Quick wins
Transformative scoring	5
Feasibility scoring	2

Policy code	IT_EN_AD_2
Country	Italy
Sector	Energy–Housing (building energy efficiency and deep renovation)
Policy Title: FAIR RETROFIT BONUS: PROGRESSIVE INCENTIVES, ADVANCES AND ANTI-EXCLUSION Ecobonus (energy efficiency tax deduction) • Law 296/2006 (Budget Law 2007), Article 1, paragraphs 344–349 — introduced the Ecobonus for energy-efficiency works. • Decree-Law 63/2013, Article 14 — the Ecobonus is “currently regulated” under this article (as the ongoing legal framework for the deduction). Building renovation deduction (“Bonus Casa”) • Presidential Decree 917/1986 (TUIR – Income Tax Code), Article 16-bis — legal basis for the renovation tax deduction (commonly used for building works; some energy-related works may fall within its scope depending on the intervention). Superbonus (where applicable to deep retrofit packages) • Decree-Law 34/2020 (“Decreto Rilancio”), Article 119 — establishes the Superbonus framework for “qualifying” energy efficiency interventions (among others). • Decree-Law 34/2020, Article 121 — defines the key mechanisms for options such as invoice discount / tax credit assignment (where allowed under the evolving rules). Conto Termico (direct incentive scheme) • Interministerial Decree 16 February 2016 (“Conto Termico 2.0”) — updates the incentive scheme for small-scale energy-efficiency interventions and renewable thermal production; published in the Official Gazette.	
Policy Type	• Economic incentives: Progressive grants/credits; public guarantees; on-bill financing/crediting; small-scale CfD/FiT; Capital Expenditure (CapEx) grants for inner areas/marginal sites. • Public services: One-stop shops & regional technical helpdesks; aggregation platforms; comparison tools for Power purchase agreement (PPA)/CER; support for applications and grid processes. • Information / education: Simple guides; agri-PV (Photo voltaic) and condominium toolkits; training for administrators/communities; consumer-protection materials. • Regulation: Simplified authorisation tracks; dedicated registers/auctions; priority grid access rules; transparent pricing templates; minimum quality standards; landlord-tenant safeguards.
Sectoral Challenge Addressed	The measures address the energy-inefficient building stock and high household energy consumption contributing to emissions and energy poverty. The problem is that the core bases above are stable while rates, ceilings, eligibility windows and procedural details are frequently amended by later laws (e.g., annual budget laws and subsequent decrees) this causes instability.
Original Policy Objective	Promote energy efficiency in buildings, reduce emissions, stimulate the construction sector and improve energy performance of residential and condominium buildings through fiscal incentives.
Existing Gaps	Existing incentives tend to favour higher-income households who can pre-finance works and navigate complex procedures. Upfront costs, credit constraints, administrative complexity and split incentives (landlord–tenant) exclude vulnerable groups and small condominiums. Risk of fraud and uneven territorial access persists.

Policy code	IT_EN_AD_2
Country	Italy
Sector	Energy–Housing (building energy efficiency and deep renovation)
Proposed Adjustment	Introduce progressive incentive scales calibrated on income and vulnerability, with higher grant intensity for low-income households and energy-poor areas. Provide public-backed advance payments and guarantee mechanisms to eliminate upfront costs. Enable on-bill repayment linked to verified energy savings (“bill-positive” financing). Strengthen anti-exclusion safeguards: simplified error-proof forms, local one-stop shops, landlord–tenant protection rules and a public quality register of contractors with fair-pricing templates.
Transformative Category	
Transformative scoring	
Feasibility scoring	

Policy code	IT_EN_AD_3
Country	Italy
Sector	Energy - electricity wholesale and retail markets; price formation mechanisms)
Policy Title: FAIR ELECTRICITY PRICE – STRUCTURAL DECOUPLING FROM GAS FAIR ELECTRICITY PRICE: STRUCTURAL DECOUPLING FROM NATURAL GAS Electricity Market Design Framework (marginal pricing system with gas-linked wholesale price formation) Legislative Decree No. 79/1999 (“Bersani Decree”): Liberalisation of the electricity market Legislative Decree No. 210/2021 : Implementation of EU Electricity Market Directive Electricity market regulatory framework : ARERA (Italian Energy Regulator) GME (Gestore dei Mercati Energetici) – Market Rules: Day-Ahead Market (MGP) and marginal pricing rules.	
Policy Type	• Regulation (market design reform; CfD framework; transparency rules) • Economic incentives (price stabilisation instruments) • Information/education (monitoring and open data on price formation)
Sectoral Challenge Addressed	Exposure of electricity prices to gas price volatility and structural advantages for incumbent actors, limiting pass-through of low-cost renewables to final consumers.
Original Policy Objective	Ensure efficient price formation through marginal pricing, incentivise generation investments and maintain market integration at EU level. Italy’s electricity price formation is embedded in the EU internal electricity market framework, where wholesale prices in the day-ahead market are generally determined through uniform marginal pricing (“pay-as-cleared”). Under this design, the market-clearing price is set by the marginal generation unit needed to meet demand, which in Italy is frequently gas-fired generation. As a result, retail electricity costs can remain strongly exposed to gas price volatility even when a substantial share of supply comes from low-marginal-cost renewables. The framework aims to deliver efficient dispatch and investment signals, but it can also

Policy code	IT_EN_AD_3
Country	Italy
Sector	Energy - electricity wholesale and retail markets; price formation mechanisms)
	generate windfall rents for inframarginal technologies and weaken perceived pass-through of renewable cost advantages to consumers. Recent EU reforms seek to preserve market efficiency while expanding stabilisation instruments such as two-way contracts for differences (CfDs) and long-term contracting. CfD means Contracting for Difference: a public authority (or market body) sets a “strike price” for electricity. • If the market price is below the strike price → the public counterparty pays the producer the difference. • If the market price is above the strike price → the producer pays back the difference to the public authority. This makes revenues predictable while protecting consumers from excessive windfall profits. The 2024 EU Electricity Market Reform strengthens the role of two-way CfDs for new renewable projects to balance market efficiency with price stability.
Existing Gaps	The marginal pricing model links electricity prices to the most expensive marginal unit (often gas), even when a large share of generation comes from low-marginal-cost renewables. This creates price volatility, windfall profits and weak consumer trust. Distributed producers and small actors face structural disadvantages in accessing stable revenue streams.
Proposed Adjustment	Introduce structural decoupling mechanisms, including a dual-basket system separating renewable and fossil generation pricing. Expand the use of two-way CfDs for renewable generation to stabilise prices and ensure fair pass-through. Strengthen transparency obligations for suppliers and introduce monitoring of price pass-through to retail tariffs, with penalties for unjustified margins. The key is to work on how electricity is procured, hedged, and passed through to consumers, rather than changing the EU day-ahead clearing itself. 1) Expand price-stabilisation instruments for new Renewable Energy Sources 2) Make long-term contracting easier and safer 3) Act on the retail layer (pass-through, transparency, conduct) 4) Use revenues to protect consumers and credibility When national measures generate public revenues (e.g., CfD paybacks, certain levy designs), earmark part for: • targeted bill credits for vulnerable households, • energy efficiency support (especially for tenants/condominiums), • and “cash-flow smoothing” tools (on-bill crediting / faster settlements for prosumers/CER). 5) Increase competition and entry for small/distributed producers 6) Cut the “gas link” by reducing gas at the margin (energy produced by gas is used to saturate the market when the demand is high if

Policy code	IT_EN_AD_3
Country	Italy
Sector	Energy - electricity wholesale and retail markets; price formation mechanisms)
	renewable increase the gas demand is reduced even at pick hours) Remove grid bottlenecks that force reliance on gas.
Transformative Category	Structural shifts
Transformative scoring	5
Feasibility scoring	1

Policy code	IT_EN_AD_4
Country	Italy
Sector	Energy — Energy (electricity distribution networks, smart metering, distributed generation integration)
Policy Title: PROSUMER-READY GRID: SMART METERS AND LOW-VOLTAGE REINFORCEMENT Smart Meter Roll-out and Local Distribution Grid Reinforcement Programme. Legislative Decree 210/2021 transposes Directive 2019/944 (Articles 19–21: smart metering systems, Consumer access to data, Interoperability and digitalisation) and Regulation (EU) 2019/943 On the internal market for electricity (Grid operation, non-discrimination, flexibility, congestion management) into Italian law and provides basis for smart metering development and grid modernisation. The Italian Authority for electricity pricing ARERA Smart Meter roll-out monitoring. Second-generation smart meter (2G) roll-out framework, has issued several key resolutions governing: • Second-generation smart meters (2G roll-out plans) • Data access rules • Distribution tariffs • Hosting capacity transparency Connection timelines	
Policy Type	Regulation Public services Information/education
Sectoral Challenge Addressed	Limitations of local distribution networks and non-smart meters that constrain accurate measurement of self-production, delay grid connections and limit distributed renewable uptake.
Original Policy Objective	The policy objectives were to modernise the electricity distribution system, improve measurement accuracy, enable remote management and facilitate integration of renewable energy sources.
Existing Gaps	Uneven roll-out of second-generation smart meters; insufficient hosting capacity at low-voltage (LV) level; slow grid reinforcement in rural and peri-urban areas; limited transparency on available grid capacity. These constraints slow down prosumer participation and energy community development.
Proposed Adjustment	Accelerate replacement of non-smart meters with interoperable second-generation smart meters. Introduce mandatory local hosting-capacity mapping (LV-level transparency per neighbourhood). Prioritise grid reinforcement in high-renewable-potential areas and establish simplified, time-bound connection procedures. Integrate smart-meter data with on-bill crediting and flexibility markets.

Policy code	IT_EN_AD_4
Country	Italy
Sector	Energy — Energy (electricity distribution networks, smart metering, distributed generation integration)
Transformative Category	
Transformative scoring	
Feasibility scoring	

Policy code	IT_EN_AD_5
Country	Italy
Sector	Energy (renewable energy deployment, grid management, technical workforce and reskilling for the energy transition)
Policy Title: SOLAR SKILLS & GRID 2035 — YOUTH, SKILLS, PROSUMER NETWORK Green Skills Training and Energy Workforce Development Programmes (VET/ITS and sectoral reskilling schemes). Currently these types of programmes already exist and are grounded in a bundle of legal frameworks that fund and regulate VET/ITS training and reskilling, including for the green/energy transition. The most relevant is the ITS Academy (higher technical education – key for “green technicians”), based on Law No. 99 of 15 July 2022 (ITS Academy system reform) and PNRR/MIM implementing framework for strengthening ITS Academy offer (investment + operational acts published by the Ministry of Education). Additionally, in Italy there is the Active labour market reskilling (including “green transition” upskilling) in the Ministerial Decree 5 November 2021 adopting the GOL Programme (Garanzia di Occupabilità dei Lavoratori), published in the Official Gazette. It provides the legal basis for large-scale reskilling/upskilling pathways that Regions implement, including profiles linked to ecological transition.	
Policy Type	Public services Economic incentives Information/education
Sectoral Challenge Addressed	Despite existing programmes there is a limited availability of qualified installers and grid technicians, especially in rural and peripheral areas, and risk of job losses in fossil fuel-related sectors.
Original Policy Objective	Support employment and workforce adaptation to the energy transition through vocational education, technical training and reskilling initiatives in renewable energy and energy efficiency sectors.
Existing Gaps	Existing training programmes are fragmented, not always aligned with local labour-market needs and insufficiently connected to grid modernisation and distributed renewable deployment. Reskilling pathways for fossil-sector workers are not systematically integrated into territorial transition plans.
Proposed Adjustment	Develop a coordinated 'Solar Skills & Grid 2035' programme linking vocational training, technical institutes and universities with local labour-market demand. Introduce targeted reskilling pathways for fossil-sector workers (electrical systems, PV installation, storage, grid operation and maintenance), with recognition of prior competences. Include incentives for firms that train and hire locally (training-first public procurement criteria). Strengthen grid-management skills to support increasing prosumer participation and distributed generation.

Policy code	IT_EN_AD_5
Country	Italy
Sector	Energy (renewable energy deployment, grid management, technical workforce and reskilling for the energy transition)
Transformative Category	5
Transformative scoring	3
Feasibility scoring	Transformative bets

Policy code	IT_EN_AD_6
Country	Italy
Sector	Energy (electricity distribution networks, local flexibility markets, storage integration, prosumer planning)
GRID 2030: PROSUMER PLANNING, LOCAL FLEXIBILITY MARKETS AND QUALITY STANDARDS Local Flexibility Market Pilots and Distribution Network Planning Framework This depends on Regulation (EU) 2019/943, the Directive (EU) 2019/944, Regulation (EU) 2017/2195 Electricity Balancing Guideline (EBGL). The most relevant is the Directive (EU) 2019/944 Internal electricity market directive which defines role of Distribution System Operator (DSO), the company responsible for operating, maintaining and developing the local electricity distribution network (low and medium voltage) and encourages flexibility procurement by system operators. This is transposed in the Italian legislation by: • Legislative Decree No. 210/2021 strengthening DSO responsibilities and opening framework for flexibility services procurement; • Legislative Decree No. 79/1999 (“Bersani Decree”), market liberalisation framework and system operation responsibilities	
Policy Type	Regulation Economic incentives Public services
Sectoral Challenge Addressed	Growing distributed renewable generation and electrification create congestion and hosting-capacity constraints at local level, while flexibility and storage resources remain underutilised
Original Policy Objective	Enable efficient grid operation, reduce congestion costs and facilitate renewable integration through market-based flexibility mechanisms and improved distribution planning.
Existing Gaps	Existing flexibility pilots are fragmented and not fully integrated into long-term distribution planning. Limited transparency on local hosting capacity and insufficient incentives for storage and demand response constrain prosumer participation. Coordination between DSOs, aggregators and municipalities remains weak.
Proposed Adjustment	Institutionalise local flexibility markets with clear remuneration schemes for storage, demand response and community batteries. Introduce mandatory neighbourhood-level hosting-capacity plans integrated into distribution planning cycles. Promote community-scale storage and micro-grid solutions in rural and urban districts. Align tariff structures and incentives to reward flexibility provision and local balancing.
Transformative Category	

Policy code	IT_EN_AD_6
Country	Italy
Sector	Energy (electricity distribution networks, local flexibility markets, storage integration, prosumer planning)
Transformative scoring	
Feasibility scoring	

9.1.2. New policies

Policy code	IT_EN_N_1
Country	Italy
Sector	Energy (cross-cutting to housing and mobility; embedded in regional/local planning cycles)
Policy title: TERRITORIAL JUST TRANSITION TABLES (TJT): MULTI-LEVEL CO-GOVERNANCE WITH DECISION POWER	
Policy Type	• Regulation (obligatory tables, minimum participation standards, SLA, public minutes, shared indicators) • Public services (technical secretariats, digital platform, facilitation) • Information/education (co-design & evidence toolkits, training) • Economic incentives (micro-grants for participation; performance awards)
Sectoral Challenge Addressed	Fragmented governance across national/regional/local levels and weak structured participation of territorial actors; need to integrate evidence and shared monitoring.
Description of the Proposal	Establish permanent decision-capable tables at national, regional and sub-regional level to co-decide priorities, criteria and roadmaps for energy plans. Each table includes social partners, civil society, local authorities, universities/research centres and grid operators, with rules on quorum, representation and public minutes. Common indicators, data transparency and biannual reviews ensure iterative corrections and accountability.
Transformative Category	
Transformative scoring	
Feasibility scoring	

Policy code	IT_EN_N_2
Country	Italy
Sector	Energy (distributed/community solar; links to jobs/training, grid integration and local development)
Policy title: LOCAL SOLAR PLAN: SKILLS, QUALITY, JUST TRANSITION AND INCLUSIVE FINANCE	
Policy Type	• Regulation (training-first criteria; quality standards/register; DSO protocols) • Economic incentives (training vouchers; co-financed hiring; public guarantees for microloans)

Policy code	IT_EN_N_2
Country	Italy
Sector	Energy (distributed/community solar; links to jobs/training, grid integration and local development)
Policy title: LOCAL SOLAR PLAN: SKILLS, QUALITY, JUST TRANSITION AND INCLUSIVE FINANCE	
	• Public services (one-stop shops; purchasing consortia; reskilling hubs) • Information/education (short technical courses; guides; SME mentoring)
Sectoral Challenge Addressed	Shortage of installers/support in remote/rural areas; grid and procedural frictions; employment risks for fossil workers; fragmented purchasing power and limited access to credit.
Description of the Proposal	An integrated territorial programme with four pillars: skills & local capacity (training-first tenders, one-stop technical helpdesks), quality & grid (qualified supplier register, streamlined DSO procedures), just transition (targeted reskilling; territorial transition plans), and finance & aggregated demand (purchasing consortia; bill-positive microloans; six-month feedback loops).
Transformative Category	
Transformative scoring	
Feasibility scoring	

Policy code	IT_EN_N_3
Country	Italy
Sector	Electricity (individual/collective self-consumption, Renewable energy communities, household prosumers and SMEs)
Policy title: NETTING PACKAGE: ON-BILL CREDIT, BRIDGE ADVANCES, FASTER SETTLEMENTS AND TRANSPARENCY	
Policy Type	• Regulation (settlement rules; supplier obligations; data/API standards; standard contracts) • Economic incentives (revolving/guarantee fund; possible service-premium for on-bill) • Public services (helpdesk; reporting platforms; transparent dashboards) • Information/education (simple guides; training for REC managers; bill literacy)
Sectoral Challenge Addressed	Without on-bill compensation and with ex-post payments, families and small producers face cash-flow frictions and fail to perceive benefits ('value dispersion').
Description of the Proposal	Four coordinated measures: on-bill crediting (monthly/quarterly settlement), bridge advances via a revolving/guarantee fund, higher settlement frequency (monthly GSE–supplier–user cycles), and standard contracts with transparent dashboards showing production, expected revenues and anticipated bill discounts.
Transformative Category	5
Transformative scoring	5
Feasibility scoring	Short-term stabiliser

Policy code	IT_EN_N_4
Country	Italy
Sector	Energy (communication/consumer protection; supports solar/retrofit/REC uptake; retail markets, energy efficiency and renewables).
Policy title: “RENEWABLES, STABLE BILLS”: TRUST & TRANSPARENCY CAMPAIGN	
Policy Type	• Information/education (public campaign; guides; bill literacy; awareness on risks & rights) • Public services (local desks; hotlines; rapid reporting & response; public registers of qualified operators) • Regulation (mandatory codes of conduct; minimum transparency standards; reporting & enforcement obligations; qualified installer list)
Sectoral Challenge Addressed	Distrust towards commercial offers; misinformation linking renewables to higher costs; low bill/incentive literacy; information asymmetries and fraud risks that reduce informed uptake—especially among vulnerable and elderly users.
Description of the Proposal	A national-local campaign delivered by credible messengers (consumer associations, parish/community leaders, cooperatives, municipalities) with clear local examples of €/month savings and comfort gains. Segmented content (condominiums, low-income families, artisans, seniors) and dedicated call to actions-CTAs (one-stop desks, hotlines, step-by-step guides). The campaign is coupled with consumer-protection tools: mandatory codes of conduct, minimum transparency standards, public registers of qualified operators, anti-fraud checklists and rapid-response mechanisms (reporting forms, target response ≤72h) to build trust and enable safe, informed adoption.
Transformative Category	5
Transformative scoring	5
Feasibility scoring	Quick wins

Policy code	IT_EN_N_5
Country	Italy
Sector	Energy – distributed generation (wind <1–5 MW, PV <1 MW; energy communities, SMEs and light agrivoltaics)
Policy title: Renewable energy sources (RES) DIFFUSA 2030: INCENTIVI AI PICCOLI PRODUTTORI EOLICO-SOLARE	
Policy Type	• Regulation: streamlined permitting procedures for small-scale installations; dedicated registers/auctions; priority grid connection and minimum quality standards. • Economic incentives: CfDs/feed-in schemes for small capacities; CAPEX grants for inner areas and rooftop/marginal land installations; loan guarantees and on-bill financing. • Public services: technical one-stop shops; aggregation platforms (consortium-based Energy performance contract- EPC procurement, Operation&Maintenance, insurance); comparison tools for PPA/CER offers.

Policy code	IT_EN_N_5
Country	Italy
Sector	Energy – distributed generation (wind <1–5 MW, PV <1 MW; energy communities, SMEs and light agrivoltaics)
Policy title: Renewable energy sources (RES) DIFFUSA 2030: INCENTIVI AI PICCOLI PRODUTTORI EOLICO-SOLARE	
	• Information/education: guides for small producers; condominium/agri-PV toolkits; training for administrators and energy communities.
Sectoral Challenge Addressed	Reduce the structural advantage of incumbent actors in the electricity value chain by opening up the market and the grid to new small-scale entrants, decentralising generation, and increasing the bargaining power of households, SMEs and communities.
Description of the Proposal	A package of targeted and stable incentives for small-scale wind and photovoltaic installations, including dedicated auctions/registers, all-inclusive tariffs/two-way CfDs for small capacities, on-bill crediting, and priority grid connection in suitable nodes. The package also includes public guarantees and “bill-positive” microfinance, streamlined permitting procedures, and the aggregation of demand and supply through public platforms, while supporting energy communities and condominiums.
Transformative Category	Transformative bets
Transformative scoring	5
Feasibility scoring	1

Policy code	IT_EN_N_6
Country	Italy
Sector	Energy (cross-cutting to housing and mobility)
Policy title: PEP-LAB 2035: PARTICIPATORY & EVIDENCE-BASED PLANNING WITH LEAN DELIVERY	
Policy Type	• Regulation (mandatory participation; service level agreements (SLA); minimum indicators; review clauses) • Public services (territorial technical secretariats; digital platform; helpdesks) • Information/education (science→policy toolkits; training for officials/stakeholders) • Economic incentives (micro-grants for participation/support studies; performance awards)
Sectoral Challenge Addressed	Policies misaligned with real needs; slow/opaque processes; weak integration of scientific evidence and limited mid-course correction.
Description of the Proposal	Structured co-design with permanent multi-level tables and public agendas; transparent commitment tracking and review clauses. Lean operating sequence with standard procedures, SLA and bridge financing to accelerate delivery while ensuring accountability. A scientific clearinghouse translates interdisciplinary evidence into rules (indicators, benefit transparency, sharing rules, minimum governance requirements) and accessible outreach tools.
Transformative Category	5

Policy code	IT_EN_N_6
Country	Italy
Sector	Energy (cross-cutting to housing and mobility)
Policy title: PEP-LAB 2035: PARTICIPATORY & EVIDENCE-BASED PLANNING WITH LEAN DELIVERY	
Transformative scoring	1
Feasibility scoring	Structural shifts

Policy code	IT_EN_N_7
Country	Italy
Sector	Energy — electricity market design, retail markets, distributed generation (CERs/rooftop PV), consumer protection.
Policy title: POWER GROUPS IN THE ENERGY SECTOR: FAIR-COMPETITION & CONSUMER-VALUE REFORM	
Policy Type	• Regulation (market design reform; transparency & reporting; standard contracts; supplier obligations) • Economic incentives (two-way CfDs for RES basket; guaranteed micro-finance for small producers; targeted bill support for vulnerable households) • Public services (data-hub on price formation and pass-through; one-stop support for small producers/CERs) • Information/education (clear consumer guidance on contracts; bill literacy; PPA/CER comparison tools)
Sectoral Challenge Addressed	Concentration of market power and structural advantages for incumbent actors across the electricity value chain, leading to weak pass-through of renewables' lower costs, price volatility exposure, barriers to small producers and limited consumer welfare.
Description of the Proposal	A package to curb structural advantages of dominant players and increase the share of competitive, distributed, low-marginal-cost generation. It combines market design tools (two-basket pricing/CfD-backed RES basket), access & transparency rules (standard PPAs, open data on price formation, supplier obligations), and pro-new-entrant levers (priority connection windows and simplified routes for small producers, interoperable billing for on-bill crediting). Consumer value is ensured via price-stabilisation instruments and monitoring of pass-through, with penalties for unjustified margins.
Transformative Category	
Transformative scoring	
Feasibility scoring	

9.2. Transport

9.2.1. Adjustments in existing policies

Policy code	IT_TR_AD_1
Country	Italy
Sector	Transport
Policy Title: Economic incentives and concessions for local public transport • Economic incentives and concessions for local public transport ○ National extension beyond ISEE (for example, for the elderly). ○ Forming consortia of small municipalities. • Relevant policies: Transport Bonus – Decree-Law No. 50 of 17 May 2022, converted with amendments by Law No. 91 of 15 July 2022	
Policy Type	Economic incentives
Sectoral Challenge Addressed	• Reduce private traffic by promoting the equitable use of public transport. • Address barriers to transport access related to low income and certain disadvantaged population groups. • Promote accessibility in public transport to ensure access to services and opportunities across the territory.
Original Policy Objective	• Make public transport more accessible through discounts or free access for specific age groups. • Support urban mobility through demand-side incentives. • Currently, only the under-18 bonus is in effect; during the pandemic, there were more exemptions.
Existing Gaps	Problems with current concessions The current policy design does not fully take into account the profound demographic and socioeconomic changes occurring in the country. Concessions are often limited to age-based criteria rather than actual needs: currently, incentives are restricted to people under 18, retirees, low-income workers, and the unemployed. Family structures have changed: alongside large families – which remain among the most vulnerable – there is a generation (mid-1970s to early 1980s) without children that often finds itself solely responsible for caring for elderly parents or relatives with disabilities. In addition, there is an overall increase in family care burdens: very low birth rates, demographic aging, a growing need for continuous assistance, difficulties accessing healthcare, and a weakening health system. At the same time, the labor market produces a high number of low-income workers, so household income is often insufficient to guarantee social mobility. Even instruments such as the ISEE often prove inadequate: thresholds are too low and risk excluding individuals and households who, although not formally qualifying, are in fragile economic conditions. In this context, many people give up work or reduce their working hours to take care of family members, further contributing to economic vulnerability. These factors make current transport concession policies ineffective or not fully equitable, as they do not reflect the complexity of new family structures, caregiving burdens, and the actual conditions of income and access to services.

Policy code	IT_TR_AD_1
Country	Italy
Sector	Transport
Proposed Adjustment	Expand the categories covered by transport concessions, without focusing exclusively on age groups or linking eligibility solely to ISEE: • Incentives for people over 70 • Disadvantaged categories: residents in suburban or rural areas, workers with rigid shift schedules • Low-income individuals • All young people regardless of ISEE (under 18 + university students): under 26 • People with caregiving responsibilities (not only parental care but also care for elderly or disabled family members)
Transformative Category	Quick wins
Transformative scoring	5
Feasibility scoring	4

Policy code	IT_TR_AD_2
Country	Italy
Sector	Transport
Policy Title: National governance of Local Public Transport: funding, minimum standards, accessibility, and resources from fuel excise duties • Italian Transport Regulation Authority (ART) – established by Decree-Law No. 201/2011 • Legislative Decree No. 43 of 28 March 2025 (excise reorganisation) – Article 3, paragraph 6: provision allocating additional revenue from excise realignment to finance contractual renewal of local public transport (LPT) • Law 104/1992, Presidential Decree 503/1996, and Ministerial Decree of 18 July 1991 – regulations on accessibility requirements for vehicles (wheelchairs, ramps, dedicated spaces) • National Fund for Local Public Transport (Law No. 228/2012, paragraph 301) To include: regulations on: ○ Transport schedules ○ Use of excise revenues Rules on wheelchair weight limits	
Policy Type	Economic incentives Public services
Sectoral Challenge Addressed	• Promote greater attention to public transport as a real, functional alternative to private transport. • Strengthen collective transport by establishing a widespread network that reduces traffic and emissions. • Overcome territorial inequalities in the public transport network between north and south, and between urban and rural areas, by prioritizing funding for areas with greater connectivity needs (peripheral, rural, and internal areas).
Original Policy Objective	Allocate resources to finance regional and local public transport services

Policy code		IT_TR_AD_2
Country		Italy
Sector		Transport
Existing Gaps	• Insufficient resources and investments: the existing transport network is not widespread. • Public transport schedules are not adequately aligned with user needs (e.g., early morning or late-night shifts). • Lack of a comprehensive cultural vision for transport use: car culture remains dominant even in areas served by public transport such as the metro. • Poor maintenance of light infrastructure, such as bike lanes. • Legislative changes that reduce excise duties without providing alternative funding for local public transport.	
Proposed Adjustment	• Allocate more funds with an integrated vision that promotes a widespread network: ○ Targeted investments focusing on transport for peripheral and internal areas. ○ Provide alternative coverage for local public transport, which risks underfunding due to changes in excise management in the budget law. ○ Guaranteed funding for the maintenance and protection of bike lanes. • Attention to users: ○ Extended schedules: expand the local transport network to 24-hour service (modelled on the Vienna metro, which guarantees night service). ○ Revise incentives that prevent drivers from accommodating wheelchairs (e.g., maximum weight limits for wheelchairs). • Transparency in the use of funds and impact assessment.	
Transformative Category	Structural shifts	
Transformative scoring	5	
Feasibility scoring	4	

Policy code		IT_TR_AD_3
Country		Italy
Sector		Transport – Private green mobility
Policy Title: Incentives targeted at private transport (Ecobonus for the purchase of private electric vehicles)		
• Law No. 145 of 30 December 2018 (Budget Law 2019) – Article 1, paragraphs 1031–1047 • (This is the establishing law for the Ecobonus for cars) • Every year or funding cycle, the law is regulated by ministerial decrees, in particular: DM 20 March 2019; DM 6 April 2022; DM 4 August 2022; DM 2 April 2024 (latest Ecobonus car scheme)		

Policy code	IT_TR_AD_3
Country	Italy
Sector	Transport – Private green mobility
- Also by annual budget laws, in particular: Law No. 178/2020 (Budget 2021); Law No. 234/2021 (Budget 2022); Law No. 197/2022 (Budget 2023); Law No. 213/2023 (Budget 2024) - PNRR – National Recovery and Resilience Plan, in particular: Mission 2 – Green Revolution and Ecological Transition Incentives and support for the development of charging infrastructure - Directive 2014/94/EU – DAFI - Implemented in Italy through Legislative Decree No. 257 of 16 December 2016 (establishes national planning obligations, definition of publicly accessible charging points, interoperability, and technical standards) - PNIRE – National Infrastructure Plan for Electric Vehicle Charging, established by Legislative Decree 257/2016 and updated by MIT / MIMIT decrees (regulates national coverage targets, criteria for the installation of charging stations, coordinates State-Regions-Municipalities) PNRR – National Recovery and Resilience Plan, Mission 2 – Green Revolution and Ecological Transition	
Policy Type	Economic incentives
Sectoral Challenge Addressed	- Reduce pollution from the vehicle fleet by promoting more sustainable mobility. • Redirect incentives that currently prioritize individuals with high purchasing power.
Original Policy Objective	Encourage the green transition of private transport. Meet private demand for vehicles that are more environmentally sustainable. Reduce emissions from the private vehicle fleet.
Existing Gaps	- Incentives are focused on private cars and are easily accessible to individuals with medium-to-high income or VAT registration (leasing is predominantly used by VAT holders). - Current incentives benefit a wealthy class/target group → regressive effect; people with limited resources continue using diesel cars, which are later excluded from circulation. - Insufficient integration of charging infrastructure, and apps/providers are not interoperable. - Contradiction with traffic reduction objectives (PNRR): incentivizing electric cars does not necessarily reduce car usage.
Proposed Adjustment	Remove barriers to accessing electric vehicles by introducing additional eligibility criteria, such as: family or caregiving responsibilities, residence in suburban/rural areas, work with inconvenient shifts (e.g., night work), low income, and persons with disabilities.

Policy code	IT_TR_AD_3
Country	Italy
Sector	Transport – Private green mobility
	• Introduce specific fiscal measures (e.g., VAT reductions or additional deductions for vehicles serving people with disabilities). • Potential inclusion as part of corporate welfare policies: ○ Tax deductions for purchasing electric vehicles ○ Mobility managers (a position required by law in companies with more than 50 employees) promoting the use of electric vehicles • Regulate the integration and interoperability of charging infrastructure (mandatory standards / integration between providers).
Transformative Category	Quick wins
Transformative scoring	3,5
Feasibility scoring	3

Policy code	IT_TR_AD_4
Country	Italy
Sector	Transport – Digitalisation of mobility services
Policy Title: Digital literacy policies (single platform and support for the elderly)	
National policies supporting digital literacy: 1. National Strategy for Digital Skills (coordinated by the Department for Digital Transformation) ○ Objectives: universal digital literacy, reduction of the digital divide, skills development for public administration, workers, and citizens ○ Target groups: adults, workers, elderly, NEET youth 2. PNRR – Mission 1 (Digitalization) ○ Relevant measures: digital training for citizens, digital skills development within the public administration	
Policy Type	Public services Information/education
Sectoral Challenge Addressed	Bridge the digital gap and the fragmentation of apps and providers, ensuring access for all age groups, with a focus on the elderly population.
Original Policy Objective	Promote access to devices and basic digital skills to enable participation in public services and shared mobility.
Existing Gaps	• Bonuses are focused on purchasing devices rather than on platform integration or supporting services. • Fragmentation of apps (e.g., ATAC subscriptions do not unlock Lime; different charging providers are not interoperable). • Fragmentation of digital systems and operators: each actor uses its own system without integration (“everyone works in their own garden”), making coordinated management of digital mobility services difficult.

Policy code	IT_TR_AD_4
Country	Italy
Sector	Transport – Digitalisation of mobility services
	• Absence of a single data aggregator: there are no unified rules or tools for data collection, interoperability, and use across services, providers, and institutions. • MaaS (Mobility as a Service) project is insufficient: the current setup has favoured different operators without addressing data protection, investment, or true service integration issues. • Lack of a data-driven culture: limited practice in conducting social impact assessments and using data to improve the organisation and accessibility of digital services. • Platform design is not rights-oriented: app and digital tool designers do not systematically include accessibility, equity, and inclusion requirements. • Weakness of coordination bodies: digital task forces and governance structures operate intermittently, without a clear or stable role. • Issues related to digital identity (SPID): reconsideration of the tool may reduce accessibility; specific difficulties exist for elderly users without smartphones or limited in using SMS. • Absence of a coherent national digital strategy: inconsistent choices between the digital department, transport sector, and other administrations; lack of a systemic vision. • Time pressure from the PNRR: the “race to spend funds” compromises the quality of digital design and the creation of sustainable long-term solutions.
Proposed Adjustment	• Simplify access to digital services for sustainable transport; current policies focus only on bonuses. • Develop an integrated public platform (one-login/one-app) that allows access to multiple providers without additional costs (currently additional fees exist) and to public transport services. • Implement targeted digital literacy programs: Learning points offering support through civil service personnel to assist elderly users with digital tools. • Simplify access procedures (reduce additional costs and technical barriers). • Conduct periodic social impact assessments on the platforms
Transformative Category	Quick wins
Transformative scoring	3,5
Feasibility scoring	5

Policy code	IT_TR_AD_5
Country	Italy
Sector	Transport
Policy title: Transparency on traffic rules and ZTL (Limited Traffic Zones) 1. Highway Code – Legislative Decree No. 285 of 30 April 1992 ○ Key articles: ▪ Art. 6 – Regulation of traffic in urban areas ▪ Art. 7 – ZTL, pedestrian areas, restrictions and prohibitions ▪ Art. 38 – Road signage ▪ Art. 45 – Approval of control devices ○ Relevance for transparency: Traffic restrictions must be justified; ZTL areas must be clearly signposted; rules must be known before access. Without adequate signage and information, penalties are illegitimate.	
2. Implementation Regulation of the Highway Code – DPR No. 495 of 16 December 1992 Contains: technical specifications for vertical and horizontal signage, methods of prior notice, information panels for ZTLs, and electronic controls	
Policy Type	Regulation Information/education
Sectoral Challenge Addressed	Reduce traffic and pollution
Original Policy Objective	The policy was design with the same purpose, the integration aims at making information in traffic easier to find
Existing Gaps	Lack of clear information on restrictions: Circulation restrictions (ZTLs, pedestrian areas, time slots, exemptions) are often fragmented across different regulations, written in technical-administrative language, and difficult to access before entering the area. This reduces the social acceptability of the measures. Inadequate signage: In many urban contexts, signage is poorly visible or provided too late, does not clearly distinguish between ZTLs, pedestrian areas, and reserved lanes, and does not indicate alternatives (public transport, park-and-ride facilities). Data not accessible: Essential information is often unavailable in open data formats, not updated, and difficult to interpret (PDFs, scans, legal language). Absence of unified portals: Each municipality uses different portals and structures information in non-standard ways, making consultation difficult for commuters, tourists, and logistics workers. This creates systemic impacts, information fragmentation, increases unintentional errors, and penalises intermodal mobility. Hotline and direct assistance: In many municipalities, there is no rapid clarification channel; users often learn about violations only after the fact.
Proposed Adjustment	Unified portals; digital signage; information hotlines.
Transformative Category	Quick wins
Transformative scoring	3,5
Feasibility scoring	5

Policy code	IT_TR_AD_6
Country	Italy
Sector	Transport
Policy Title: Restriction of freight traffic, urban logistics, and tourist vehicle traffic • Highway Code – Legislative Decree 285/1992 ○ Key articles: ▪ Art. 6 – Traffic restrictions outside urban areas ▪ Art. 7 – Regulation of traffic within urban areas ▪ Art. 158 – Parking and stopping (freight loading/unloading) • Urban and planning policies – Urban Traffic Plans (PUT) ○ Mandatory for municipalities with over 30,000 inhabitants. They regulate freight access, loading/unloading hours, and rationalization of urban distribution. • Sustainable Urban Mobility Plans (PUMS)	
Policy Type	Regulation Public services
Sectoral Challenge Addressed	Reduce freight traffic, congestion, and emissions in urban centers while ensuring logistical efficiency, equity among operators, and quality of public space.
Original Policy Objective	Regulate freight traffic in urban areas
Existing Gaps	Need for further limitation
Proposed Adjustment	Adjust criteria, destinate a specific logistic area in the cities Limit the circulation of vehicles dedicated to tourist transport in historic city centers. Establish Urban Logistics Areas (ALU), a network of Urban Consolidation Hubs (HUC), and smart time windows. Improve public transport services to meet the demand for tourist mobility
Transformative Category	Structural shifts
Transformative scoring	5
Feasibility scoring	5

9.2.2. New policies

Policy code	IT_TR_N_1
Country	Italy
Sector	Transport (Freight Transport – Logistics and Intermodality)
Policy title: National Plan for Intermodal Freight Transport – Rail and Road	
Relevant policies: General Transport and Logistics Plan (PGTL) Policies on road transport and logistics – Law 298/1974 (road transport) Direct incentives for intermodality – Ferrobonus PNIEC – National Integrated Energy and Climate Plan PNRR – Mission 2, Component 2 (M2C2) and measures related to fleet renewal and logistics infrastructure Integration with EU policies (essential): a national plan must be consistent with TEN-T, the Green Deal, Fit for 55, and the EU Regulation on rail freight transport	
Policy Type	Public services: infrastructure Issues with requirements: standards, specifications
Sectoral Challenge Addressed	Freight transport service is inefficient, highly polluting, and contributes to traffic congestion.
Description of the Proposal	- Improve intermodality of international freight transport across sea, rail, and road, including at the transnational level. - Develop infrastructure and regulations to promote the transfer of freight traffic from road to rail using intermodal systems (e.g., trucks directly loaded onto railcars, use of high-speed rail for freight), relaunch mixed and dedicated freight services, and provide incentives to operators to create intermodal hubs. Best practices from Eastern European countries, discussed in the consultation, are used as a reference. - Cross-border workers: ensure paid protection and economic safeguards agreed upon by both countries.
Transformative Category	Transformative bet
Transformative scoring	5
Feasibility scoring	1

Policy code	IT_TR_N_2
Country	Italy
Sector	Transport – Governance and public planning
Policy title: Awareness campaign for policymakers and users on public transport.	
Policy Type	Information/education
Sectoral Challenge Addressed	Guide investment decisions toward sustainable infrastructure solutions.
Description of the Proposal	Information campaigns; awareness campaigns; comparative data; workshops for administrators; independent observers.
Transformative Category	Structural Shift
Transformative scoring	5
Feasibility scoring	5

Policy code	IT_TR_N_3
Country	Italy
Sector	Transport – social inclusion
Policy title: Transport safety for women and LGBTQIA+ people	
Policy Type	Public services Information/education
Sectoral Challenge Addressed	Ensure safe access to services
Description of the Proposal	Raise awareness among users and transport staff about gender-based violence and harassment to support safe access to public transport for women, LGBTQIA+ people, and persons with disabilities.
Transformative Category	Structural Shift
Transformative scoring	5
Feasibility scoring	5

10. Spanish policy portfolio

10.1. Energy

10.1.1. Adjustments in existing policies

Policy code	ES_EN_AD_1
Country	Spain
Sector	Energy
Policy Title: National Energy Poverty Strategy	
Policy Type	Regulation Economic incentives
Sectoral Challenge Addressed	Addressing energy inefficiency from different perspectives
Original Policy Objective	The National Strategy against Energy Poverty in Spain (ENPE), currently in its 2025-2030 phase, is a government plan to reduce the lack of access to affordable energy through structural measures and protection for vulnerable consumers. It focuses on the energy-efficient renovation of homes, improvements to the social energy voucher, a ban on disconnections during the winter months, and the creation of a "social tariff" or "minimum vital consumption" to guarantee energy as a basic right, seeking a fairer and more efficient system. Key Components of the Strategy: 1) Energy-efficient renovation of homes. 2) Specific programs to improve insulation and efficiency in vulnerable households, using subsidies and energy audits. 3) Protection of vulnerable consumers. 4) Social Energy Voucher (improved accessibility and coverage). 5) Minimum vital supply (guaranteeing basic consumption without interruptions, especially in winter). 6) Social tariff (proposal for a reduced VAT rate for minimum vital consumption to incentivise savings). Structural Measures: addressing long-term energy poverty through educational policies and intergovernmental coordination. Multisectoral approach: involving civil society, businesses, academia, and the third sector in its design and implementation.
Existing Gaps	1) The need to address structural problems (helping people afford the price is one thing, but helping them understand how their money is going is another. 2) The legislation focuses not on the most vulnerable, but on the middle class who can afford to pay upfront for installation, renovations, etc. 3) Problems with the allocation of aid (general public lack of awareness) and with payments (beneficiaries initially have to pay upfront, and not all families can afford that).
Proposed Adjustment	1) The social bonus should be automatic as soon as a vulnerable group is identified (by "automatic" they mean reducing the bureaucracy to access it). 2) The policy must be more profound in its intended changes (the changes must be structural, not merely cosmetic). 3) The law must explicitly include an ambition to end energy poverty (the current draft lacks medium-term indicators for eradicating it (what the participants were saying is that there needs to be a political ambition more

Policy code	ES_EN_AD_1
Country	Spain
Sector	Energy
	powerful than any regulation, and that is the intention to eradicate poverty at its root. No concrete ideas were offered on how to achieve this, but all participants agreed that such a commitment from public authorities would be very beneficial). 4) In Spain, 43% of homes are not energy efficient, and policies must address this problem at its root. 5) The policy should consider not only residential buildings but also public and service buildings (e.g., schools, hospitals, nursing homes, community centers, etc.). It is limited to households affected by, or at risk of being affected by, energy poverty.
Transformative Category	
Transformative scoring	
Feasibility scoring	

Policy code	ES_EN_AD_2
Country	Spain
Sector	Energy
Policy Title: Hydrogen Roadmap	
Policy Type	Regulation Public services
Sectoral Challenge Addressed	Pollution is a problem that affects the quality of life, so we need to find new ways to deal with it.
Original Policy Objective	The Hydrogen Roadmap in Spain is a strategic plan to promote renewable hydrogen, key to decarbonization and climate neutrality by 2050, setting ambitious targets for 2030 (4 GW of electrolyzers) and 2050. With legislative and incentive measures (taxes, guarantees of origin), it tries to decarbonise industry, mobility and electricity, seeking to create a high-value national industry, supported by European frameworks such as the H2Med corridor.
Existing Gaps	Regulation Public services
Proposed Adjustment	1) Improve the technology for its storage and transport. 2) Adapt the scale of impact (this law may work on a small scale—small communities—but not on a large scale—cities). 3) Proposal for the generation of energy clusters (also useful for transportation).
Transformative Category	
Transformative scoring	
Feasibility scoring	

10.1.2. New policies

Policy code	ES_EN_N_1
Country	Spain
Sector	Energy
Policy title: Rehabilitation of homes for vulnerable people.	
This housing renovation policy was explained as parallel to or complementary to the National Strategy against Energy Poverty. While the National Strategy against Energy Poverty is a strategic framework of a social nature, whose main objective is to ensure that vulnerable households can meet their basic energy needs, the housing renovation policy for energy efficiency was conceived, above all, as an investment policy in the existing building stock.	
Policy Type	- Regulation (laws, standards, bans, requirements) - Economic incentives (subsidies, taxes, grants, penalties) - Public services (programmes, infrastructure investments)
Sectoral Challenge Addressed	Energy poverty
Description of the Proposal	Funds for the rehabilitation of homes for vulnerable households, with comprehensive insulation, accessibility and energy generation. Some of the criteria discussed were: geographical areas with more extreme climates (for example, the center of the peninsula has more temperature variations than the coast), age of the houses, special needs (unspecified or, at least, we do not remember them).
Transformative Category	Quick wins
Transformative scoring	4
Feasibility scoring	5

Policy code	ES_EN_N_2
Country	Spain
Sector	Energy
Policy title: Progressive and automatic energy tariff for vulnerable households	
Policy Type	- Regulation (laws, standards, bans, requirements) - Economic incentives (subsidies) - Public services (subsidies control programmes)
Sectoral Challenge Addressed	Current energy pricing structures often do not account for household income, size, or specific needs, leaving low-income or dependent households at a financial disadvantage.
Description of the Proposal	Implement a progressive energy tariff, where the price per kWh depends on: 1) Income level; 2) Number of people in the household; 3) Specific needs (age, dependency, illness). The application would be automatic, cross-referencing tax and social data, without requiring a specific request.
Transformative Category	Transformative bets
Transformative scoring	4
Feasibility scoring	3

Policy code	ES_EN_N_3
Country	Spain
Sector	Energy
Policy title: Social energy communities for vulnerable households	
Policy Type	- Regulation (laws, standards, bans, requirements) - Economic incentives (subsidies) - Public services (subsidies control programmes and information and education plan for local and regional authorities) - Information/education: public campaigns.
Sectoral Challenge Addressed	Traditional energy systems often fail to address the specific needs of vulnerable populations, leading to inequality in energy consumption and high energy bills. The challenge lies in developing inclusive, community-driven energy solutions that enable vulnerable households to actively benefit from renewable energy sources and reduce energy costs.
Description of the Proposal	The aim is to democratise access to renewable energy. This would be achieved through the creation of community-based energy communities, promoted by local councils or public entities. The National Strategy Against Energy Poverty 2021-2024 (ENPE) emphasises the importance of community-based solutions, such as shared energy projects, to empower local populations.
Transformative Category	Transformative bets
Transformative scoring	4
Feasibility scoring	2

Policy code	ES_EN_N_4
Country	Spain
Sector	Energy
Policy title: Public fund for the stabilization of energy prices for vulnerable households	
Policy Type	- Regulation (laws, standards, bans, requirements) - Economic incentives (taxes to companies) - Public services (subsidies control programmes)
Sectoral Challenge Addressed	Electricity and other energy bills are linked to crises and contextual events. Electricity isn't always expensive, but sometimes it's excessively so and can financially impact many families.
Description of the Proposal	With the aim of protecting people from energy crises and sudden price increases, the creation of a public fund is proposed to automatically offset bills when prices exceed certain thresholds. It would be financed by taxes on windfall profits in the energy sector.
Transformative Category	Transformative bets
Transformative scoring	5
Feasibility scoring	2

Policy code	ES_EN_N_5
Country	Spain
Sector	Energy
Policy title: Personalised public energy support service	
Policy Type	- Regulation (laws) - Economic incentives (high inversion and payments of salaries) - Public services (control and subsidies programmes)
Sectoral Challenge Addressed	Subsidies to combat energy poverty are difficult to understand and the bureaucracy is complex.
Description of the Proposal	With the aim of transforming passive aid into lasting solutions, the creation of local energy advisory offices is planned. These offices would offer: 1) contract optimization; 2) support with aid applications; and 3) basic training in energy efficiency. Services could be provided in person or digitally, but must always be adapted to the needs of older adults or those with limited digital literacy.
Transformative Category	Transformative bets
Transformative scoring	5
Feasibility scoring	3

Policy code	ES_EN_N_6
Country	Spain
Sector	Energy
Policy title: Integrating Energy Poverty into Public Health Policies	
Policy Type	- Regulation (Legal recognition of energy poverty as a social determinant of health, which would entail public health laws and national and regional health strategies). - Economic incentives (specific subsidies, economic incentives for energy rehabilitation based on health criteria, penalties for non-compliance) - Public services (energy poverty screening programs in primary care, energy support services prescribed by health professionals, investment in efficient social and healthcare infrastructure) - Information/education (mandatory training for healthcare and social services personnel, clinical and social guidelines with clear protocols for action, public awareness campaigns)
Sectoral Challenge Addressed	Energy poverty affects health, as the inability to heat your home in winter can lead to respiratory problems, and inadequate heating in summer can cause heat-related illnesses. It also has negative effects on mental health.
Description of the Proposal	This involves recognizing the health dimension of energy poverty. To achieve this, protocols for detecting energy poverty are needed, starting with primary care and social services. This would work by automatically activating energy assistance in cases of health risk. Its contribution to ending energy poverty: 1) preventing illnesses associated with cold, extreme heat, or poor indoor air quality; 2) reducing healthcare costs in the medium term.

Policy code	ES_EN_N_6
Country	Spain
Sector	Energy
Policy title: Integrating Energy Poverty into Public Health Policies	
Transformative Category	Transformative bets
Transformative scoring	5
Feasibility scoring	2

10.2. Transport

10.2.1. Adjustments in existing policies

Policy code	ES_T_AD_1
Country	Spain
Sector	Transport
Policy Title: Plan MOVE	
Policy Type	Regulation Economic incentives Public services Information/education
Sectoral Challenge Addressed	- Electric and plug-in hybrid vehicles still have higher purchase prices than conventional internal combustion vehicles, which discourages adoption among households and small businesses. - Limited availability of charging points—especially in residential buildings and rural or semi-urban areas—creates “range anxiety” and slows EV adoption. - Uncertainty around demand, resale value, and infrastructure discourages manufacturers, dealers, and consumers from committing to electric mobility.
Original Policy Objective	The MOVES III Plan is a subsidy program in Spain to promote sustainable mobility, offering grants for the purchase of electric/plug-in hybrid cars (up to €9,000 or €7,000 with scrapping) and for the installation of charging points. Valid until the end of 2025, it is managed by the Autonomous Communities and has limited funding, meaning applications are processed on a first-come, first-served basis and may be taxable under Personal Income Tax (IRPF). What is subsidised? 1) Purchase of vehicles: Electric, hydrogen (FCEV), or plug-in hybrid (PHEV). 2) Up to €9,000 for zero-emission vehicles (or €7,000 with the trade-in of an old car). 3) Minimum discount of €1,000 from the manufacturer/dealer. 4) Installation of charging points (both public and private, including pre-installations in apartment buildings). 5) There is also aid available for electric motorcycles, electric cargo bikes, and transportation systems for commuting to workplaces. Key Points and Considerations: 1) Limited funds (applications are processed in the order they are received. Funds are exhausted by

Policy code	ES_T_AD_1
Country	Spain
Sector	Transport
	regions, and there may not be enough for everyone). 2) Advance payment: Brands can advance the amount, but it is a loan that is repaid with the official aid, which takes time to arrive. 3) Income Tax (the aid received must be declared on your income tax return as a capital gain, with different percentages depending on your income level). 4) Administration (the Autonomous Communities manage the applications and payments).
Existing Gaps	1) Economic: Disability is very expensive, and not everyone has the money to invest in buying an electric vehicle. 2) Civic: If people don't see the importance of owning an electric car, they won't understand the urgency of acquiring one. 3) Inequalities: The regional management of aid creates inequalities between regions and individuals. 4) Negative effects on multiple disabilities: Those who are focused on survival are not concerned about electric cars. 5) Lack of awareness of individual rights to obtain financial aid to finance the purchase of an electric car. 6) The focus is solely on private cars; the renewal of the public transport fleet is not taken into account. 7) Imbalance between coercive measures against using non-electric cars and measures that incentivise the purchase of electric vehicles.
Proposed Adjustment	1) Change of purpose: shifting from the private sphere (promoting electric private cars) to the public sphere (promoting improvements to public transport with zero-emission vehicles). This would imply a greater focus on public transport (nothing was said about eliminating private transport, but rather about placing a more precise emphasis on public transport). 2) This change could allow for the consideration of adaptability measures for everyone, including vulnerable groups.
Transformative Category	
Transformative scoring	
Feasibility scoring	

10.2.2. New policies

Policy code	ES_T_N_1
Country	Spain
Sector	Transport
Policy title: Universal Mobility and Transport Program	
Policy Type	Regulation: include training in educational plans -Economic incentives: fines for non-compliance -Public services: educative programmes. -Information/education: public campaigns.
Sectoral Challenge Addressed	

Policy code	ES_T_N_1
Country	Spain
Sector	Transport
Policy title: Universal Mobility and Transport Program	
	- Address barriers to transport access related to low income and certain disadvantaged population groups. - Promote accessibility in public transport to ensure access to services and opportunities across the territory.
Description of the Proposal	Raising awareness through education about the need for universal transportation for all.
Transformative Category	Quick wins
Transformative scoring	4
Feasibility scoring	4

Policy code	ES_T_N_2
Country	Spain
Sector	Transport
Policy title: Guarantee universal access to means of transport	
Policy Type	Ensuring universal access to public transportation requires: -Regulation: legislation -Economic incentives: likely aid or subsidies -Public services: increased public services (in number, frequency, staffing, etc.) -Information/education: to help people choose between public and private options.
Sectoral Challenge Addressed	- Reduce reliance on private vehicles by encouraging fair and widespread use of public transport. - Address transport access barriers linked to low income levels and disadvantaged population groups. - Promote accessibility in public transport to ensure equal access to services and opportunities across the territory.
Description of the Proposal	-To achieve an urban transport network accessible to all (subway, bus, scooters, bicycles, etc.). -To achieve full accessibility in intercity transport (plane, ship, etc.).
Transformative Category	Transformative bets
Transformative scoring	5
Feasibility scoring	2

Policy code	ES_T_N_3
Country	Spain
Sector	Transport
Policy title: Taxes on private transport	
Policy Type	-Regulation -Economic incentives

Policy code	ES_T_N_3
Country	Spain
Sector	Transport
Policy title: Taxes on private transport	
Sectoral Challenge Addressed	Progressive taxes on private transport to obtain funds to finance public transport and contribute to its improvement and universal adaptability. Note: Taxes on private transport already exist (for example, vehicle registration tax, road tax, etc.). The proposal is to make these taxes progressive (i.e., more polluting cars pay more, families with more than one car pay more, etc.).
Description of the Proposal	-To achieve an urban transport network accessible to all thanks to obtaining extra funding derived from the extra tax on private transport.
Transformative Category	Transformative bets
Transformative scoring	5
Feasibility scoring	3

Policy code	ES_T_N_4
Country	Spain
Sector	Transport
Policy title: Exclusive lane for public transport on tensioned (congested) national roads and urban accesses	
Policy Type	-Regulation: providing high priority status for public transport use -Economic incentives: fines for non-compliance for invading public transport lane or for not providing exclusive public transport lane in tensioned roads and urban accesses -Public services: educative programmes. -Information/education: public campaigns.
Sectoral Challenge Addressed	Congestion and traffic saturation on key roads and urban access points limit the efficiency and reliability of public transport, preventing it from operating with priority over private vehicles.
Description of the Proposal	Prioritizing the use of public transport and ensuring its accessibility value
Transformative Category	Transformative bets
Transformative scoring	5
Feasibility scoring	3

Policy code	ES_T_N_5
Country	Spain
Sector	Transport
Policy title: Progressive cost of public transport monthly pass	
Policy Type	-Regulation: establishing a progressive cost of public transport monthly pass (from minimal cost for people at risk of social exclusion to cost according to monthly rent) -Economic incentives: subsidies for most vulnerable citizens to enable their access to work, health or education opportunities. -Public services: educative programmes.

Policy code	ES_T_N_5
Country	Spain
Sector	Transport
Policy title: Progressive cost of public transport monthly pass	
	-Information/education: public campaigns.
Sectoral Challenge Addressed	High congestion in urban areas and the affordability of transport costs reduce the attractiveness and accessibility of public transport, encouraging continued reliance on private vehicles—particularly among lower-income users.
Description of the Proposal	Prioritizing the use of public transport and ensuring its accessibility value for most vulnerable cohorts
Transformative Category	Quick wins
Transformative scoring	5
Feasibility scoring	4

Policy code	ES_T_N_6
Country	Spain
Sector	Transport
Policy title: Tax on single occupancy of private vehicles	
Policy Type	-Regulation: establishing a penalty / tax for people who drive their car alone -Economic incentives: introducing a new tax can limit the number of circulating private vehicles, reduce energy consumption and promote more responsible and sustainable use of transport options (car sharing). -Public services: educative programmes. -Information/education: public campaigns.
Sectoral Challenge Addressed	High levels of single-occupancy private vehicle use contribute to traffic congestion, inefficient use of road space, and increased emissions, while shared transport options remain underutilised.
Description of the Proposal	Limiting the use of private vehicles by single persons
Transformative Category	Quick wins
Transformative scoring	4
Feasibility scoring	5

Policy code	ES_T_N_7
Country	Spain
Sector	Transport
Policy title: Tax on corporate parkings in urban areas (parking provided by companies for their employees)	
Policy Type	-Regulation: establishing a corporate tax on companies located in cities that provide parking for their employees and thus facilitate /encourage the use of private vehicles -Economic incentives: introducing a new corporate tax can make companies redefine their transport policies for employees (providing a

Policy code	ES_T_N_7
Country	Spain
Sector	Transport
Policy title: Tax on corporate parkings in urban areas (parking provided by companies for their employees)	
	collective transport option instead of facilitating the use of personal vehicles) -Public services: educative programmes. -Information/education: public campaigns.
Sectoral Challenge Addressed	The widespread provision of free or subsidised corporate parking in urban areas encourages commuting by private car, exacerbating congestion and undermining the competitiveness and use of public transport in already congested city centers.
Description of the Proposal	Prioritizing the use of public transport and limiting the use of private vehicles
Transformative Category	Quick wins
Transformative scoring	5
Feasibility scoring	4

Policy code	ES_T_N_8
Country	Spain
Sector	Transport
Policy title: Toll for all private vehicles to access and circulate in urban areas	
Policy Type	-Regulation: establishing a toll for all private vehicles entering urban areas -Economic incentives: introducing a new toll can make drivers reconsider their transport options -Public services: educative programmes. -Information/education: public campaigns.
Sectoral Challenge Addressed	Unrestricted access of private vehicles to urban areas leads to severe congestion, inefficient use of limited road space, and reduced speed and reliability of public transport, undermining its priority in densely populated areas.
Description of the Proposal	Prioritizing the use of public transport and limiting the use of private vehicles
Transformative Category	Quick wins
Transformative scoring	5
Feasibility scoring	5

Policy code	ES_T_N_9
Country	Spain
Sector	Transport
Policy title: Ban on the development of new parking lots in city centres	
Policy Type	-Regulation: establishing a ban for the development of new parking lots

Policy code	ES_T_N_9
Country	Spain
Sector	Transport
Policy title: Ban on the development of new parking lots in city centres	
	-Economic incentives: incentives to use public transport -Public services: educative programmes. -Information/education: public campaigns.
Sectoral Challenge Addressed	The continued expansion of parking supply in city centres encourages private car use, increases traffic congestion, and weakens the priority and efficiency of public transport in already saturated urban areas.
Description of the Proposal	Prioritizing the use of public transport and limiting the use of private vehicles
Transformative Category	Quick wins
Transformative scoring	5
Feasibility scoring	5

11. Portuguese policy portfolio

11.1. Energy

11.1.1. Adjustments in existing policies

Policy code	PT_EN_AD_1
Country	Portugal
Sector	Energy
Policy Title: SOLENERGE	
Policy Type	Economic incentives
Sectoral Challenge Addressed	Improve access to renewable energy solutions by reducing upfront financial barriers through direct support via certified suppliers, advance payments, and coverage of a percentage of eligible costs.
Original Policy Objective	Provide financial support for the acquisition and installation of photovoltaic solar panels, covering up to 100% of eligible costs, with a maximum support of €1,500.
Existing Gaps	Low-income and vulnerable households face difficulties accessing the scheme due to the need to pre-finance installation costs before reimbursement, leading to exclusion of those without sufficient liquidity.
Proposed Adjustment	Introduce upfront payment mechanisms and direct supplier payments to eliminate pre-financing requirements for beneficiaries, expand eligibility criteria, and simplify administrative procedures.
Transformative Category	Short-term stabiliser
Transformative scoring	2
Feasibility scoring	4

11.1.2. New policies

Policy code	PT_EN_N_1
Country	Portugal
Sector	Energy
Policy title: Accessible Zero-Emission Public Transport for All	
Policy Type	Public services
Sectoral Challenge Addressed	Lack of accessible and electric public transport solutions.
Description of the Proposal	Ensure that each municipality has at least one zero-emission and fully accessible bus, taxi, and minibus, supporting inclusive mobility and emissions reduction.
Transformative Category	Quick win

Policy code	PT_EN_N_1
Country	Portugal
Sector	Energy
Policy title: Accessible Zero-Emission Public Transport for All	
Transformative scoring	4
Feasibility scoring	4

Policy code	PT_EN_N_2
Country	Portugal
Sector	Energy and agriculture
Policy title: Agrivoltaics Azores	
Policy Type	Economic incentives (subsidies and grants) Public programmes
Sectoral Challenge Addressed	Sustainable agricultural production combined with renewable energy generation.
Description of the Proposal	Reduce the agricultural sector's energy footprint through non-repayable financial support for equipment and integrated digital and sustainable technologies.
Transformative Category	Transformative bet
Transformative scoring	4
Feasibility scoring	3

Policy code	PT_EN_N_3
Country	Portugal
Sector	Energy and transport infrastructure.
Policy title: The New Roads	
Policy Type	Public services (infrastructure investment)
Sectoral Challenge Addressed	Energy-efficient and low-emission road infrastructure.
Description of the Proposal	Promote sustainable road requalification by reducing polluting elements and integrating clean energy production and electric vehicle charging infrastructure.
Transformative Category	Non-starter
Transformative scoring	2
Feasibility scoring	2

11.2. Housing

11.2.1. New policies

Policy code	PT_HO_N_1
Country	Portugal
Sector	Housing and public infrastructure.
Policy title: Accessible Public Spaces	
Policy Type	Public services (investment in accessible infrastructure).
Sectoral Challenge Addressed	Lack of accessible infrastructure and public equipment for all users, including people with reduced mobility and visual impairments.
Description of the Proposal	This policy promotes universal access to public spaces by ensuring mobility-friendly infrastructure and adapted access systems, enabling equal use of public facilities.[S
Transformative Category	Quick wins
Transformative scoring	4
Feasibility scoring	4

Policy code	PT_HO_N_2
Country	Portugal
Sector	Housing and social inclusion
Policy title: Integrated Neighbourhoods	
Policy Type	Public services Information/training programmes
Sectoral Challenge Addressed	Social stigma and exclusion of residents in social housing, and the need to reposition social housing as a temporary and enabling solution.
Description of the Proposal	The policy provides direct support and shared responsibility with residents, combined with training across multiple areas, to prepare households for a transition towards personal and family independence beyond social housing.
Transformative Category	Transformative bets/Structural shifts
Transformative scoring	3
Feasibility scoring	2

11.3. Cross-cutting: Energy and Housing

11.3.1. New policies

Policy code	PT_ENHO_N_1
Country	Energy and Housing
Sector	Portugal
Policy title: Zero VAT on Energy-Efficient Equipment	
Policy Type	Economic incentives (tax reduction and subsidies).
Sectoral Challenge Addressed	Improving affordability and access to energy-efficient household equipment.

Policy code	PT_ENHO_N_1
Country	Energy and Housing
Sector	Portugal
Policy title: Zero VAT on Energy-Efficient Equipment	
Description of the Proposal	This policy introduces a 0% VAT rate on the purchase of domestic energy-efficiency equipment, reducing upfront costs and accelerating the adoption of more efficient technologies in households.
Transformative Category	Short-term stabiliser
Transformative scoring	2
Feasibility scoring	3

Policy code	PT_ENHO_N_2
Country	Energy and Housing
Sector	Portugal
Policy title: Renewable Neighbourhoods	
Policy Type	Public services (investment programmes in renewable energy infrastructure).
Sectoral Challenge Addressed	Limited access to renewable energy solutions for financially disadvantaged households.
Description of the Proposal	The policy supports the acquisition and installation of domestic renewable energy systems in social housing neighbourhoods, ensuring equitable access to clean energy and reducing energy poverty.
Transformative Category	Short-term stabiliser
Transformative scoring	2
Feasibility scoring	4

